

When Statistical Trust Layers Go Silent: Feasibility and Padding Evasion in ML Intrusion Detection

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Abstract—Statistical trust layers can combine conformal evidence, grouped security hypotheses, and online false-discovery-rate control to turn ML detector scores into calibrated alerts. We show that this composition creates two system-level failure modes and characterize their security consequences.

First, we derive an ex-ante feasibility law for bounded conformal evidence under two families of arrival-time controllers. It yields a finite discovery horizon beyond which a rejection-free run cannot alert regardless of detector score, and an optimal calibration-to-horizon scaling; at $k = 1$, LOND/e-LOND require 20 calibration units per tested hypothesis even under the best cold-start allocation.

Second, we formalize adversarial robustness when an attacker can enlarge its own grouped hypothesis and prove a class-wide impossibility: no alert-capable symmetric e-merging family is robust to arbitrary zero-evidence padding. For the arithmetic mean, we derive the exact suppression cost.

Finally, we validate these effects on LSPR23 and AIT-LDSv2.0. Real-flow replay confirms padding suppression, while joint controller-state reruns show that suppressing early alerts can reduce campaign cost by roughly two orders of magnitude relative to independently pricing alerts.

These results show that statistical validity of individual components does not guarantee feasibility or adversarial robustness of the composed alerting system.

I. Introduction

A network intrusion detector produces scores; a security operations centre (SOC) acts on alerts. AUROC, AUPRC, and threshold recall describe a classifier, but they do not tell an operator how much of the resulting alert queue is false. At the low base rates typical of intrusion detection, even a strong classifier can produce a queue dominated by false positives [1], [2].

Online multiple testing provides a natural control layer. At each time t , a procedure tests one hypothesis, makes an immediate decision, and updates future testing levels while controlling an online false-discovery criterion [3], [4], [5]. Conformal calibration can turn detector scores into distribution-free evidence under exchangeability [6], [7]; e-values can support false-discovery-rate (FDR) control under arbitrary dependence [8], [9]. These components form a modular statistical trust layer: detector $\rightarrow$ conformal evidence $\rightarrow$ security grouping $\rightarrow$ online controller $\rightarrow$ alert (Fig. 1). Prior work combines subsets of this chain: anomaly scores with online FDR control, conformal evidence with online or batch decisions, or calibrated streaming intrusion alerts [10], [11], [12], [13], [14]. We

assemble the full composition as an evaluated design and study a failure mode that is absent from the guarantee of any single module.

The first problem is feasibility. A finite calibration set places a hard upper bound on deterministic conformal evidence, while an uninterrupted online controller distributes a finite error budget over a potentially long stream. Along a rejection-free run, its offered levels can eventually become too small for even the maximum admissible evidence. At that point, no detector output can trigger an alert. This structural inability to report a discovery and its onset can be calculated before deployment.

A natural repair is to test fewer hypotheses. SOC already aggregate flows into host-, service-, or time-based episodes; coarser grouping reduces the horizon that the controller must cover. Yet the same operation determines which observations share a hypothesis. When group membership depends on endpoints, services, and time, an attacker can often directly influence it by generating additional traffic.

We connect these facts into one causal chain. Our contributions are:

A deployment-level feasibility law (Section III). We turn the interaction between finite conformal resolution and online error spending into an attained, ex-ante deployment criterion: given $|\mathcal{C}|$, k , and T , one can decide before deployment whether a covered controller can possibly report an alert throughout the intended horizon, independently of the detector. Grouping restores feasibility at exactly the criterion’s exchange rate. Switching the primary window from a host-pair to a source-host key raises canonical detections from 3 to 46 and the median exact suppression cost, from 24 to 496 added flows. So one grouping choice sets feasibility, alert semantics, and attack cost together (Section IV).

A cross-arity security impossibility (Section V). E-merging theory studies a fixed set of e-values; our adversary can enlarge its own hypothesis. We formalise robustness across arities and prove that, under attacker-controlled, unbounded membership, no symmetric e-merging family can both attain an alert threshold and remain above it after arbitrary zero-evidence padding. For the arithmetic mean, the bound is tight and gives the exact suppression cost.

An operational attack boundary (Sections V and VI).

TABLE I

The contribution boundary: inherited mathematics, our security formalisation, and the resulting system-level consequences.

Inherited result	Our formalisation $\rightarrow$ consequence
Finite conformal resolution [6], [15], [16]; alpha-death and low power at small levels [17], [4]	Formalise: cold-start feasibility for specified controller-sequence pairs. Derive: an absorbing horizon for two families and an attained ex-ante sizing law (Theorems 1 and 2 and Corollary 1).
Fixed-arity domination of symmetric e-merging rules [8]	Formalise: cross-arity robustness when an attacker enlarges its own hypothesis (Definition 1). Derive: a class-wide impossibility and the mean's exact cost bound (Theorem 3).
Batching can recover online-testing power [18]; zero inputs dilute an arithmetic mean	Formalise: grouping, attacker-controlled membership, and controller state jointly. Derive: one ratio for best-case feasibility and universal integer dilution under Corollary 2, plus the joint-cost feedback in Section V-D.

We evaluate the attack along a capability ladder: a sequential oracle that sizes each pad exactly from the realised controller state, a retrospective attacker that replays real benign-to-victim traffic, and a temporally restricted, state-free attacker limited to traffic the victim had already received and to its own episode size. The distance between the rungs is a measured result rather than an assumption: on AIT-LDSv2.0 the retrospective attacker suppresses 78 of 79 flow-only true detections under the canonical order, while the victim-prior and state-free arms leave 24 to 65 standing, because template availability and attacker knowledge bind before the algebra does. Controller allocation raises the price of an individual alert by two orders of magnitude, but a joint rerun on LSPR23 shows a controller-state feedback: once early alerts are suppressed, later alerts remain near their cheaper cold-start levels. A capacity-constrained attacker shows where a membership bound would interrupt the attack; we do not present that constraint as an end-to-end admission policy.

The key distinction is between weak evidence and an impossible decision. Once evidence cannot exceed M , level decay produces a calculable step after which no admissible detector output can trigger a discovery. Maximising that horizon over spending sequences yields an attained deployment boundary in $|\mathcal{C}|$, k , and T (Corollary 1), rather than another post-hoc measure of detection power.

Our main evaluation uses 16,353,511 flows from LSPR23 [19]. The chronological position 0.55 defines the primary window, and 0.62 provides a secondary check. Position 0.85 has a benign tail that violates marginal e-validity, so we use it only as an explicitly marked stress window for mechanism and cost measurements. AIT-LDSv2.0 provides a second, benign-inclusive setting. The appendices report the full windows, seeds, ordering variants, and validity diagnostics; Fig. 1 maps the argument to the paper structure.

II. Statistical Trust Layer and Evaluation Model

A. Bounded evidence and online decisions

Flows $x_1, x_2, \dots$ arrive in timestamp order, and a detector f assigns each flow a score $s_i = f(x_i)$. We convert the score to threshold conformal evidence using a benign calibration set $\mathcal{C}$ that is disjoint from training and precedes deployment. Let $K_i = 1 + \#\{c \in \mathcal{C} : s_c \geq s_i\}$ denote the rank of s_i from the top of the calibration scores, with ties resolved against the test flow. At rank depth $1 \leq k \leq |\mathcal{C}| + 1$, the flow e-value is

$$e_i = \frac{|\mathcal{C}| + 1}{k} \mathbf{1}\{K_i \leq k\}, \quad (1)$$

with $\mathbb{E}[e_i] \leq 1$ under the null and the relevant exchangeability premise, and $p_i = \min(1, 1/e_i)$. The construction is two-point: a flow either reaches the calibration tail or receives zero. It therefore imposes the hard ceiling

$$M = \frac{|\mathcal{C}| + 1}{k}. \quad (2)$$

No detector score can produce evidence above M . This ceiling follows from the finite resolution of deterministic rank-based conformal evidence [6], [15]; it does not depend on the detector. Randomised smoothing removes the corresponding floor, and Appendix C-B evaluates that alternative.

We use α for the target error level and α_t for the level offered at step t ; in particular, α_T denotes the offered level at step T , not a product. The procedure rejects hypothesis t when

$$\text{Ev}(G_t) \geq 1/\alpha_t. \quad (3)$$

Together, (2) and (3) give the paper's central feasibility condition: a rejection at step t requires $\alpha_t \geq 1/M$, regardless of the detector score. We use e-LOND as the principal controller because it controls FDR under arbitrary dependence when the group e-values are marginally valid [5]. We instantiate the horizon results for LOND, e-LOND, e-LORD, and LORD++; Appendix C covers additional procedures and escape mechanisms. All experiments use $q = \alpha = 0.05$, $w_0 = 0.025$, and $k = 1$.

We compare two spending regimes. Let $(\gamma_j)_{j \geq 1}$ be the spending sequence. The horizon-free regime uses $\gamma_j = j^{-1.6}/\zeta(1.6)$, abbreviated $\gamma_j \propto j^{-1.6}$, and represents uninterrupted operation without a known terminal horizon. The horizon-aware regime uses the horizon-uniform allocation $\gamma_j = T^{-1} \mathbf{1}\{j \leq T\}$, abbreviated $\gamma_j = 1/T$. It assumes a known stream length and maximises the smallest cold-start level over $1, \dots, T$, making it the strongest cold-start benchmark covered by Corollary 1. This comparison separates the merger's structural exposure from the controller's choice of allocation.

B. Grouped security hypotheses

Because SOC investigations rarely operate at flow granularity, we group flows into episodes G_j using pre-committed metadata and a time bucket. Our main unit is

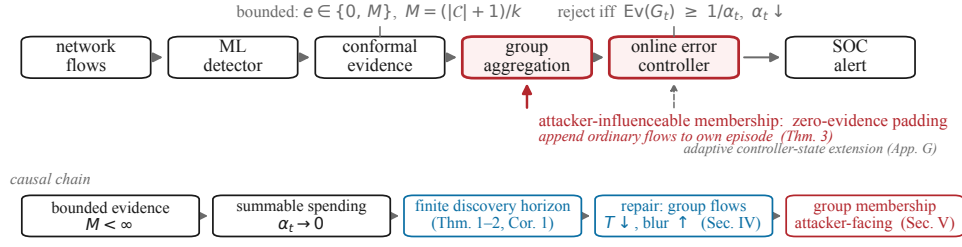


Fig. 1. The statistical trust layer and the causal chain studied in this paper. Bounded evidence and decaying levels create a finite discovery horizon (Section III). Grouping reduces the number of tests and restores feasibility (Section IV), but a co-occupiable grouping key lets an attacker pad its own hypothesis with ordinary traffic (Section V). The malicious flows and their detector scores remain unchanged. Exact sizing uses the realised controller state; the state-free attack over-provisions from the attacker’s episode size alone.

a source–destination host pair within a two-hour bucket. The controller receives the arithmetic mean

$$\text{Ev}(G_j) = \frac{1}{m_j} \sum_{i \in G_j} e_i, \quad m_j = |G_j|. \quad (4)$$

At every fixed arity, the mean is a valid e-merging rule under arbitrary dependence. Unlike the maximum, it does not systematically increase with group size, so it is the symmetric rule we evaluate.

Guarantee scope. The group-level premise is stated as Assumption 1. Neither the discovery-horizon results nor Theorem 3 requires it; it is needed only to read the empirical e-LOND runs as nominally FDR-controlled. The nominal FDR guarantee for an empirical e-LOND run requires more than unconditional flow-level validity; what it needs is a marginally valid e-value for every true-null group. The premise we adopt is one sufficient condition for that, not a necessary one: for each true-null group G_j , every benign member remains e-valid conditional on the pre-committed local metadata that fixes the key, membership, arity, order position, and horizon; writing $\mathcal{M}_j$ for the σ -field that metadata generates, $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$ for every benign $i \in G_j$. Endpoint and service metadata can correlate with the features driving the score, so the unconditional conformal statement does not imply this conditional one. Under this premise, linearity gives marginally valid group e-values even with random arity and arbitrary dependence within and between groups; other constructions of valid group e-values are possible, and Appendix A-A gives one in which the premise holds by construction. Appendix F and Table VI audit and track its use.

C. Threat model

We assume that the defender correctly implements the detector, calibration, grouping, aggregation, and controller. The adversary controls traffic from an attacker-operated or compromised host and can open ordinary connections to the destination it is already attacking before the same bucket closes. Under the key (SrcIP, DstIP, bucket), those connections join the attacker’s tested hypothesis. The adversary neither changes

the detector or calibration code nor modifies stored scores or malicious flows; its goal is to suppress the alert on its own grouped activity.

We test, rather than assume, that suitable ordinary traffic contributes zero evidence. Section V-E uses real LSPR23 flows chosen before deployment by network-observable service frequency alone. Section V-F repeats the attack on AIT with genuine benign traffic sent to attacked victims, then separates a retrospective deployment-wide template pool from traffic that had reached the victim before the attacker’s last flow in the target episode. The latter restricts temporal availability, but it does not by itself make every template observable to a compromised source: such a host need not see other benign traffic received by the victim. We therefore report these pools as attacker-capability boundaries, not as demonstrated online attacks.

We distinguish two knowledge models. The exact attack cost r_t^* uses the realised group evidence and live controller level, and is therefore an oracle lower bound. State-free over-provisioning instead uses a fixed multiplier of the attacker’s own episode size (Section V-C).

Groups that close in the same bucket still require an order. Our main results use a canonical order: a deterministic hash of each group’s metadata key, with collisions broken by the key itself. Because it ignores arrival time, appending flows to one group does not move any other hypothesis; this preserves the group-validity premise on the attacked stream. Canonicalisation is not a security mechanism (Appendix D-A). We also audit first-flow arrival order, which exposes an attacker timing lever and gives a detector-favourable upper bound on LSPR23 detections. Every such contrast is labelled explicitly.

D. Data and chronological protocol

LSPR23 [19] contains 16,353,511 flows from the 161.5-hour Locked Shields live-fire exercise; 1,644,599 flows are malicious (10.06%, or 0.48–0.81% of episodes). We sort the data by timestamp and construct chronological training, benign calibration, and deployment blocks, with calibration and test each occupying 15% of N . No data are randomly mixed across time.

The five deployment positions are overlapping chronological views of one exercise, not five datasets and not independent replications. Adjacent windows share 46.7–53.3% of their deployment flows; each deployment block coincides, to within one flow, with the calibration block of the position 0.15 later; and the training pools $[0, i_1)$ are nested. We therefore report each window separately and use position 0.62 as a robustness check rather than confirmatory replication. Appendix B-D gives the exact geometry. Two detector seeds change the scores but not any feasibility quantity, and we never pool them.

The primary detector is HistGradientBoosting over the protocol and 32 flow timing and volume features. Using a standard classifier keeps the analysis focused on the trust layer. A second detector adds six strictly causal host-context features (Section V-F). AIT-LDSv2.0 [20] provides an external, benign-inclusive transfer setting in which attacked hosts also receive ordinary traffic. Appendix B gives the features, preprocessing, and implementation checks.

We fixed positions 0.55 (primary) and 0.62 (secondary) before running any validity diagnostic. At $k = 1$, only 0–3 benign deployment flows reach the calibration tail in each of the four non-stress windows, too few to establish validity. At position 0.85, 46 benign flows fire—50.9× the nominal rate—which refutes marginal e-validity ($\mathbb{E}[e] \approx 51$). We therefore use 0.85 only as a known-invalid stress window for algebraic, mechanism, and cost results that do not require valid evidence, and label every use accordingly.

Unless stated otherwise, measured results use detector seed 0, canonical within-bucket order, and the horizon-free regime at the named window. Appendix B-C gives the complete reporting conventions.

III. Bounded Evidence Creates a Finite Discovery Horizon

Detection power asks whether malicious activity produces strong evidence. Feasibility asks a more basic question: could the controller accept even the strongest admissible evidence? Combining (3) and (2) gives the necessary condition

$$\alpha_t \geq 1/M, \quad (5)$$

which is determined by $|\mathcal{C}|$, k , and the offered level before detector performance enters. We call step t evidence-feasible when (5) holds. On a rejection-free run, the final such step is the discovery horizon t^* . Beyond it, the run is absorbed: even the maximum admissible evidence cannot produce a rejection. We call this operational state structural silence. A procedure is cold-start feasible through T if every $t \leq T$ remains evidence-feasible before the first rejection.

Weak detectors, rare attacks, and unlucky orderings can all reduce recall. Structural silence is different: $|\mathcal{C}|$, k , and T determine it independently of the observed scores. Accordingly, the results in this section concern

the rejection rule itself and require neither valid e-values nor Assumption 1; those conditions matter only for the nominal FDR interpretation (Table VI).

A. One headline result, two controller forms

Theorem 1 (Finite discovery horizon, multiplicative family). Let the evidence be bounded by $M < \infty$ and the controller run without restart, with $\alpha_t = \alpha \gamma_t g(\mathcal{H}_{t-1})$, $\gamma \geq 0$, $\sum_t \gamma_t \leq 1$ and $g \leq c_g(R_{t-1} + 1)^d$ for the rejection count R and some $d \geq 0$ (which the step below uses; every procedure we cover has $d \in \{0, 1\}$). Since $R_{t-1} \leq t - 1$, a rejection is possible at t only if $\gamma_t t^d \geq 1/(\alpha c_g M)$. If $\gamma_t t^d \rightarrow 0$, only finitely many steps are feasible on a rejection-free run; if that set is non-empty it has a last element t^* and the run is absorbed from $t^* + 1$ onward, and if it is empty the run is absorbed from step 1. No monotonicity of γ is needed: a set of steps that is finite already has a maximum.

Theorem 2 (Finite discovery horizon, lag-sum family). Let the evidence be bounded by $M < \infty$ and $\alpha_t = w_0 \gamma_t + \sum_j a_j \gamma_{t-\tau_j}$ over prior rejection times $\tau_j < t$, with $w_0 \geq 0$, $0 \leq a_j \leq \alpha$, $\gamma \geq 0$ and $\sum_t \gamma_t < \infty$. During a rejection-free run of length Δ every lag is distinct and at least Δ , so $\alpha_t \leq (w_0 + \alpha) \sum_{u \geq \Delta} \gamma_u$. The tail sum tends to zero, so $\Delta^* = \min\{\Delta : (w_0 + \alpha) \sum_{u \geq \Delta} \gamma_u < 1/M\}$ is finite and the run is absorbed once $\Delta \geq \Delta^*$, with no monotonicity assumption on γ .

The two theorems capture the same mechanism for the two common forms of arrival-time levels, but their scope differs. Theorem 1 uses $R_{t-1} \leq t - 1$ to give a bound uniform over the preceding rejection history. Theorem 2 is a rejection-gap result: its tail bound applies after a gap of length Δ , and a new rejection starts a new gap. Along the relevant rejection-free prefix, the offered level eventually falls below the fixed floor $1/M$. Once that holds, no subsequent observation can trigger a rejection, because every group’s evidence is bounded by M , and since only a rejection would raise the level, the state is absorbing unless the mechanism itself changes. Theorem 1 covers LOND, e-LOND, and e-LORD, while Theorem 2 covers LORD++. The sequence $\gamma_j \propto j^{-1.6}$ satisfies both sets of conditions. Full proofs and the precise hypothesis scope appear in Appendix A.

Huo et al. [16] identify the conformal floor $1/(|\mathcal{C}| + 1)$ and observe poor online-testing behaviour for LOND, SAFFRON, and ADDIS. Our result characterises when this interaction becomes structural: for the two controller forms above, only finitely many steps are feasible and a sufficiently long rejection-free run becomes permanently silent. The result applies to both p-value and e-value procedures and identifies the relevant procedure–sequence pairs. The calibration law below then converts this mechanism into an ex-ante deployment test.

B. The best-case calibration–horizon exchange rate

Before the first rejection, a procedure can spend only its cold-start budget. Let c_0 denote the coefficient multiplying the spending weight: $c_0 = w_0$ for a level- w_0 procedure such as LORD++, and $c_0 = \alpha = 2w_0$ for LOND and e-LOND. Among non-negative sequences with total mass at most one, the horizon-uniform allocation $\gamma_t = T^{-1}\mathbf{1}\{t \leq T\}$ maximises $\min_{t \leq T} \gamma_t$, and therefore maximises the cold-start horizon. Appendix A gives the proof.

Corollary 1 (Horizon-uniform feasibility). Cold-start feasibility over a horizon T — every step $t \leq T$ evidence-feasible before the first rejection, which the ceiling decides and detection power does not — requires $|\mathcal{C}| \geq k/\alpha_T - 1$, for α_T the level offered at step T . Under the horizon-uniform allocation $\gamma_t = T^{-1}\mathbf{1}\{t \leq T\}$, where α_T is also the smallest level offered over the horizon, the condition is necessary and sufficient and reads

$$|\mathcal{C}| \geq \frac{kT}{c_0} - 1.$$

The requirement is a scale-free count ratio k/c_0 : at $k = 1$ it is 20 calibration units per hypothesis for LOND/e-LOND and 40 for LORD++, whatever the unit is — exactly $20T - 1$ and $40T - 1$ over a horizon T , the ratio being the rate rather than the count.

Corollary 1 is the best cold-start scaling attainable by any pre-committed allocation. It turns the evidence ceiling into a sizing rule: the count ratio k/c_0 is independent of both the dataset and the alerting unit. Reducing the number of tested hypotheses therefore reduces the required calibration size in direct proportion, which motivates the grouping analysis in Section IV. The same invariance does not hold in calendar time because calibration and deployment produce units at different rates (Appendix C). Krönert et al. [21] derive an algebraically related equality for a windowed procedure; Section VIII distinguishes the two settings.

We summarise feasibility by the allocation-optimal, horizon-uniform ratio

$$\rho_p = \frac{M c_{0,p}}{T} = \frac{(|\mathcal{C}| + 1) c_{0,p}}{kT}. \quad (6)$$

For procedure p under horizon-uniform allocation, $\rho_p \geq 1$ is necessary and sufficient for cold-start feasibility through T by Corollary 1. Under another pre-committed allocation γ , the corresponding minimum-level ratio is $\rho_p(\gamma) = M c_{0,p} \min_{t \leq T} \gamma_t \leq \rho_p$. Throughout the paper, ρ denotes the horizon-uniform ratio, with feasibility at 1.

At network scale, the requirement is substantial (Fig. 2). Testing all 16,353,511 flows in one uninterrupted run requires about 3.3×10^8 calibration flows for LOND/e-LOND and 6.5×10^8 for LORD++. The smaller value is already 20 times the entire exercise, compared with the 1.8–2.4 million benign calibration flows available before each deployment window. Scaling calibration and

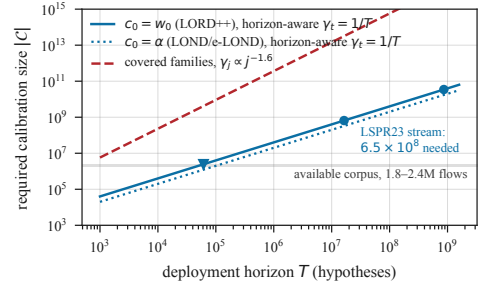


Fig. 2. Calibration size needed to keep a rejection possible through horizon T ($k = 1$, $w_0 = 0.025$). The horizon-uniform allocation is cold-start optimal and scales linearly in T , with the controller entering through c_0 ; linear scaling is therefore the best case. Horizon-free $\gamma \propto j^{-1.6}$ scales as $T^{1.6}$. The grey band marks the 1.8–2.4M calibration flows available in our windows. Procedures outside the two theorem families are compared in Appendix C.

deployment together does not close the gap because the ratio k/c_0 remains fixed. Moreover, these figures assume the cheapest allocation: under horizon-free $\gamma \propto j^{-1.6}$, the same horizon requires about 1.6×10^{13} calibration flows for LOND/e-LOND and 3.2×10^{13} for LORD++.

C. Feasibility is not detection power

Feasibility is necessary for detection, but it does not guarantee detection. Two-hour host-pair episodes reduce T to 31,568–57,368, making horizon-uniform e-LOND feasible in all five windows ($\rho = 2.13$ –3.34). Under horizon-free spending, however, useful level remains concentrated in a short prefix: before the first discovery, only the first 748–902 hypotheses can reject, or 1.6–2.6% of the stream. A near-ceiling malicious episode must land inside that prefix. Under the canonical order, e-LOND therefore detects three of 275 malicious episodes in the primary window and eleven in the secondary window (Table II).

The evidence ceiling and the spending sequence play different roles. At flow granularity, the shortfall is allocation-optimal, so reallocation cannot remove it. At the grouped unit, allocation matters: horizon-aware uniform spending is feasible ($\rho = 2.13$) and detects 105 of 275 malicious episodes at the primary window with no false discoveries, versus three detections under $\gamma \propto j^{-1.6}$ (Table II). Two additional configurations show the same separation (Appendix C-A). Horizon-free spending exposes the short feasible prefix of an anytime deployment; horizon-aware spending or restart restores substantial power. Under first-flow order on both legs at the primary window and seed 0, e-LOND with horizon-aware weighting reaches 101 detections with no false discoveries, and a two-hour periodic restart that keeps the deployment-wide budget takes e-LOND from 18 to 55 rejections (Appendix C-B). We price the padding attack under both regimes in Section V-D.

Across the five canonical-order positions, e-LOND detects 3, 11, 0, 0, and 34 episodes. The two zero counts arise because no episode clears its offered level, not because

the controller is absorbed. The 34 detections occur in the known-invalid stress window.

A detector with better ranking can even produce fewer alerts. Adding six strictly causal host-context features raises primary-window AUROC from 0.916 to 0.954 but reduces e-LOND detections from 18 to two under first-flow order across both legs. Because $|\mathcal{C}|$, k , and T are unchanged, so is ρ : ranking quality affects power but cannot move the feasibility boundary. Appendix C reports the full matrices, an Isolation Forest comparison, and a low-prevalence analysis in which the probability that e-LOND ever rejects falls to 0–40% while ρ remains fixed.

D. Procedures outside the theorems

The horizon theorems cover arrival-time controllers whose effective spending index advances with elapsed hypotheses or rejection lags. ADDIS can avoid elapsed-time decay by selectively advancing its index. History-wide procedures such as online e-BH also fall outside the predicate because they reconsider earlier hypotheses instead of fixing a decision level at arrival. These alternatives move the constraint rather than making evidence scale irrelevant. Appendix C provides the full taxonomy and measurements, while Appendix G shows how an evidence-conditioned index creates a separate attacker-controlled state variable.

IV. Grouping Restores Feasibility by Changing the Security Unit

Corollary 1 makes the alerting unit a design variable. When the calibration corpus is fixed, feasibility can be restored by reducing T : test groups of flows instead of individual flows. Replacing flow-level hypotheses with episodes changes the horizon from T_{flow} to T_{group} and gives the allocation-optimal requirement—the minimum calibration size over pre-committed spending allocations—

$$|\mathcal{C}|_{\min} + 1 = \frac{kT_{\text{group}}}{c_0}, \quad (7)$$

Thus $|\mathcal{C}| + 1$ falls in direct proportion to the number of tested hypotheses. For the covered procedures under uninterrupted control, grouping restores cold-start feasibility at this exact exchange rate.

Grouping changes more than T . Moving from a host-pair key to a source-host key raises the primary window’s canonical detections from 3 to 46 under horizon-free spending, while the median exact suppression cost—an oracle lower bound—rises from 24 to 496 added flows. The grouping key therefore changes feasibility, alert semantics, and attack cost together. It becomes a membership defence only when the attacker cannot cheaply co-occupy it; our operational keys include an endpoint that the attacker controls.

We evaluate five SOC-relevant grouping families over six bucket widths and a whole-deployment bucket, for 35 configurations per window and seed (Appendix D).

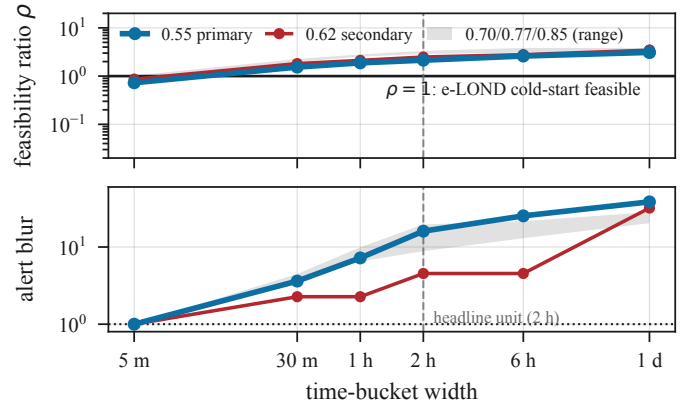


Fig. 3. Effect of coarsening the source–destination bucket. The feasibility ratio ρ ((6)) rises through 1, while alert blur—the number of fixed five-minute malicious atoms represented by one alert—increases. Lines show the primary and secondary windows; shading covers the other three. Seed 0; the two-hour headline unit is marked.

Every family follows the predicted linear relationship between hypothesis count and required calibration, and each becomes feasible once the bucket is sufficiently coarse. Source-host grouping is feasible at every bucket, reducing each window to 763–24,043 episodes. Table XVII reports the primary window at seed 0; feasibility itself depends only on T .

The cost is lower semantic resolution. Episode recall cannot be compared directly across bucket widths because both the positive unit and its count change. We therefore fix an atomic reference: every malicious source–destination episode in a five-minute bucket is one atom, independent of the controller’s alerting bucket. For an issued alert, alert blur is the number of distinct malicious atoms that it represents.

We measure blur under first-flow order, which produces alerts at every width and holds ordering fixed as granularity changes. In the primary window, mean blur grows from 1.0 at five minutes to 7.2 at one hour, 16.1 at the two-hour headline unit, and 38.5 at one day (Fig. 3). Fixed atomic coverage increases at the same time, but each alert identifies a progressively broader span of malicious activity.

The same operation that restores feasibility decides which flows share a hypothesis. When endpoints, services, or timing determine membership, the attacker can influence the composition of the alerting unit directly.

V. Padding Evasion through Attacker-Influenceable Membership

A. Padding the tested hypothesis

An attack episode is a source–destination pair within a two-hour bucket. Before that bucket closes, the attacker opens additional ordinary connections to the same destination, placing them in the same hypothesis. The malicious flows and all existing detector scores remain

unchanged. Under the two-point evidence construction, each added flow that misses the calibration tail contributes exactly zero, so r additions change the aggregate from $F_m(e_1, \dots, e_m)$ to $F_{m+r}(e_1, \dots, e_m, 0^r)$.

The following theorem shows that, under attacker-controlled, unbounded membership, no alert-capable symmetric e-merging family can ignore arbitrary padding while preserving validity. Like the feasibility results, suppression of a specified alert is a property of the aggregation rule and realised controller level; it does not require valid e-values or Assumption 1 (Table VI). Nor does validity prevent it: Appendix A-A gives a four-hypothesis construction in which every true-null e-value is exactly valid by exchangeability and one appended zero-evidence flow still erases the only alert.

B. Symmetric alerting and arbitrary padding invariance are incompatible

Definition 1 (θ -padding-robustness, attainment). A family $\{F_N\}_{N \geq 1}$ of e-merging functions is θ -padding-robust, for $\theta > 1$, if for every $m \geq 1$, every finite $x \in [0, \infty)^m$ with $F_m(x) \geq \theta$ and every integer $r \geq 0$, $F_{m+r}(x, 0^r) \geq \theta$. It attains θ if some finite input has $F_m(x) \geq \theta$.

Attainment excludes rules that can never fire, such as $F \equiv 1$ (Appendix A-C).

Theorem 3 (Symmetric padding impossibility). No family of symmetric e-merging functions that attains θ is θ -padding-robust, for any $\theta > 1$. Quantitatively, for every symmetric e-merging family, every $m \geq 1$, every finite $x \in [0, \infty)^m$, every $\theta > 1$ and every integer $r \geq 0$,

$$F_{m+r}(x, 0^r) < \theta \quad \text{whenever} \quad r > \frac{\sum_i x_i}{\theta} - m.$$

Proof idea. Vovk and Wang [8, Prop. 3.1] show that the arithmetic mean essentially dominates every symmetric e-merging function: $F(e) > 1$ implies $\text{mean}(e) \geq F(e)$. A case split therefore gives $F_N(x) \leq \max\{1, \text{mean}(x)\}$ on finite inputs. Padding gives $\text{mean}(x, 0^r) = \sum_i x_i / (m + r)$, which falls below θ once $r > \sum_i x_i / \theta - m$, and attainment supplies an x to which this applies. The mean meets the bound with equality, so the quantitative statement is tight. Appendix A gives the full proof, the scaled-rule edge case, and an independent negatively-dependent special case. $\square$

Vovk and Wang’s domination result [8, Prop. 3.1] compares e-merging rules at a fixed arity: it asks how best to combine a given set of e-values. Our adversary changes that set. Definition 1 captures the required cross-arity property, and Theorem 3 establishes an exact incompatibility under attacker-controlled, unbounded membership: a symmetric e-merging family can withstand arbitrary zero-evidence additions or attain an alert threshold, but not both. Whenever a symmetric merger issues an alert, the attainment condition holds at that threshold (Appendix A-C). The arithmetic mean reaches the quantitative bound, yielding the exact cost in (8). Substituting

another symmetric merger therefore cannot eliminate the vulnerability while membership remains unbounded.

C. Exact and state-free attack costs

For the arithmetic mean, let $S_t = \sum_{i \in G_t} e_i$ and $m_t = |G_t|$ for the group tested at step t . After r zero-evidence additions, its evidence is $S_t / (m_t + r)$. If the original group is rejected at level α_t , the minimum number of additions needed to suppress that rejection is

$$r_t^* = \lfloor S_t \alpha_t \rfloor - m_t + 1, \quad (8)$$

where we write r^* when the step is clear. The traffic construction is black-box—ordinary connections on a common service to the host already under attack—but exact sizing requires S_t , the live level α_t , and the unperturbed controller trajectory. Thus r_t^* is the oracle per-alert cost defined in Section II-C.

We report three complementary cost summaries. For the baseline true detections $\mathcal{D}$, r_t^* prices one alert on the unperturbed trajectory and $\sum_{t \in \mathcal{D}} r_t^*$ adds those per-alert costs. The campaign-level quantity J_{seq} is the total paid by a greedy sequential oracle that re-evaluates each episode on the attacked trajectory. A state-free attacker instead uses a fixed multiplier c of its own episode size. Table II reports all three summaries.

A state-free attacker can over-provision from its own episode size alone. Multiplying an episode by c , through $(c - 1)m_t$ additions, suppresses its alert whenever $c \geq 1 + r_t^* / m_t$. No live access to S_t , α_t , or $|\mathcal{C}|$ is required; on LSPR23, $c = 10$ is the smallest tested multiplier in $\{2, 3, 5, 10, 100\}$ that suppresses every canonical, horizon-free alert at seed 0 in the primary and stress windows, the two arms with stored per-episode arities. The measured cost is higher than the exact minimum. At the secondary window, one canonical alert with $m_t = 22$ and $r_t^* = 859$ needs $c = 41$. For horizon-aware spending, the joint rerun in Section V-D yields an exact, sharper result.

Corollary 2 (Horizon-uniform feasibility ratio as a dilution threshold). Run e-LOND under the horizon-uniform allocation, so that $\alpha_t = (\alpha/T)(R_{t-1} + 1)$ with $R_0 = 0$, on arithmetic-mean group evidence whose flow e-values are bounded by M , and write $\rho = \rho_{\text{e-LOND}} = M\alpha/T$, the horizon-uniform ratio (6) for e-LOND. Suppose no hypothesis other than the attacker’s own episodes carries evidence at or above the cold-start threshold T/α , and the attacker multiplies every one of its own episodes by an integer $c \geq 1$, appending $(c - 1)m_t$ zero-evidence flows to an episode of arity m_t . If $c > \rho$, the attacked trajectory is rejection-free. If $c \leq \rho$ and some attacker episode attains M , it is not rejection-free. Hence the smallest integer multiplier that suffices for every stream satisfying the hypotheses is $c_{\text{int}} = \lfloor \rho \rfloor + 1$.

For a particular stream, the smaller integer $\lfloor c_{\text{crit}} \rfloor + 1$ suffices, where $c_{\text{crit}} \leq \rho$ is the largest attacker evidence multiplied by α/T . Equality holds when an attacker

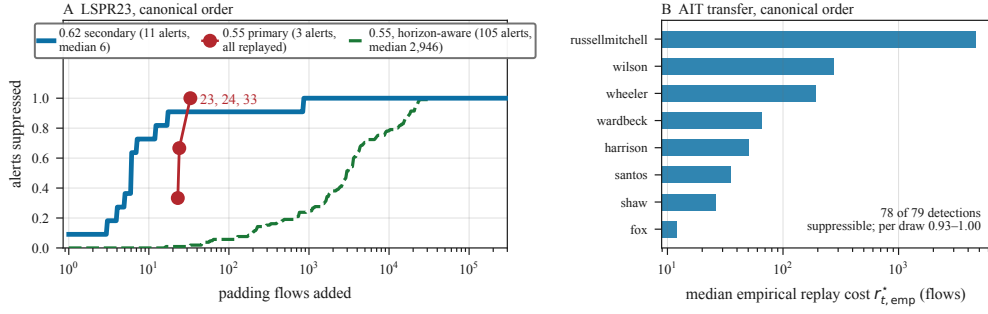


Fig. 4. Per-alert suppression cost under canonical order, seed 0. A: LSPR23. Points mark the three horizon-free primary-window alerts suppressed by real-flow replay (23/24/33 added flows); the solid curve shows the eleven secondary-window alerts. The dashed curve shows the same primary window under horizon-aware allocation (Table II). B: median empirical replay cost by AIT organisation. Empirical cost is at least the zero-pad bound and equals it in seven of eight organisations; replay success is reported separately. Fig. 5 gives the stress-window and first-flow results.

episode attains the ceiling, as occurs in every evaluated window; the proof is in Appendix A. Under the corollary’s assumptions, the same ratio that governs cold-start feasibility also gives the universal dilution threshold for attacker-owned cold-start alerts. Appendix E-D measures the cases in which the assumptions fail.

In the primary and secondary windows, ρ is 2.13 and 2.49, so $c = 3$ suffices and $c = 2$ does not: against the horizon-aware controller, multiplying every own episode by $c = 3$ silences it at both windows. The one horizon-aware false discovery in the secondary window occurs only after earlier detections have raised its level, and disappears when those detections are suppressed, so the corollary’s condition on other hypotheses holds. The attacker needs only its own arity m_t . On LSPR23, all 674 detected episodes lie on host pairs with no benign traffic, so that arity is simply the number of flows the attacker generated. No estimate of S_t , α_t , or $|C|$ is needed. Appendix E-B reports the over-provisioning factors, tested variants, and the one first-flow arm for which no fixed multiplier suffices.

D. Controller allocation prices alerts, not robustness

Because (8) increases with α_t , the offered level determines the cost of suppressing an individual alert. Under horizon-free $\gamma \propto j^{-1.6}$, the primary window produces three canonical alerts near the evidence ceiling, with a median suppression cost of 24 added flows. Under the max-min-optimal horizon-aware allocation, the same detector, groups, and order produce 105 true alerts with no observed false discoveries; the median cost rises to 2,946. The corresponding feasibility ratios are $\rho = 2.13$ and 2.49 for the primary and secondary windows.

Summing the separately priced primary-window pads gives $\sum_{t \in \mathcal{D}} r_t^* = 627,495$. This is an upper bound on the cost of jointly suppressing that set, not a joint minimum: removing an early alert lowers e-LOND’s rejection count, which lowers the level and cost of later alerts. The secondary horizon-aware arm’s one false discovery is not included in its per-alert sum. Within the unbounded-membership model of Theorem 3, allocation changes these

finite costs but cannot remove padding vulnerability. A hard membership limit would change that model; below we measure the resulting attack boundary, not a deployed admission policy.

We measure this feedback on the canonical, seed-0 trajectories for both windows by rerunning e-LOND over the fully padded stream, using the zero-evidence model validated by the real-flow replay in Section V-E. Applying all per-alert pads and rerunning the controller under canonical order leaves no true detection standing in either regime or window; the secondary window’s horizon-aware false discovery also disappears. The greedy sequential attacker is cheaper because it sizes each pad against the level actually offered on the attacked run. With earlier alerts removed, later levels stay at or near cold start.

Under canonical order at the primary window, this attacker pays 5,650 flows in total rather than the per-alert sum of 627,495; at the secondary window, it pays 9,992 against 977,567. Across the alerts it pads, the median pad is 48.5 and 33 flows rather than the per-alert horizon-aware medians of 2,946 and 3,908, and 13 and 4 alerts need no pad at all because they existed only at levels raised by earlier detections. Thus the high horizon-aware cost is an individual-alert price created by the controller’s own detection history. The greedy joint costs remain one to two orders of magnitude above the horizon-free totals of 30 and 859, and they are upper bounds on the minimum joint cost because they describe one oracle strategy.

The resulting group sizes locate the membership-cap boundary. Under horizon-free spending, the median padded primary-window episode has arity 63, and 5% of deployment episodes are at least that large. Per-alert horizon-aware pads have median arity 2,989, reached by only 0.22% of episodes. Once controller state is propagated, however, the attacker has median padded arity 91 at the primary window and 55 at the secondary, sizes reached by 3.8% and 4.3% of episodes.

We therefore rerun the sequential attacker under a capacity constraint: it pads an episode only when $m_t + r_t \leq n$ and otherwise leaves the original alert in place. This

TABLE II

Detection and padding cost with detector, grouping, canonical order, and seed 0 fixed. r_t^* is the oracle cost of one alert; $\mathcal{D}$ is the set of true detections; and J_{seq} is the cost of a greedy sequential oracle on the attacked trajectory (Section V-D and Table XXXVIII). The per-alert sum upper-bounds the joint cost; FDP is realised.

Window	allocation	true det.	FDP	median r_t^*	$\sum_{t \in \mathcal{D}} r_t^*$	J_{seq}
0.55 primary	horizon-free	3	0.000	24	80	30
	horizon-aware	105	0.000	2,946	627,495	5,650
0.62 secondary	horizon-free	11	0.000	6	926	859
	horizon-aware	107	0.009	3,908	977,567	9,992

experiment does not truncate or rescore the complete clean and attacked streams, so it is not an end-to-end evaluation of an admission-control policy. Under canonical order and $n = 100$, 96 of the 105 horizon-aware primary-window alerts still fire: 44 because their pad would not fit even at the cold-start level, and 52 because those first alerts raise R enough to price later pads above the cap; at the secondary window 79 of 107 fire (33 and 46) by the same mechanisms. At a cap of 300, we observe 19 and 60 alerts firing, whereas the per-alert reading where $m_t + r_t^* > n$ predicts 91 and 95, and at 1,000 the sequential attacker silences both windows. A cap of 100 would alter 3.4% and 2.9% of benign episodes as currently formed. This is the share an implemented cap would alter, not a measurement of its effect on evidence, clean detection, or benign service.

The same constraint does not stop the horizon-free attacker: all three primary-window alerts remain suppressible at $n \geq 100$. Reaching them requires $n = 30$, which would alter 7.4% of benign episodes. Thus the experiment identifies a mitigation boundary under an enforced capacity model; it does not establish the performance of a working defence.

E. Real-flow replay validates the cost model

At the running levels received by each episode, the three canonical primary-window alerts require 23, 24, and 33 zero-evidence additions; the eleven canonical secondary-window alerts have a median cost of six (Fig. 4 and Table III). Those costs are arithmetic on stored evidence. To test them against the implemented detector, we build a weak-attacker pool using only network-observable service frequency. Specifically, we select the most common protocol-port pair in the training prefix, before either calibration or deployment. The selection uses no attack labels, detector outputs, calibration scores, or deployment traffic.

None of 20,000 sampled pool flows reaches the primary window’s conformal tail, giving a one-sided 95% upper bound of 1.5×10^{-4} on the firing probability. Injecting these real flows suppresses all 3/3 canonical primary-window detections at exactly the predicted 23–33-flow thresholds. It also suppresses all 11/11 secondary-window detections, with measured cost equal to the closed form for every episode and median cost six.

The horizon-aware regime replays the same way: per-alert replay suppresses 105/105 and 107/107 true detections in the primary and secondary windows at their original levels, on every draw. The baseline runs have no false discovery in the primary window and one in the secondary (realised FDP 0.009). A firing pad flow would contribute the full evidence ceiling, but none of the 845,691 available primary-window pool flows reaches the tail. This full-pool zero count tightens the one-sided 95% upper bound to 3.5×10^{-6} and makes the replay cost equal, rather than approximate, to the zero-evidence closed form for this finite pool. The interval treats future pads as draws from a common firing rate; it is not a distribution-free guarantee for future operational traffic. Appendix E-C quantifies the remaining risk. Across both windows, replay covers fourteen horizon-free alerts at recalls 0.01 and 0.03, and 212 horizon-aware true detections at recalls 0.38 and 0.27.

Replay evaluates each alert separately at its level on the unperturbed trajectory. It measures e-values with real traffic and the implemented detector, then applies the aggregation and controller arithmetic at that fixed level. Its purpose is to establish the zero-evidence pad model; the joint rerun in Section V-D propagates those pads through the altered e-LOND state.

LSPR23 provides no ordinary traffic on attacked host pairs, which are 100% malicious. Its pool is therefore service-level: flows matching the modal training-prefix protocol and port across the window, rather than flows sent to a particular victim. The AIT experiment below closes this gap using actual benign-to-victim traffic. Results are largely insensitive to the eligible pool (Appendix E-C), but they do depend on when the pool is selected: choosing the modal service from the deployment window can select the attacker’s own high-scoring flood, which does not dilute evidence.

Ordering changes which alerts are issued, not the conditional result. Every alert produced under any of the fifty audited pre-committed orders has a finite exact cost under (8) (Appendix E-A).

F. Transfer to benign-inclusive hosts and host context

A host-aware detector might make repetitive padding anomalous. We test this possibility in two stages. First, on LSPR23, we add six strictly causal host-context features:

TABLE III

Padding results under horizon-free spending at seed 0. Order is canonical unless stated. LSPR23 costs are exact zero-pad oracle costs from (8); AIT costs marked “emp.” are per-organisation empirical replay medians and are no smaller than the closed form.

Suppression uses per-alert real-flow replay; AIT uses the retrospective deployment template pool. For the canonical flow-only AIT row, per-draw success is 0.93–1.00. Appendix E gives all arms.

Setting	order	det. cost	suppr.
LSPR23 0.55 primary	canonical	323, 24, 33	3/3
LSPR23 0.62 secondary	canonical	11 med. 6	11/11
AIT flow-only, 8 orgs	canonical	79 med. 12–4,650 (emp.)	78/79
AIT host-cond., 8 orgs	first-flow	63 med. 43–8,471.5 (emp.)	57/63

prior connection counts, distinct-peer counts, and failed-connection fractions. We recompute the context causally as each pad arrives. The pads still contribute zero evidence, and both host-conditioned targets remain suppressible in the primary window under seed 0, horizon-free spending, and first-flow order, as well as in the stress window (Table XXX). Because every attacked LSPR23 host pair is 100% malicious, this experiment cannot test replay from ordinary traffic already sent to the victim.

Second, AIT-LDSv2.0 provides eight organisations in which attacked hosts also serve benign traffic. For each held-out organisation, we train on the other seven, calibrate on in-organisation benign flows that precede the first attack, and run the same two-hour host-pair/e-LOND pipeline. We then replay real benign-to-victim traffic into every detected episode using 200 draws.

Under canonical order, the flow-only detector yields 79 true detections, of which 78 are suppressible when each alert is replayed separately using templates drawn from the full deployment pool. Per-draw success ranges from 0.93 to 1.00, and the per-organisation median empirical cost is 12–4,650 flows. Wherever a pad fires, this cost exceeds the zero-pad closed form. The eight runs issue 83 rejections in total, including four false discoveries, for pooled realised FDP = 0.048. First-flow resequencing produces 85 true detections and suppresses 84, at higher cost (Appendix E).

With host context recomputed throughout accumulation, the same chain at all eight organisations uses first-flow order, issues 63 host-conditioned true detections and suppresses 57 of them (Table III). For the other six, no draw succeeds within twice the zero-pad cost; this is a replay-budget limit, not evidence of impossibility. Host context changes the channel because a pad can itself reach the evidence ceiling instead of diluting the group.

The per-alert results above price each target on the unperturbed trajectory. Joint reruns propagate both the sampled pads and the changed e-LOND state (Table V). Under canonical order, with the retrospective deployment pool, the sequential oracle spends 244,250 and 91,713 flows—40% and 15% of the corresponding per-alert sums—and leaves one and zero of the 79 and 80 baseline alerts.

Under the same canonical order, restricting templates to traffic received by the victim before the attacker’s last flow in each target episode leaves 24 and 27 alerts. Across the 95 attacker-owned episodes, 30 have fewer than 20 such prior flows and 18 have none.

Removing live controller-state knowledge narrows the attack further. With the victim-prior pool, the fixed multiplier $c = \lfloor \rho \rfloor + 1$ from Corollary 2 leaves 65 and 60 alerts; under horizon-free spending this multiplier is a heuristic transfer, since the corollary is proved for horizon-uniform spending. Thus the deployment-pool arm establishes suppression when suitable templates are available, whereas the prior-pool and state-free arms expose the practical dependence on temporal availability and attacker information. A compromised source does not automatically observe every benign flow received by its victim, so even the prior-pool rows are capability bounds rather than a fully demonstrated low-information attack. The shortfall is a direct test of the corollary’s hypotheses (Table XXXVI). Its premise on the other hypotheses, that no non-attacker episode carries evidence at or above the cold-start threshold T/α , fails in three of the ten cells: one non-attacker episode on russellmitchell reaches $1.50 T/\alpha$, and on shaw a non-attacker episode attains the ceiling. It holds in the other seven, including santos, whose horizon-aware non-attacker alert fires only after nine earlier rejections have raised the level, so a baseline non-attacker alert is not by itself a premise failure. The hypothesis that every own episode is multiplied fails wherever an episode has no usable prior pool, two to five per organisation: those episodes fire, raise R , and price the remaining pads above the cold-start level, which is why the multiplier leaves alerts standing even where the premise holds. And sampled templates do not all carry zero evidence: the median share of multiplier pads that fire is 0–6% by organisation. Unknown co-resident arity is not the binding issue in this arm—sizing from the full group rather than the attacker’s own flows leaves the same 65 and 60 alerts (Table XXXIV).

VI. Design Guidance and Evaluated Boundaries

A. Where a mitigation must act

Theorem 3 identifies the design boundary. Under attacker-controlled, unbounded membership, replacing one alert-capable symmetric merger with another cannot provide padding robustness. A mitigation must instead control membership or admission, change the calibrated unit, or introduce asymmetry. Table IV evaluates one representative from each route; each exchanges one property for another. The membership-cap experiment makes this boundary especially clear: under horizon-aware spending, preventing a required pad from fitting can leave an early alert in place, and that rejection raises the cost of later pads. Under horizon-free spending, the tested caps do not stop the attack. Because we constrain only the attacker’s additions and do not apply a truncation policy

TABLE IV

Representative design directions after Theorem 3, evaluated on LSPR23 two-hour host-pair episodes. Aggregation and weighting use first-flow order. The membership-cap row uses a capacity-constrained attacker under canonical order, not a complete admission policy. Supporting measurements appear in Section V-D and Appendix E-E.

Design direction	Gain	Cost / boundary
Capped normalisation $\sum_i e_i/n_0$	Invariant to members appended below the cap n_0	Leaves the symmetric e-merging class above n_0 ; validity-preserving caps detect nothing (Table XXVIII)
Pre-committed asymmetric weights [22]	Valid invariance to appended flows	A few leading flows defeat every scheme that keeps the mean’s recall; a larger reach costs about 58% of it (Table XXVII)
Group-calibrated maximum	Padding-robust aggregate	The smaller calibration unit destroys feasibility: no alerts at the primary or secondary window (Appendix F-B)
Per-host-pair membership cap (outside the merging class)	Makes pads above the cap infeasible in the capacity model; an early unpadded alert can raise later costs	Not evaluated as an end-to-end policy: baseline traffic is not truncated or rescored; the constraint leaves residual suppression and does not stop the horizon-free attacker (Section V-D)

TABLE V

AIT flow-only joint-attack boundary under canonical order, seed 0: attacker-owned alerts remaining (lower is more effective). Baselines are exact sums across eight organisations; attacked counts sum the per-organisation medians over 100 sampled trajectories and are neither one combined trajectory nor the median of an aggregate total; Table XXXV gives the per-trajectory reliability. The deployment pool is retrospective; the prior pool contains only victim traffic observed before the attacker’s last flow in the target episode.

Attacker arm	horizon-free	horizon-aware
Baseline	79	80
Sequential oracle, deployment pool	1	0
Sequential oracle, victim-prior pool	24	27
State-free multiplier, victim-prior pool	65	60

to the complete stream, these results locate a capacity boundary; they do not demonstrate the clean or attacked performance of an operational admission-control defence.

B. A deployment checklist

The evaluated boundaries reduce to five checks a deployment can make before and during operation.

- 1) Compute ρ before deployment. Given $|\mathcal{C}|$, k , and T , Corollary 1 decides whether a covered controller can report an alert at all across the intended horizon. If calibration cannot support the horizon, a better-ranking detector cannot repair the shortfall, and switching among the procedure–sequence pairs of Theorems 1 and 2 leaves the boundary where it is.
- 2) Do not group on a key the attacker can cheaply co-occupy. The grouping key sets the horizon, the meaning of an alert, and the attack cost together (Section IV); report the calibration-to-hypothesis ratio and a fixed-denominator resolution measure such as alert blur alongside recall.
- 3) Treat membership limits and admission as part of the threat model. Any field that determines group membership is attacker-facing, as is state used to

set future testing levels (Appendix G). A per-host-pair cap is a capacity boundary, not a demonstrated defence, until the clean and attacked streams have been evaluated end to end under it.

- 4) If restarting, say which guarantee holds. A pre-committed split $\sum_i q_i \leq q$ keeps the deployment-wide budget and recovers less power; a fresh q per epoch recovers more but promises only a per-epoch FDR (Appendix C).
- 5) Do not expect a different symmetric merger to fix padding. Under attacker-controlled, unbounded membership, Theorem 3 rules that out; a mitigation has to act on membership, on the calibrated unit, or through asymmetry, and each route trades something away (Table IV). A metadata-keyed order stops one padded group from shifting unrelated hypotheses but does not stop padding, and a keyed hash only changes the cost of insertion (Appendix D-A).

VII. Scope and Limitations

A. Validity and claim scope

Table VI separates the algebraic results, labelled empirical measurements, and nominal guarantee. The nominal FDR interpretation of the empirical e-LOND runs is conditional on the group-validity premise in Assumption 1. The diagnostics in Appendix F neither establish nor refute that premise. Appendix F-B evaluates the constructive alternative of calibrating groups directly. The feasibility and padding theorems do not depend on this premise.

B. Limitations

Empirical breadth. LSPR23 supplies the main chronology, with correlated windows from one live-fire exercise (Section II-D). Position 0.85 is the known-invalid stress window and supports only mechanism and cost measurements. AIT-LDSv2.0 adds a benign-inclusive transfer setting, but not broad cross-domain validation. Because LSPR23 has no reliable flow-to-campaign identifier, our

TABLE VI

Assumptions used by the main results. “Valid e ” means valid e -values; A1 denotes Assumption 1. Labels identify malicious and attacker-owned episodes. FDP is the realised false-discovery proportion. Table XXXIX gives the corresponding evaluation windows.

Result	valid e	A1	labels / ownership
Finite horizon, exchange rate (§III)	no	no	no
Grouping buys feasibility (§IV)	no	no	no
Padding impossibility (Theorem 3)	no	no	no
Suppression and r_t^* of a specified alert (§V)	no	no	no
Aggregates over $\mathcal{D}$, incl. $\sum_{t \in \mathcal{D}} r_t^*$ (§V)	no	no	yes
Joint rerun J_{seq} , whole-window multiplier (§V)	no	no	yes
Replay suppression counts, recall, FDP (§V)	no	no	yes
Nominal e-LOND FDR reading of the runs	yes	yes	no

operational units are episodes; the five-minute atom is a stable evaluation denominator, not externally validated incident ground truth.

Attack cost. Per-alert cost depends on the configuration that produced the alert because (8) scales with the offered level. The joint rerun covers two windows and one seed, and assumes that the attacker knows which episodes it owns and that its pads carry zero evidence. On LSPR23, labels provide ownership and attacked host pairs contain no benign traffic. A positive pad firing rate would increase J_{seq} ; the complete primary-window pool contains no firing flow, giving a one-sided 95% upper bound of 3.5×10^{-6} , but the joint rerun does not sample that residual risk. The class-wide result is finite cost, not equal cost or available cover in every deployment. On AIT, a state-free attacker sees only a lower bound on m_t ; the deployment pool is retrospective, and the temporally restricted prior pool can still contain traffic invisible to a compromised source. Joint totals sum per-organisation medians, not one campaign trajectory; Table XXXV gives the per-trajectory reliability behind them.

Detector and grouping scope. The host-conditioned experiment covers six causal features and our specific accumulation model for an already-active attacker-victim pair with fixed distinct-peer counts. It does not characterise every host-aware detector, padding from many new peers, or a high-volume flood. The AIT flow-only result uses canonical order with first-flow sensitivity; the host-conditioned arm covers all eight organisations under first-flow order only (Table III). LSPR23 joint reruns use zero-evidence pads, whereas AIT joint runs sample real pads; the wilson host costs are upper bounds.

Capacity model. The cap constrains attacker additions; it is not applied to the full stream. The benign shares count episodes an implemented policy would alter, but

we do not recompute evidence, clean detection, false discoveries, or disruption. This is a capacity boundary, not a demonstrated defence.

Theorem scope. The finite horizon covers the procedure–sequence pairs in Section III-A under uninterrupted operation, together with the sparse-evidence closure variants. Smoothing, restart, deferred decisions, different evidence constructions, and faster-growing history terms change either the mechanism or its guarantee. The result does not imply that online FDR control is unusable in general. Appendix G separately extends the analysis to ADDIS’s evidence-conditioned spending index.

VIII. Related Work

Online FDR and finite conformal resolution. LOND, LORD, SAFFRON, ADDIS, e-LOND, online e-BH, and e-GAI provide online error guarantees under different evidence and dependence assumptions [3], [23], [4], [24], [25], [5], [9], [26], [27]. They build on α -investing [17], safe testing [28], and offline FDR control [29]. Alpha-death and power loss at small levels are well known [4], [17]; mem-e-LORD [27], for example, matches e-LORD before the first rejection.

Huo et al. [16] observe how the conformal floor affects LOND, SAFFRON, and ADDIS, while Hennhofer and Preisach [15] study resolution collapse and smoothing in low-data batch tests. Our theorems turn this interaction into a deployment boundary: given $|\mathcal{C}|$, k , and T , they determine whether any admissible observation can be rejected at the intended horizon for two controller families. Corollary 1 shows that the horizon-uniform boundary is attained, and the resulting criterion also classifies recent closure and donation constructions [30], [31]. Krönert et al. [21] derive the related equality $n = \nu m / \alpha - 1$ to tune calibration for an exact realised level in a windowed procedure. Our result instead marks the infeasibility boundary of one uninterrupted controller; their windowing corresponds to the restart alternative in Appendix C.

Trustworthy intrusion detection and temporal evaluation. TESSERACT’s temporal evaluation principles [32] and conformal drift systems such as Transcend(ent) [33], [34] motivate our chronological design. The closest systems implement subsets of our pipeline. Rebjock et al. [10] apply online FDR to time-series anomaly statistics; C-PP-COAD [11] adds conformal p-values and active statistics, but not attacker-influenceable metadata groups. Online multi-layer FDR [12] partitions hypotheses rather than merging observations. CALIBURN’s streaming intrusion pipeline uses empirical calibration, cost-sensitive thresholds, within-window conformal risk control, and burn-rate alerts [13]; Gao et al. [14] combine conformal novelty detection with batch BH and boundary-FDR procedures. Neither is an arrival-time FDR controller over merged e -values. We therefore treat the full stack as an evaluated design assembled from established components, not as an existing deployment. Because label quality is a persistent

concern [35], [36], we also audit against an external record (Appendix F).

E-merging and adversarial testing. E-merging theory characterises admissible symmetric [8] and asymmetric [22] rules at fixed arity. Our padding adversary acts across arities. It neither changes detector scores nor corrupts reports, unlike attacks based on manipulated scores, perturbed p-values, or Byzantine inputs [37], [38], [39], [40]. Instead, it adds legitimate observations to its own tested hypothesis. Theorem 3 lifts this mechanism from one implementation to the entire alert-capable symmetric class.

Batching can improve online-testing power [18], but standard results require within-batch independence or positive dependence that co-located flows sharing a calibration tail do not provide. We therefore use decision deadlines [41], [30], which are compatible with arbitrary dependence; batching under one deployment-wide budget still obeys Corollary 1. Feedback-aware procedures incorporate revealed labels with finite-sample control [42], [43]; Appendix H evaluates simpler threshold controllers under analyst delay.

IX. Conclusion

A statistically valid trust layer can still be operationally silent or adversarially suppressible. For the controller families studied here, finite evidence and arrival-time spending create a calculable, absorbing discovery horizon; calibration must grow at least linearly with the number of tested hypotheses. Grouping restores feasibility at that best-case rate, but an attacker-co-occupiable key turns group membership into a control surface. Under unbounded, attacker-controlled membership, every alert-capable symmetric e-merging family yields to sufficient zero-evidence padding.

Controller allocation sets the price of an individual alert, not the existence of the vulnerability. In our joint reruns, suppressing early alerts keeps later levels near cold start and makes campaign cost far smaller than the per-alert sum. The AIT capability arms also distinguish finite algebraic cost from a temporally restricted, low-information attack: template availability and observability determine the realised suppression. Evidence scale, controller spending, alert granularity, and membership control therefore require joint design; a membership cap requires end-to-end evaluation before it can be called an effective defence.

Open Science

We will release the complete set of artifacts needed to reproduce every number and figure in this paper; they exist and have been run in full at the time of submission: a single executable notebook that carries the paper’s derivations and measurements alongside the prose, the library of measurement scripts it imports, the cached

results of a full run, and a pinned environment. Cached-result mode executes in about half a minute, renders every table and figure directly from the shipped result objects (the figure-drawing code is part of the library, and the script that writes this paper’s figures is a wrapper over it) and requires no raw dataset. Full mode recomputes those objects from the original flow data; we executed that mode end to end and verified that it reproduces the shipped results.

The obstacle is third-party data, on which the artifact depends in two places. LSPR23 is not ours to redistribute but is publicly available under CC-BY-4.0 (Zenodo record 8042347), and the artifact ships the exact preprocessing commands that turn the published CSV into the pipeline’s inputs. The transfer test additionally uses AIT-LDSv2.0 [20] (Zenodo record 5789064, CC-BY-4.0): the artifact ships the per-organisation tstat netflow extraction and the exact malicious/benign label mapping (Appendix B) that turn its released packet captures into the eight-organisation netflow inputs, following the dataset’s own recipe. Full-mode reproduction therefore requires downloading both records; the derivations, procedure implementations, attack code and cached results carry no such restriction. They will be deposited in a fully anonymised repository within three days of submission, kept accessible for the duration of review and not edited after that deadline, and, if accepted, deposited on Zenodo by the venue’s final-artifact deadline.

LLM Usage Considerations

Hosted LLMs were accessed through Claude Code and Codex CLI for editorial refinement, code generation, evaluation support, and read-only code audits. LLMs were used for editorial purposes in this manuscript, and all outputs were inspected by the authors to ensure accuracy and originality. They are not part of the proposed method, the released pipeline’s operation or the dataset labelling. Their use was motivated by the volume of automation, evaluation and consistency-checking code the project required within its timeframe. Code was validated through unit tests, per-alert real-flow replay against the closed forms, closed-form cross-checks of the derivations, and paper-to-artifact consistency gates that recompute every reported number from the shipped result objects; every number and reference was verified by the authors. The released artifact contains the exact code used to produce the reported results. The authors take responsibility for all content and any remaining errors.

Necessity, model choice and footprint. We used the general-purpose assistants already available in our development environment; no model was trained or fine-tuned, and no LLM is required to execute or reproduce the released pipeline. The models were chosen because the tasks were editorial, code-generation and code-audit tasks for which a general model suffices. The clients’ session logs for this project record Claude Code (client

versions 2.1.241–2.1.261) with the model identifiers claude-opus-5, claude-opus-4-8, claude-fable-5-1 and claude-fable-5, and Codex CLI (client versions 0.142.3 and 0.152.1) with the model identifiers gpt-5.5, gpt-5.6-terra and gpt-5.6-sol; neither client records the provider-side model snapshot date, so we do not report one. We limited query volume by working in long sessions that batch related requests, by reusing generated code and the cached result objects rather than regenerating them, by running every consistency gate locally as a script rather than by querying a model, and by keeping the released pipeline free of any model call: reproducing every number and figure in this paper requires no access to an LLM, and the artifact imports no LLM client. Provider-side energy measurements for the assistants were not available to us, so we do not estimate their emissions. The reported experiments themselves are CPU-only scikit-learn computations on a single workstation, a MacBook Pro with an Apple M5 Pro system-on-chip (18 CPU cores, 48GB unified memory, no discrete GPU; the pipeline uses no GPU): a full recomputation from the flow data takes about two to three hours and the cached-result reproduction about half a minute.

Ethical Considerations

This paper describes working attacks against a statistical trust layer for intrusion detection, so we state our reasoning against the Menlo Report framing [44].

Respect for persons and justice. All measurements are made on public research datasets released for this purpose, captured during authorised cyber-defence exercises on purpose-built ranges rather than on production networks carrying third-party traffic. We rely on the publishers’ own ethical review and release terms and did not conduct a separate consent audit, so we make no claim about individual consent beyond what those releases state. No attack was executed against any live network, and no personal data is used, disclosed or re-identified.

Beneficence and responsible disclosure. This work does not target a deployed product, and the class of system it studies is, to our knowledge, not yet widely deployed, which is the point at which disclosure is cheapest and most useful. It introduces no new attack on the underlying detector: it identifies a security consequence that emerges when public statistical components are composed — bounded evidence, attacker-influenceable grouping and online error spending together produce an alert layer that can become infeasible or suppressible. We report both the inexpensive and the expensive attack regimes and evaluate representative mitigations, so that designers can assess the exposure before deploying such a composition.

Law and public interest. We identify a small set of host pairs whose labels we believe to be wrong and deliberately do not publish corrected labels: rewriting a public dataset’s labels from our own detector’s tail would

be circular, and the pairs were identified using test-split information. We report the direction of the error instead.

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Appendix A Proofs and Exact Scope

Question answered. What exactly do the body’s four formal results assume, and what does the FDR reading of an empirical e-LOND run assume on top of them? Takeaway. The feasibility theorems need only a finite evidence ceiling and the stated level form; the padding impossibility needs only the symmetric e-merging class; the group-validity premise below is needed by neither, and only by the nominal guarantee.

The body states Theorems 1 to 3 and Corollary 1. This appendix adds the group-validity premise, the exact scope of each body result, and five further results the body cites; all proofs follow, verified numerically in the released artifact.

A. The group-validity premise

Using the mean at variable, metadata-selected arity needs a group-level premise, and different controllers need different things of it: e-LOND and online e-BH control FDR under arbitrary dependence given only marginally valid group e-values, whereas the p-value procedures need conditional super-uniformity, independence or monotonicity. The guarantee-bearing claims therefore use e-LOND.

The conditioning is stated per hypothesis. For group j let $\mathcal{M}_j$ be the σ -field generated by the pre-committed metadata that determines that hypothesis and nothing else: the grouping key, the membership indicators and arity m_j , the group’s position in the pre-committed within-bucket order (hence the spending weight γ_j it is offered), and the horizon T . It carries no information about any other group’s composition. Pre-committed means chosen before the scores and not score-adaptive.

Assumption 1 (Group-local metadata-conditional validity). For every benign flow i of a true-null group G_j , $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$; for the two-point e-value, equivalently $\Pr(K_i \leq k \mid \mathcal{M}_j) \leq k/(|C| + 1)$.

Lemma 1 (Group-level marginal e-validity). Under Assumption 1, every true-null group with $m_j > 0$ has a marginally valid group e-value, $\mathbb{E}[\text{Ev}(G_j)] \leq 1$, for random membership and arity and under arbitrary within- and between-group dependence.

Conditioning on $\mathcal{M}_j$ fixes membership and m_j , so linearity gives $\mathbb{E}[\text{Ev}(G_j) \mid \mathcal{M}_j] \leq 1$ and the tower property removes the conditioning (proof below). The premise is stronger than the unconditional validity split conformal supplies, it is a genuine modelling assumption (endpoint and service metadata can correlate with the timing and volume features that drive the score), and it is not

established by the dataset. Table XXXIX separates every claim by what it needs.

A construction in which the premise holds and the attack still succeeds. The coexistence the paper claims—valid true-null evidence and a suppressible alert in the same run—can be exhibited without data. Draw $|\mathcal{C}| = 99$ calibration scores and the benign test scores independently from $\text{Uniform}(0, 1)$, take $k = 1$, so $M = 100$, and fix the groups and the within-bucket order in advance, independently of every score. Run e-LOND under the horizon-uniform allocation with $T = 4$ hypotheses and $\alpha = 0.05$: the cold-start level is $\alpha/T = 0.0125$ and the threshold is 80. Place a singleton attack flow with score 2 first. It exceeds every calibration score, so $K = 1$, $e = 100 \geq 80$, and it is rejected. Now let the attacker append one flow of its own with score -1 to that episode: the flow ranks below every calibration score, $K = 100 > k$, its e-value is 0, and the group mean becomes $50 < 80$, so the alert is suppressed; nothing else in the stream has changed. Assumption 1 holds by construction before and after the pad. Every true-null group is untouched by the attack, its membership, arity and position were fixed before the scores were drawn, so conditioning on $\mathcal{M}_j$ conditions on nothing that depends on the scores, and each benign score is exchangeable with the calibration scores, giving $\Pr(K_i \leq 1 \mid \mathcal{M}_j) = 1/(|\mathcal{C}| + 1)$ and $\mathbb{E}[e_i \mid \mathcal{M}_j] = 1$ exactly. By Lemma 1 every true-null group e-value is marginally valid and the nominal FDR guarantee is intact, while the one alert the guarantee was meant to deliver has been removed by a single flow. The validity of the nulls and the suppressibility of the alert are properties of different objects, and neither constrains the other; Section V measures the same separation on real streams.

B. Exact scope of the finite-horizon theorems

Theorem 1 covers LOND, e-LOND and e-LORD, the last through the equivalence e-LORD = e-LOND under $\gamma_t = \omega_t \prod_{j < t} (1 - \omega_j)$ [27]. That equivalence is a hypothesis and not a remark, and it needs two things of the weights, not one. They must be deterministic functions of the index, fixed before deployment — predictability in the filtration, which is all online FDR normally asks, would make γ a history-dependent random object, and Theorem 1’s condition is a statement about a sequence. And they must satisfy $\omega_j \in [0, 1]$: pre-commitment alone gives fixedness only, and a pre-committed $\omega_1 = 2$ yields $\gamma_1 = 2$ and $\gamma_2 < 0$, breaking both $\gamma \geq 0$ and the budget. Under $\omega_j \in [0, 1]$ the partial sums telescope, $\sum_{t \leq n} \gamma_t = 1 - \prod_{j \leq n} (1 - \omega_j) \leq 1$, which gives Theorem 1 the non-negativity and the budget. It does not give the decay condition, and the sparse sequence of the next paragraph is itself induced by a deterministic $\omega \in [0, 1]$, so $\gamma_t t^d \rightarrow 0$ has to be checked for the induced γ , as it is for the $\gamma_j \propto j^{-1.6}$ pair we run.

Theorem 2 covers LORD++ ($a_1 = \alpha - w_0$, $a_j = \alpha$ for $j \geq 2$). What is covered is a procedure together with its spending sequence, not a named procedure in the abstract: summability alone is not enough for Theorem 1 at $d = 1$, since $\gamma_{2^n} = n^{-2}/\zeta(2)$ for $n \geq 1$, with $\gamma_t = 0$ elsewhere, sums to exactly one yet has $2^n \gamma_{2^n} \rightarrow \infty$. That example refutes the hypothesis, not the conclusion: the same sequence has $\gamma_t \rightarrow 0$ and so is still absorbed at $d = 0$. The decay condition therefore has to be checked per pair, and it is asymptotic: for $\gamma_j \propto j^{-1.6}$, $t \gamma_t \rightarrow 0$ so $d = 1$ holds, whereas a level growing quadratically in the rejection count would escape the argument. A horizon-uniform γ is supported on $t \leq T$, so an asymptotic decay condition is met trivially and decides nothing; it needs none, because before any rejection the level is the constant c_0/T across the horizon, feasibility there is all-or-nothing, and Corollary 1 decides it exactly.

The two families are not equally strong, and the difference is worth stating. Theorem 1’s bound $\alpha_t \leq \alpha c_g \gamma_t t^d$ uses only $R_{t-1} \leq t-1$, so it holds whatever the history: the feasible set is a fixed set of indices and absorption does not depend on the run being rejection-free. Theorem 2’s bound is driven by the length Δ of the current rejection-free gap, so its absorption is conditional on such a gap occurring. A lag-sum procedure that keeps rejecting keeps its level up — for LORD++ under $\gamma_j \propto j^{-1.6}$ with a rejection at every step, α_t climbs from its cold-start value towards 0.039, and every step of that run is feasible once $|\mathcal{C}| \geq 91$, the binding step being the cold start itself. This is why the paper’s claims are cold-start claims: the horizon is the horizon of a rejection-free prefix, which is the state a deployment starts in and, on these streams, mostly stays in. The theorems are scoped deliberately: they do not say every online FDR procedure dies, and Appendix C locates the procedures that escape.

C. Attainment and the symmetric class

Every rejection threshold in our setting is $\theta = 1/\alpha_t > 1$, and attainment is the hypothesis the deployment supplies whenever an alert is possible at all: if a symmetric family ever issues an alert then some input it receives has $F \geq \theta$, so it attains θ and Theorem 3 applies to it. Not being θ -padding-robust and attaining θ are therefore the same condition, since a family reaching θ nowhere — $F \equiv 1$, which is padding-invariant but can never fire, because rejection needs evidence $\geq 1/\alpha_t > 1$ — satisfies Definition 1 vacuously. For the mean rule the body evaluates the witness is explicit: the all-firing group $x = (M, \dots, M)$ attains θ at every evidence-feasible step (5). The implication runs only that way. Attainment is strictly the weaker condition, since it quantifies over all of $[0, \infty)^m$ — a scaled rule such as $\frac{1}{2}\text{mean}$ is a valid symmetric e-merging family that attains θ somewhere yet, on two-point evidence with $\theta > M/2$, can never fire — so what attainment excludes is only the rules that reach θ nowhere, which is all the theorem needs. Ordinary

domination by the mean is false ($\kappa + (1 - \kappa)$ mean, for $\kappa \in (0, 1)$, is a counterexample), which is why the proof splits cases.

D. Further results used in the body

Proposition 1 (Donation is bounded before the first rejection). For donation e-LOND with parameter $\delta \in (0, 1)$ and a spending sequence with $\gamma \geq 0$ and $\sum_t \gamma_t \leq 1$ — the normalisation the bound relies on — on a rejection-free prefix $\alpha_t \leq \delta \gamma_t / (1 - \delta)$. The restriction $\delta < 1$ is necessary, not cosmetic: at $\delta = 1$ the donated wealth can drive the denominator to zero and hold α_t away from zero along a rejection-free run, so neither the bound nor absorption survives (Appendix A). Hence donation e-LOND satisfies Theorem 1’s hypotheses along a rejection-free prefix, with $c_g = \delta / [\alpha(1 - \delta)]$ — which is $1/(1 - \delta)$ exactly when $\delta = \alpha$, the setting we run, since Theorem 1 writes the level as $\alpha \gamma_t g$ — and $d = 0$, because that bound holds uniformly in t — g itself is not constant along the prefix, since the donated wealth grows, but $d = 0$ asks for a constant bound, not a constant g . The instantiation matters: $d = 0$ asks only for $\gamma_t \rightarrow 0$, which summability supplies, whereas $d = 1$ would ask for $t \gamma_t \rightarrow 0$, which it does not (take $\gamma_{2^j} = 2^{-j}$ and $\gamma_t = 0$ elsewhere). That is all Theorem 1 needs, since its conclusion is itself a statement about rejection-free runs — so the horizon stays finite and the run is absorbed. It is not a global member of the family: past $1/\delta$ rejections no finite $c_g(R+1)^d$ bounds the level. The bound is a cold-start statement only — past $1/\delta$ rejections the denominator can vanish — and it is sharp where it applies. On our streams the realised donation wealth stays at most 0.12 of a budget of 1.

Proposition 2 (Closure is bounded by its zero-evidence subset). For closed e-LOND whose subset aggregator is zero on all-zero input — $E_S = 0$ whenever $E_i = 0$ for every $i \in S$, including $S = \emptyset$ — let $Z_t = \#\{i < t : E_i = 0\}$. Then $\alpha_t \leq \delta \gamma_{Z_t+1} (R_{t-1} + 1)$. The hypothesis is not implied by e-merging validity: $\kappa + (1 - \kappa)$ mean is symmetric and valid yet returns $\kappa > 0$ on all-zero input, and the zero-evidence subset then certifies nothing.

Corollary 3 (When the closure is absorbing). Let $H = \min\{j : \gamma_j < 1/(\delta M)\}$ be the first index at which the spending weight dips below the floor — Theorem 1’s index horizon when γ is decreasing, as ours is — and $P_T = \#\{i \leq T : E_i > 0\}$ the number of hypotheses carrying any evidence. (i) Along a rejection-free run, closed e-LOND is absorbed by step $H + P_T$ whenever $H + P_T \leq T$: an additive offset, holding pathwise with no assumption beyond counting. (ii) Asymptotically, if $\liminf_t Z_t/t > 0$ then for every $\rho' < \liminf_t Z_t/t$ there is an onset t_1 beyond which the closure’s effective spending index is at least $\rho' t$, so absorption occurs by $\max\{t_1, \lceil (H - 1)/\rho' \rceil\}$ — the index is rescaled by ρ' and the absorption time therefore stretched by $1/\rho'$, not shortened. A positive $\liminf$ permits an arbitrarily long initial transient, which is why t_1 cannot

be dropped. On our streams $P_T \in \{0, \dots, 152\}$, at most 0.48% of T , so form (i) applies and the offset is at most 152 hypotheses.

Theorem 4 (Reach). Fix one hypothesis, offered level α_t , whose episode has flow positions i . For a pre-committed weight sequence with $\sum_i w_i \leq 1$, the number P of positions at which a lone attack flow can fire the rule $F(e) = \sum_i w_i e_i$ is at most $\alpha_t M$, the quantity that governs Corollary 1.

Theorem 5 (Front-load cost). At that same hypothesis, detection is impossible once $L \geq L^*(w) := \min\{L : \sum_{i>L} w_i < \beta\}$, with $\beta = 1/(\alpha_t M)$; L^* is finite for every summable w . The threshold is sufficient for impossibility, and exact only when every non-prepended position carries firing evidence: with fewer firing flows detection can already be impossible at smaller L , so L^* is an upper bound on what the attacker must prepend.

E. Group-level marginal e-validity (Lemma 1)

Proof. For a tested group G_j , recall $\mathcal{M}_j$ is the σ -field generated by the pre-committed metadata that determines this hypothesis: the grouping key, the membership indicators $\mathbf{1}\{i \in G_j\}$, the arity m_j , the group’s position in the pre-committed stream order (and hence the spending weight γ_j it is offered), and the horizon T . It carries no information about any other group’s composition or arity, and — as the scope note below records — none about the realised calibration scores. Fix a true-null group G_j (all constituent flows benign) with $m_j > 0$. By linearity of conditional expectation and $\mathcal{M}_j$ -measurability of the membership indicators and of m_j ,

$$\begin{aligned} \mathbb{E}[\text{Ev}(G_j) \mid \mathcal{M}_j] &= \mathbb{E}\left[\frac{1}{m_j} \sum_{i \in G_j} e_i \mid \mathcal{M}_j\right] \\ &= \frac{1}{m_j} \sum_{i \in G_j} \mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1, \end{aligned}$$

the inequality by Assumption 1. Taking expectations and applying the tower property gives $\mathbb{E}[\text{Ev}(G_j)] = \mathbb{E}[\mathbb{E}[\text{Ev}(G_j) \mid \mathcal{M}_j]] \leq 1$. No independence or exchangeability among the $\{e_i\}$ is used, so within-group dependence is unrestricted; and because e-LOND and online e-BH control FDR under arbitrary dependence given marginally valid per-hypothesis e-values, arbitrary between-group dependence is permitted as well. An empty group ($m_j = 0$) tests no hypothesis; assigning it the trivial $\text{Ev} = 1$ is harmless. For the two-point conformal e-value $e_i = M \mathbf{1}\{K_i \leq k\}$ the premise $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$ is exactly $\Pr(K_i \leq k \mid \mathcal{M}_j) \leq k/(|C| + 1)$, since $M = (|C| + 1)/k$. $\square$

Why the conditioning is per hypothesis. A single global metadata σ -field $\mathcal{M}$ will not support the attack analysis of Section V. $\mathbb{E}[e_i \mid \mathcal{M}] \leq 1$ does not imply $\mathbb{E}[e_i \mid \mathcal{M}'] \leq 1$ for a richer $\mathcal{M}'$ — take $e = 2\mathbf{1}_A$ with $\Pr(A) = \frac{1}{2}$: against the trivial σ -field the conditional expectation is 1, against $\sigma(A)$ it is 2 on A — so an adversary who enriches the stream’s metadata anywhere would, on the global reading, void the premise even for groups it never

touched. With $\mathcal{M}_j$ the statement is local and survives: padding adds flows carrying the attacked group's own key, so it changes no other group's key, membership or arity, and adds no hypothesis, leaving T fixed. Provided the within-bucket order is a function of group metadata alone (the deterministic hashed-key canonical order of Appendix D-A, with collisions broken on the key itself), every true-null group's position and γ_j are likewise fixed, so $\mathcal{M}_j$ is literally the same σ -field before and after the attack. Under an arrival-ordered tie-break this fails: a pad placed before the attacked group's first flow moves that group within its bucket and can shift a true-null group's index, hence γ_j , hence $\mathcal{M}_j$. Creating a new host pair is likewise outside the padding claim — it adds a hypothesis and changes T , which the horizon-uniform $\gamma_j = 1/T$ reads.

Scope. The lemma certifies the e-value controllers (e-LOND, online e-BH) only. The p-value procedures carry their own conditions — conditional super-uniformity for mFDR, and independence with monotonicity for FDR — which this argument does not supply, and which Section III-D shows ADDIS violates on this stream. The premise $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$ is marginal over the calibration draw and test score; it is not validity conditional on the realised calibration scores, which are deliberately excluded from $\mathcal{M}_j$ (the Bates adjustment supplies that stronger, calibration-conditional guarantee at a power cost and is not required here). $\mathcal{M}_j$ is also smaller than a global metadata σ -field, so Assumption 1 is the weaker of the two conditional premises, while remaining stronger than unconditional $\mathbb{E}[e_i] \leq 1$.

F. Feasibility (Theorems 1 and 2, Corollary 1)

Theorem 1. With $\alpha_t = \alpha \gamma_t g(\mathcal{H}_{t-1})$, $g \leq c_g(R_{t-1} + 1)^d$, and $R_{t-1} \leq t - 1$, we have $\alpha_t \leq \alpha c_g \gamma_t t^d$. A rejection at t needs $\alpha_t \geq 1/M$, hence $\gamma_t t^d \geq 1/(\alpha c_g M)$. If $\gamma_t t^d \rightarrow 0$ then only finitely many t satisfy that inequality, since a sequence tending to zero exceeds a fixed positive constant finitely often. A finite set of indices has a maximum, so there is a last feasible index t^* (or none at all), and every $t > t^*$ fails the necessary condition: the run is absorbed from $t^* + 1$. No monotonicity of γ is needed. Requiring permanence from the first infeasible step would be a stronger property — false for non-monotone γ , since feasibility can fail once and return — and it is not what Section III defines. $\square$

Theorem 2. Let $\alpha_t = w_0 \gamma_t + \sum_j a_j \gamma_{t-\tau_j}$ with $w_0 \geq 0$ and $0 \leq a_j \leq \alpha$ over rejection times $\tau_j < t$. During a rejection-free run of length Δ the last rejection is at least Δ before t , so every lag $t - \tau_j \geq \Delta$ and the lags are distinct. With $\Gamma(\Delta) = \sum_{u \geq \Delta} \gamma_u$ we have $\gamma_{t-\tau_j} \leq \Gamma(\Delta)$ termwise and $\gamma_t \leq \Gamma(t) \leq \Gamma(\Delta)$, so, using $a_j \leq \alpha$ and the distinctness of the lags, $\alpha_t \leq w_0 \Gamma(\Delta) + \alpha \Gamma(\Delta) = (w_0 + \alpha) \Gamma(\Delta)$; the first term is where $w_0 \geq 0$ is used, since $\gamma_t \leq \Gamma(\Delta)$ may only be multiplied through by a non-negative w_0 . Since Γ is non-increasing and $\sum \gamma < \infty$ forces $\Gamma(\Delta) \rightarrow 0$, the

quantity $\Delta^* = \min\{\Delta : (w_0 + \alpha) \Gamma(\Delta) < 1/M\}$ is finite; Δ only grows during a rejection-free run, so the state is absorbing. No monotonicity of γ is used. $\square$

Corollary 1 and max-min optimality of $\gamma_t = 1/T$.

Before any rejection, family I has $\alpha_t = \alpha \gamma_t$ and family II has $\alpha_t = w_0 \gamma_t$; write the cold-start coefficient $c_0 \in \{w_0, \alpha\}$, so $\alpha_T = c_0 \gamma_T$. Feasibility at T requires $\alpha_T \geq 1/M = k/(|\mathcal{C}| + 1)$, i.e. $|\mathcal{C}| \geq k/\alpha_T - 1 = k/(c_0 \gamma_T) - 1$. Under $\gamma_T = 1/T$ this is $|\mathcal{C}| \geq kT/c_0 - 1$: $kT/w_0 - 1$ for a level- w_0 procedure such as LORD++, and $kT/\alpha - 1$ for LOND/e-LOND. For optimality, a cold-start rejection is possible at every $t \leq T$ iff $\min_{t \leq T} \gamma_t \geq k/((|\mathcal{C}| + 1)c_0)$; among sequences with $\sum_{t \leq T} \gamma_t \leq 1$ the value $\min_{t \leq T} \gamma_t$ is maximised, at $1/T$, by the uniform allocation: $\min_{t \leq T} \gamma_t \leq T^{-1} \sum_{t \leq T} \gamma_t \leq 1/T$, since a minimum is at most an average, and $\gamma_t = 1/T$ attains it. Constraining the mass over all t rather than over $t \leq T$ changes nothing, since $\sum_t \gamma_t \leq 1$ implies $\sum_{t \leq T} \gamma_t \leq 1$, and the attaining sequence satisfies the global constraint too. Hence $\gamma_t = 1/T$ maximises the feasible horizon. That same criterion supplies the corollary's sufficiency, which is otherwise easy to miss: under $\gamma_t = 1/T$ it reads $1/T \geq k/((|\mathcal{C}| + 1)c_0)$, i.e. $|\mathcal{C}| \geq kT/c_0 - 1$ once more, so this one inequality is not merely necessary at step T but sufficient for feasibility at every $t \leq T$. The two directions coincide here only because a cold-start uniform level is constant over the horizon; for a general γ the necessary condition at T is strictly weaker than feasibility throughout, which is why the corollary states the equivalence for $\gamma_t = 1/T$ alone. $\square$

G. The UAI 2026 procedures (Propositions 1 and 2)

Proposition 1. Donation e-LOND sets $\alpha_t = \delta \gamma_t (|R_{t-1}| + 1) / (1 - (\delta(|R_{t-1}| + 1) \bar{W}_t \wedge 1))$ with $\bar{W}_t = \sum_{i \in R_{t-1}} \gamma_i ((E_i - \frac{1}{\delta \gamma_i (|R_{t-1}| + 1)}) \wedge 1) + \sum_{i \notin R_{t-1}} \gamma_i (E_i \wedge 1)$, with both sums running over $i < t$. Every summand is γ_i times a quantity capped at 1, so $\bar{W}_t \leq \sum_{i < t} \gamma_i \leq 1$ always. On a rejection-free prefix $R_{t-1} = \emptyset$, so for $\delta < 1$ the denominator is at least $1 - \delta > 0$ and $\alpha_t \leq \delta \gamma_t / (1 - \delta)$, which is Theorem 1 with $c_g = \delta / [\alpha(1 - \delta)]$ and $d = 0$: Theorem 1 writes the level as $\alpha \gamma_t g(\mathcal{H}_{t-1})$, so the constant must absorb the ratio δ/α , and it reduces to $1/(1 - \delta)$ exactly in the $\delta = \alpha$ setting we run. $d = 0$ is the right instantiation, not merely a valid one: on a rejection-free prefix $R_{t-1} = \emptyset$, so the displayed bound $g \leq c_g$ holds uniformly in t — g itself rises with $\bar{W}_t$ and is not constant — and Theorem 1's conclusion then needs only $\gamma_t \rightarrow 0$, which $\sum_t \gamma_t \leq 1$ gives. Taking $d = 1$ instead would demand $t \gamma_t \rightarrow 0$, which summability does not imply ($\gamma_{2^j} = 2^{-j}$, zero elsewhere, sums to one with $t \gamma_t = 1$ infinitely often).

Why $\delta < 1$ is required. At $\delta = 1$ the denominator is $1 - \bar{W}_t$, which the donated wealth drives to zero, and the conclusion fails outright. Take $\delta = 1$, $\gamma_t = 2^{-t}$ and $E_t = 1$ for every t . Then $\bar{W}_t = \sum_{i < t} \gamma_i (E_i \wedge 1) = \sum_{i < t} \gamma_i = 1 - 2^{-(t-1)}$ and $\alpha_t = \gamma_t / (1 - \bar{W}_t) = 2^{-t} / 2^{-(t-1)} = \frac{1}{2}$ for

every t . The run is rejection-free because rejection needs $E_t \geq 1/\alpha_t = 2$ and $E_t = 1$ — this is why the evidence is pinned at exactly 1 rather than merely bounded below by it — yet the offered level never decays, so with any $M \geq 2$ every step stays evidence-feasible and the run is never absorbed. The proposition therefore quantifies over $\delta \in (0, 1)$; our runs use $\delta = \alpha = 0.05$, where $1/(1 - \delta) = 1.05$.

The restriction to the rejection-free prefix is not cosmetic. With $R = |R_{t-1}|$ the denominator is $1 - (\delta(R + 1)\bar{W}_t \wedge 1)$, which vanishes once $\delta(R + 1)\bar{W}_t \geq 1$; since $\bar{W}_t$ can approach 1, that is reachable at $R + 1 \geq 1/\delta$. The bound is therefore a cold-start statement, which is exactly the state Corollary 1 concerns. It is also sharp: taking $E_i \geq 1$ for every i makes $\bar{W}_t = \sum \gamma_i \rightarrow 1$ and the bound tight. $\square$

Proposition 2. Closed e-LOND sets $\alpha_t = \min_{S \subseteq [t-1]: D_t(S) > 0} \delta \gamma_{|S|+1} (|R_{t-1}| + 1) / D_t(S)$ with $D_t(S) = 1 + |S \cap R_{t-1}| - \delta E_S (|R_{t-1}| + 1)$, where E_S aggregates the evidence on S . Because α_t is a minimum over admissible subsets, any single admissible S furnishes an upper bound.

Take $S = \{i < t : E_i = 0\}$, of size Z_t . A hypothesis carrying zero evidence is never rejected, so $S \cap R_{t-1} = \emptyset$; and $E_S = 0$ because the aggregator is zero on all-zero input, including the empty subset that arises when $Z_t = 0$. Hence $D_t(S) = 1 > 0$, S is admissible, and $\alpha_t \leq \delta \gamma_{Z_t+1} (|R_{t-1}| + 1)$.

This is the decay of Theorem 1 with the time index rescaled from t to $Z_t + 1$. The bound itself is unconditional and pathwise; the conclusion that the state is absorbing is not, and Corollary 3 supplies its hypothesis. $\square$

Corollary 3. By Proposition 2, $\alpha_t \leq \delta \gamma_{Z_t+1} (|R_{t-1}| + 1)$ along any run. On a rejection-free run $|R_{t-1}| = 0$, so $\alpha_t \leq \delta \gamma_{Z_t+1}$. If $\gamma_{Z_t+1} \rightarrow 0$ then $\alpha_t \rightarrow 0$, and since the evidence is bounded above by M (Section III) the offered level falls below $1/M$ after finitely many steps and no rejection is possible thereafter: the state is absorbing, which is Theorem 1's conclusion.

Form (i), exact and finite-horizon. Along the rejection-free run $|R_{t-1}| = 0$, so $\alpha_t \leq \delta \gamma_{Z_t+1}$. Our $\gamma_j \propto j^{-1.6}$ is decreasing, so no rejection is possible once $\gamma_{Z_t+1} < 1/(\delta M)$, i.e. once $Z_t \geq H - 1$ for $H = \min\{j : \gamma_j < 1/(\delta M)\}$, which is exactly Theorem 1's index horizon. Write $P_T = \#\{i \leq T : E_i > 0\}$. Of the $t - 1$ hypotheses before step t , at most P_T carry positive evidence, so $Z_t \geq t - 1 - P_T$ for every $t \leq T$ — a counting statement, with no assumption on the stream. Hence $Z_t \geq H - 1$ as soon as $t \geq H + P_T$, and

$$t_{\text{closed}} := \inf\{t : Z_t \geq H - 1\} \leq H + P_T.$$

The closure therefore reaches the absorbing state within an additive P_T steps of the horizon Theorem 1 gives, whenever $H + P_T \leq T$. Nothing is asymptotic and no density is assumed. For a non-monotone γ the same conclusion follows without the decreasing step, because

α_t is a minimum over admissible subsets: any $H - 1$ zero-evidence indices furnish an admissible S and hence the same bound.

Form (ii), asymptotic. Suppose $\liminf_t Z_t/t = \rho > 0$ and fix any $\rho' < \rho$. By the definition of the $\liminf$ there is a t_1 with $Z_t \geq \rho't$ for all $t \geq t_1$; so $Z_t \rightarrow \infty$ and, $\gamma_j \propto j^{-1.6}$ being decreasing to 0, $\gamma_{Z_t+1} \leq \gamma_{\lceil \rho't \rceil} \rightarrow 0$. Substituting into Theorem 1's discovery-horizon computation replaces the index t by $\rho't$: the effective spending index is rescaled by ρ' , so absorption is reached once both $t \geq t_1$ and $\rho't \geq H - 1$, i.e. by $\max\{t_1, \lceil (H - 1)/\rho' \rceil\}$. The onset t_1 cannot be dropped: a positive $\liminf$ constrains only the eventual density and permits an arbitrarily long initial transient. The time to absorption is therefore stretched by $1/\rho'$, not shortened — a sparser zero-evidence stream buys the closure more steps, not fewer. This holds for every $\rho' < \rho$ and not, from this argument, at ρ itself: the $\liminf$ gives eventual domination below its value, not at it.

On our streams form (i) is the operative one and needs no density at all. The number of hypotheses carrying any evidence is $P_T \in \{0, \dots, 152\}$ over the ten window-seed cells, at most $\hat{p} = 4.82 \times 10^{-3}$ of T , attained at position 0.85, and $P_T = 0$ at 0.70 seed 1; so the horizon offset is at most 152 hypotheses and the cap is attained (ratio 1.000000 over every configuration measured). The argument is specific to sparse evidence: a stream in which most hypotheses carry positive evidence has P_T close to T , Z_t small, and this bound says little — which is the regime the closure was designed for, and not a security stream. $\square$

H. Symmetric padding impossibility (Theorem 3)

We give a full proof (Route A) and an independent lemma (Route B) that needs no domination result. Route B's scope is narrow and worth stating exactly: it shows that a symmetric family maps any permutation of a mean-one two-point vector to at most 1. That yields the impossibility only for families whose unpadded value on that vector already reaches θ — the arithmetic mean among them — and not for the class. A symmetric valid family such as $F_N(e) = \text{mean}(e) \mathbf{1}\{\text{not all } e_i \text{ equal}\}$ attains θ elsewhere yet returns 0 on Route B's unpadded vector, so Route B produces no violation for it and Route A must. The class-wide statement and the quantitative bound are Route A's alone.

Route A (essential domination). Vovk and Wang's Proposition 3.1 [8] states that the arithmetic mean M_K essentially dominates every symmetric e-merging function: $F(e) > 1 \Rightarrow M_K(e) \geq F(e)$. Ordinary domination is false ($\kappa + (1 - \kappa)M_K$, for any $\kappa \in (0, 1)$, is a symmetric valid counterexample), so we use it through a case split: for finite x , either $F_N(x) \leq 1$, or $F_N(x) > 1$ and then $M_K(x) \geq F_N(x)$; in both cases $F_N(x) \leq \max(1, \text{mean}(x))$.

We need that inequality at $(x, 0^r)$, a boundary point of $[0, \infty)^N$, so we prove it directly rather than rely on

the domain of the cited proposition. Let $y \in [0, \infty)^N$, $\bar{y} = \text{mean}(y)$, and let Π be a uniformly random permutation of y ; each coordinate of Π has mean $\bar{y}$, and $F_N(\Pi) \equiv F_N(y)$ by symmetry. If $\bar{y} \leq 1$ then Π is an admissible input, $F_N(\Pi)$ is a constant e-variable, so $F_N(y) \leq 1$. If $\bar{y} > 1$, take $E = \Pi$ with probability $1/\bar{y}$ and $E = 0$ otherwise; every coordinate of E then has mean exactly 1, so validity gives $F_N(y)/\bar{y} + (1 - 1/\bar{y})F_N(0) \leq 1$, and $F_N(0) \geq 0$ yields $F_N(y) \leq \bar{y}$. Either way $F_N(y) \leq \max(1, \bar{y})$ on all of $[0, \infty)^N$, zeros included, which is what the padding step uses. This also removes any need for monotonicity or measurability assumptions on F_N . Padding $x \in [0, \infty)^m$ with r zeros gives $\text{mean}(x, 0^r) = (\sum_i x_i)/(m+r)$, so

$$F_{m+r}(x, 0^r) \leq \max\left(1, \frac{\sum_i x_i}{m+r}\right) < \theta \text{ once } r > \frac{\sum_i x_i}{\theta} - m.$$

Attainment supplies an x with $F_m(x) \geq \theta$ to which this applies, so no attaining symmetric family is θ -padding-robust for any $\theta > 1$. $\square$

Route B (a negatively-dependent special case). Fix $\theta > 1$ and m . Draw a uniformly random size- m subset $S \subseteq [N]$ and set $e_i = (N/m)\mathbf{1}\{i \in S\}$. Then $\mathbb{E}[e_i] = (N/m)\Pr(i \in S) = 1$, so each e_i is an e-variable, and the vector is a permutation of $((N/m)\mathbf{1}_m, 0^{N-m})$. A symmetric F_N maps every realisation to the same constant, and a constant e-variable is ≤ 1 . Hence $F_N((N/m)\mathbf{1}_m, 0^{N-m}) \leq 1 < \theta$ for all $N \geq m$: appending $N - m$ zeros to the m firing positions drives any symmetric rule below θ . $\square$

Tightness and attainment. The bound is tight: the mean itself has $F_{m+r}(x, 0^r) = (\sum x)/(m+r) \geq \theta$ exactly while $r \leq (\sum x)/\theta - m$. Attainment (Section V) is necessary, not cosmetic: the constant rule $F \equiv 1$ is symmetric, valid and padding-invariant but never reaches any $\theta > 1$, satisfying θ -padding-robustness vacuously; since rejection needs evidence $\geq 1/\alpha_t > 1$, restricting to attaining families excludes exactly the rules that cannot fire.

I. Feasibility ratio as a dilution threshold (Corollary 2)

Corollary 2. Under $\gamma_t = 1/T$ the level is $\alpha_t = (\alpha/T)(R_{t-1} + 1)$, so on a rejection-free prefix $\alpha_t = \alpha/T$ and (3) rejects G_t exactly when $\text{Ev}(G_t) \geq T/\alpha$; the rule is inclusive. Sufficiency. Let $c > \rho$ be an integer; we show by induction on t that no rejection occurs. Suppose none has occurred before t , so $R_{t-1} = 0$ and the level is α/T . If G_t is not an attacker episode, $\text{Ev}(G_t) < T/\alpha$ by hypothesis. If it is, with unpadded sum S_t and arity m_t , each of its flow e-values is at most M by (2), so $S_t \leq Mm_t$; its padded arity is the integer cm_t and the padded mean is $S_t/(cm_t) \leq M/c < M/\rho = T/\alpha$, using $c > \rho$ and $\rho = M\alpha/T$. In either case (3) fails at t and the induction closes. Necessity. Let $c \leq \rho$ be an integer and let some attacker episode G_u attain the ceiling, $S_u = Mm_u$. If the attacked trajectory were rejection-free, G_u would be tested at α/T with padded mean $Mm_u/(cm_u) = M/c \geq M/\rho = T/\alpha$ and rejected by the inclusive rule, a contradiction. Hence

no integer $c \leq \rho$ suffices for every stream satisfying the hypotheses, whereas the integer $\lfloor \rho \rfloor + 1 > \rho$ does. Two remarks. The hypothesis on the other hypotheses cannot be dropped: a benign group rejected at the cold-start level raises R and with it every later level, and the attacker's pads are then priced above α/T . And the threshold is uniform over streams: for one given stream with at least one attacker episode, the same two arguments, with M replaced by $\max_u \text{Ev}(G_u)$ over the attacker's episodes, show that the smallest sufficient integer is $\lfloor c_{\text{crit}} \rfloor + 1$ with $c_{\text{crit}} = \max_u \text{Ev}(G_u) \alpha/T \leq \rho$, and $c_{\text{crit}} = \rho$ exactly when some attacker episode attains M . A real-valued c would make $(c - 1)m_t$ non-integral; the statement is therefore made for integers, which is what an attacker can append. $\square$

J. Reach and front-load (Theorems 4 and 5)

Let $w = (w_1, w_2, \dots)$ be precommitted with $w_i \geq 0$ and $\sum_i w_i \leq 1$, let $F(e) = \sum_i w_i e_i$ with each $e_i \in \{0, M\}$, and write $\beta = 1/(\alpha_t M)$ for the weight mass a detection needs: a lone flow firing at position i gives $F = w_i M$, which clears $1/\alpha_t$ iff $w_i \geq \beta$.

Theorem 4. The positions at which a lone attack flow fires are $\{i : w_i \geq \beta\}$. Since $\sum_i w_i \leq 1$, at most $1/\beta$ indices can carry mass $\geq \beta$, so $P = |\{i : w_i \geq \beta\}| \leq 1/\beta = \alpha_t M$, the quantity that governs Corollary 1. $\square$

Theorem 5. Prepending L zero-evidence flows moves every attack flow to a position $> L$. Every $e_i \leq M$, so $F(e) \leq M \sum_{i>L} w_i$, and detection requires $\sum_{i>L} w_i \geq \beta$; it is impossible once $L \geq L^*(w) = \min\{L : \sum_{i>L} w_i < \beta\}$. Summability gives $\sum_{i>L} w_i \rightarrow 0$, so the set is non-empty and L^* is finite for every summable w .

Sufficient, and exact only under a full-firing episode. The displayed bound is attained only when every position $> L$ fires, so L^* is the exact minimum in that case and an upper bound otherwise. The gap is real: for $w = (0.4, 0.3, 0.3, 0, \dots)$ and $\beta = 0.5$ the tail first falls below β at $L = 2$, yet an episode carrying a lone firing flow is already undetectable at $L = 0$, since $\max_i w_i = 0.4 < \beta$ and Theorem 4 gives $P = 0$. We therefore quote L^* as an attacker's upper bound throughout. $\square$

Consequently a precommitted scheme cannot simultaneously be padding-invariant (weights fixed by absolute position), have usable power (some $w_i \geq \beta$), and make ordering robustness unbounded ($L^* = \infty$). Concentrating weight early gives $L^* = 1$ but $P = 1$; spreading it over more than $1/\beta$ positions drives every $w_i < \beta$, so $P = 0$.

K. ADDIS state budget (Theorem 6) and the coincidence

Theorem 6. ADDIS offers $\alpha_t = \min\{\lambda, (\tau - \lambda)[W\gamma_{S^t - C_{0+}} + \dots]\}$. The theorem assumes $\lambda \geq k/(|\mathcal{C}| + 1)$, i.e. the cap sits at or above the conformal floor, so that the driving term and not the cap is what the attack has to push below $1/M$; if the cap were already below the floor no target could be rejected at $D = 0$ and the budget would be zero rather

than the displayed value. With $W = w_0$ before the first rejection and $W = \alpha R$ after R of them, and 0-indexed $\gamma_j = (j+1)^{-1.6}/\zeta(1.6)$. Precursors injected at index 0 hold the spending index at $S^t - C_{0+} = D$, so the driving level is $(\tau - \lambda)W\gamma_D$. The collapse of ADDIS's reward sum to the single term $W\gamma_D$ is exact in the pre-rejection state, where $W = w_0$ and there is one active term. After $R \geq 1$ rejections each reward term carries its own selected-non-candidate lag $\ell_j \geq 0$, sitting at index $D - \ell_j$; the coefficients sum to exactly $w_0 + (\alpha - w_0) + \alpha(R - 1) = \alpha R$, but γ is decreasing, so $\sum_j a_j \gamma_{D-\ell_j} \geq \alpha R \gamma_D$. The drive is therefore larger than the collapsed expression, feasibility survives further, and the displayed value at $W = \alpha R$ is a lower bound on B^* — equality only when every $\ell_j = 0$. Numerically, at $\lambda M = (\tau - \lambda)M = 4.5 \times 10^5$, $R = 1$ and a single lag of 100, the expression returns 313 while the true budget is 373. The attack is mounted in the pre-rejection state, since the precursors precede every target. A rejection is possible only while $(\tau - \lambda)W\gamma_D \geq 1/M$, i.e. $(D + 1)^{-1.6}/\zeta(1.6) \geq k/((\tau - \lambda)W(|\mathcal{C}| + 1))$, i.e. $D + 1 \leq [(\tau - \lambda)W(|\mathcal{C}| + 1)/(k\zeta(1.6))]^{1/1.6}$. The smallest D violating this is B^* . Each precursor is a group carrying exactly one firing flow, so its evidence is M/m and its p-value m/M ; landing in $(\lambda, \tau]$ needs $m \in (\lambda M, \tau M]$, whose smallest integer is $m^* = \lfloor \lambda(|\mathcal{C}| + 1)/k \rfloor + 1$ (the interval is non-empty because $(\tau - \lambda)M \gg 1$). The total is B^*m^* , scaling as $\lambda|\mathcal{C}| [(\tau - \lambda)w_0|\mathcal{C}|]^{0.625}/k^{1.625}$. $\square$

Coincidence. When ADDIS's cap binds, $\alpha_t = \lambda$, and the minimal suppressing pad raises the padded p-value just past λ . Because the p-value moves in steps of one conformal floor $1/(|\mathcal{C}| + 1) \ll \tau - \lambda$, it lands in $(\lambda, \tau]$ and advances the index: cap-saturation implies landing. The converse is not forced, but on this stream every unsaturated detection has a level four orders of magnitude below λ , so none can land, and the observed equality (all 101 saturated detections land) is the generic case rather than a coincidence of this window.

L. Restart FDR composition

Proof. Restart runs n controllers on disjoint epochs, each controlling its own FDR at q . The deployment-wide proportion $\text{FDP} = \sum_i V_i / (\sum_i R_i \vee 1)$ is a weighted average of the per-epoch FDP_i with data-dependent weights $R_i / \sum_j R_j$, so per-epoch control does not bound $\mathbb{E}[\text{FDP}]$. Indeed, suppose each epoch makes one false rejection with probability q and no rejection otherwise. Each epoch then has $\text{FDR}_i = \mathbb{E}[V_i / (R_i \vee 1)] = q$, yet whenever any epoch rejects every pooled rejection is false, so the deployment-wide FDP is 1 on that event and 0 otherwise, and with independent epochs $\text{FDR} = \Pr(\text{some epoch rejects}) = 1 - (1 - q)^n$. (The neighbouring construction — each epoch always rejecting once, false with probability q — instead gives pooled $\text{FDR} = q$, so non-composition requires the all-or-nothing epoch above and does not follow from per-epoch control alone.) In contrast, mFDR pools: if

$\mathbb{E}[V_i] \leq q \mathbb{E}[R_i]$ for every i , the median inequality gives $\sum_i \mathbb{E}[V_i] / \sum_i \mathbb{E}[R_i] \leq q$.

This non-composition is specific to budget reset — every epoch spends its own q , total nq . A preallocated restart — pre-committed epoch budgets $q_1, \dots, q_n$ with $\sum_i q_i \leq q$, fixed before deployment — instead preserves the deployment-wide guarantee. Since $\sum_j R_j \geq R_i$, the pooled proportion is dominated termwise,

$$\text{FDP} = \frac{\sum_i V_i}{\sum_j R_j \vee 1} \leq \sum_i \frac{V_i}{R_i \vee 1} = \sum_i \text{FDP}_i$$

(both sides 0 when no epoch rejects), so by linearity of expectation $\text{FDR} \leq \sum_i \text{FDR}_i \leq \sum_i q_i \leq q$ — no independence across epochs is required (independence enters only the $1 - (1 - q)^n$ counterexample above). Each e-LOND epoch delivers $\text{FDR}_i \leq q_i$ under the same group e-validity premise (Assumption 1); the shared calibration set only induces between-epoch dependence, which this bound tolerates, provided the epoch boundaries and the q_i are not data-dependent. Restart therefore does not simply void the deployment-wide guarantee: it forces a choice — budget reset recovers more power but controls only a per-epoch FDR, whereas preallocation keeps deployment-wide $\text{FDR} \leq q$ at a power cost (Table XIV). $\square$

Appendix B

Procedures, Data, and Reproducibility

A. Terms

TABLE VII

Terms held distinct throughout the paper. Definitions only, so order-invariant: the table reports no measurement, and the quantities defined here that are measured on a sequenced stream are reported elsewhere. Validity is an assumption; feasibility, power and resolution are separate consequences.

Term	Meaning
Statistical validity	the assumptions the stated FDR/e-value guarantee needs (Assumption 1 and Lemma 1)
Feasibility	whether any admissible evidence can cross the controller threshold
Detection power	whether malicious episodes actually generate sufficient evidence
Episode recall	recall under the current grouping definition (moving denominator)
Atomic coverage	coverage of a fixed fine-grained attack-unit denominator
Alert blur	number of fixed atomic reference units one issued alert represents
Per-alert padding cost r_t^*	zero-evidence flows that suppress one alert at the level it received on the unperturbed run, (8); an oracle lower bound (Surface A). On AIT, where a pad flow can fire, the empirical replay cost $r_{t,\text{emp}}^* \geq r_t^*$ is reported and labelled
Per-alert sum $\sum_{t \in \mathcal{D}} r_t^*$	the per-alert costs of the baseline true detections $\mathcal{D}$ added up on the unperturbed trajectory; an upper bound on the joint cost
Joint padding cost J_{seq}	total cost of the greedy sequential oracle when e-LOND is re-run over the padded stream, so that R_{t-1} responds to the pads (the joint controller-state rerun, Surface A)
State-free multiplier c	the attacker multiplies its own episode size by c and reads no controller state (Corollary 2)
Controller-state attack cost B^*	precursor hypotheses that advance ADDIS's evidence-conditioned spending index (Surface B, Appendix G); a different mechanism from the e-LOND rerun above: padding changes e-LOND's rejection state indirectly by preventing discoveries, whereas the ADDIS attack directly advances its evidence-conditioned spending index

B. Procedure definitions

Each procedure offers a level α_t and rejects H_t when the evidence exceeds $1/\alpha_t$ (p-value procedures reject when $p_t \leq \alpha_t$). We write R_{t-1} for the rejection count, τ_j for the j -th rejection time, and, for the candidate-based procedures, $C_{j+}(t)$ for the number of candidates after τ_j . Transcriptions were unit-tested against literal $O(T^2)$ implementations of the published formulas and against four identities the sources state (offline e-BH as the $t = K$ case of online e-BH; $R(\text{e-LOND}) \subseteq R(\text{online e-BH})$; $\text{e-LORD} = \text{e-LOND}$ under $\gamma_t = \omega_t \prod_{j < t} (1 - \omega_j)$; e-

SAFFRON($\lambda=0$) = e-LORD).

LOND, e-LOND: $\alpha_t = \alpha \gamma_t (R_{t-1} + 1)$,

LORD++: $\alpha_t = \gamma_t w_0 + (\alpha - w_0) \gamma_{t-\tau_1} + \alpha \sum_{j \geq 2} \gamma_{t-\tau_j}$,

SAFFRON: $\alpha_t = \min \{ \lambda, (1 - \lambda)[w_0 \gamma_{t-C_{0+}} + \dots] \}$,

ADDIS: $\alpha_t = \min \{ \lambda, (\tau - \lambda)[w_0 \gamma_{t-C_{0+}} + \dots] \}$,

online e-BH: reject H_i iff $E_i \geq 1/(k_t^* \alpha \gamma_i)$,

e-LORD (e-GAI): $\alpha_t = \omega_t (\alpha - \sum_{j < t} \frac{a_j}{R_{j-1}+1}) (R_{t-1} + 1)$.

SAFFRON (serial estimate of the alpha fraction that is futilely rationed on true null hypotheses) and ADDIS (adaptive discarding) are the adaptive descendants of LORD, and online e-BH is the e-value Benjamini–Hochberg procedure run at each step. SAFFRON's candidates are the $p \leq \lambda$; ADDIS additionally selects the $p \leq \tau$ and counts selected non-candidates. Here $k_t^* = \max \{ k \leq t : \sum_{j \leq t} \mathbf{1}\{E_j \geq 1/(k \alpha \gamma_j)\} \geq k \}$ is the online-e-BH step-up fixed point, and ω_t is the e-GAI weight update [27], which we run pre-committed — the specialisation under which the e-LORD reparametrisation of Section III yields a fixed summable γ , and the only one our feasibility coverage claims. Throughout $\alpha = q = 0.05$, $w_0 = 0.025$, $k = 1$; ADDIS uses $\lambda = 0.25, \tau = 0.5$, SAFFRON $\lambda = 0.5$, e-SAFFRON $\lambda = 0.1$, and e-GAI $\phi = \psi = 0.5$, $\omega_1 = 1/T$. The two spending sequences are $\gamma_j \propto j^{-1.6}$ (normalised by $\zeta(1.6) = 2.2858$, no horizon needed) and horizon-uniform $\gamma_j = 1/T$ (max-min optimal, [horizon-aware]). We do not extend the feasibility bound to e-SAFFRON at $\lambda = 0.1$, a testing level the equivalence does not cover. mem-e-LORD is a different case and the tables mark it covered: its level has the family-I form, so Theorem 1 bounds its feasibility exactly as for e-LORD; what does not transfer is the error-rate reading, since it controls mem-FDR rather than FDR, and we never quote its rejections as FDR-controlled. The r-LOND variants inherit the classification verbatim, their rejection sets coinciding with e-LOND's on calibrated p -values. For e-TOAD we use the interpretation of the active set and step-up limit consistent with its stated limiting cases ($d_t = t \Rightarrow \text{e-LOND}$, $d_t = \infty \Rightarrow \text{online e-BH}$); the alternate literal reading is a sensitivity analysis in Table XV. One equation of the compound-e construction admits two readings (a literal snapshot and a running union of the rejection history); we report both and default to the literal one.

Table VIII gives the full procedure matrix at the primary and stress windows, including the first-flow rejection counts the body's taxonomy (Table IX) summarises.

C. Reporting conventions

Unless a sentence, caption or table says otherwise, every measured number in this paper is detector seed 0 under the canonical within-bucket order at the stated window, in the horizon-free regime $\gamma_j \propto j^{-1.6}$; horizon-aware results are named as such or marked [horizon-aware]. Results that necessarily run under first-flow arrival —

TABLE VIII

Procedures, assumptions, and coverage by Theorems 1 and 2. Power is reported at the primary analysis window (0.55, $|C| = 2,448,993$, $T = 57,368$ episodes, 275 malicious) and the stress-test window (0.85, $|C| = 1,813,113$, $T = 31,568$, 255 malicious), two-hour grouping. Horizon-uniform γ requires the stream length in advance and is marked [horizon-aware] wherever it appears. The FDP column is the 0.85 stress-window value, measured against labels where the evidence is not a valid e-value (Appendix F-A); at 0.55 every FDP here is ≤ 0.010 .

Which procedures are covered is position-independent; the rejection counts differ because the two windows carry different traffic.

[†]SAFFRON sits outside the letter of Theorems 1 and 2 because its index counts tested hypotheses, but it does not escape in practice: it is silent for 99.3% of the stream at 0.55 and 75% at 0.85. Only ADDIS and online e-BH avoid the horizon’s practical consequence, and they do so differently: ADDIS’s index stops advancing, a genuine escape from arrival-time absorption, whereas online e-BH is decided by a fixed point over the history rather than by a level fixed at arrival, so the absorption predicate is not defined for it at all. It is still unable to reject 90.1% of hypotheses at their own arrival here, exactly as often as e-LOND, and recovers them only by revisiting (Section III-D and Table XV). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

Procedure	Evidence	Spending index	Horizon needed	Covered by C1?	Rej. 0.55 (primary)	Rej. 0.85 (stress)	FDP (0.85)	Nominal guarantee / required assumptions
LOND [3]	p-value	elapsed time	no	yes (I)	18	72	0.000	FDR (super-unif. null p ; indep.)
e-LOND [5]	e-value	elapsed time	no	yes (I)	18	72	0.000	FDR under group e-validity
LORD++ [4]	p-value	lag over rejections	no	yes (II)	19	72	0.000	FDR (conditional / indep.)
e-LORD (=e-SAFFRON $_{\lambda=0}$) [27]	e-value	elapsed time (reparam.)	[horizon-aware]	yes (I)	101	150	0.020	FDR under group e-validity
mem-e-LORD [27]	e-value	elapsed time (reparam.)	[horizon-aware]	yes (I)	87	142	0.000	controls mem-FDR
SAFFRON [24]	p-value	non-candidates	no	no [†]	0	70	0.000	assumes independence
ADDIS [25]	p-value	selected non-cand.	no	no	110	152	0.033	null-cons. violated
online e-BH [5]	e-value	fixed point over history	no	no	18	72	0.000	FDR under group e-validity

the alert-blur, smoothing, restart, asymmetric-weighting, host-detector and ADDIS contrasts, and the AIT host-conditioned arm — say so at the point of claim, and their appendix tables carry both seeds where a seed applies. Every per-alert padding cost r^* is an oracle lower bound, because exact sizing reads the realised group evidence and the live controller level; it is written r_t^* throughout, the subscript marking the step whose running level prices it, and a row priced against the static cold-start threshold instead says so in words and uses no such symbol (Tables XXIV and XXXVII); whole-detected-set padding totals and the front-load cost L^* are upper bounds, and the keyed-hash budgets $N_{50/90/99}$ are reliability figures over 400 draws (Table XXII). Windows are named by chronological position: 0.55 primary, 0.62 secondary, 0.85 the known-invalid stress window, used for mechanism and cost measurement only.

D. Chronological split geometry

The five deployment positions are overlapping chronological views of one exercise. Each deployment block spans $[\text{pos}, \text{pos} + 0.15)N$ up to integer rounding, so adjacent positions share 46.7–53.3% of their deployment flows — the primary/secondary pair shares 53.3%, the largest in the set — and the deployment block at one position coincides, to within one flow, with the calibration block of the position 0.15 later. Deployment blocks two or more positions apart are disjoint, but no two positions are independent even then: the training pools $[0, i_1)$ are nested, so every earlier position trains on a prefix of every later one’s data and the fitted detectors are correlated across all five. Each position is internally chronological — training, then calibration, then deployment — so this couples the positions to one another without crossing any single position’s own split. This is why the paper reports the five windows separately and never pools them, and

why 0.62 is a correlated robustness check rather than confirmatory replication of 0.55.

E. Datasets and preprocessing

LSPR23 [19] (Zenodo 8042347, CC-BY-4.0) is a single CSV of 16,353,511 flows; 89% of them fall in the final 25.6 hours, so window positions are chronological slices rather than random splits. The detector reads Protocol plus 32 per-flow timing and volume statistics (columns 9–40); the grouping key reads SrcIP, DstIP (columns 2–3) and, for the service and subnet families, the ports (columns 4–5); the alert audit of Appendix F-A reads the annotation columns. The dataset carries no flow-to-campaign labels: Expoid_dst is empty for 1,630,732 of the 1,644,599 malicious flows, and Category, Severity and SigID are effectively constant where present. Timestamps are in microseconds and 7,381,261 rows arrive out of order, so the pipeline sorts on load. With test-window start fraction pos, $i_2 = \lfloor \text{pos} \cdot N \rfloor$, $i_1 = i_2 - \lfloor 0.15N \rfloor$ and $i_3 = \min(N, i_2 + \lfloor 0.15N \rfloor)$ give training $[0, i_1)$, calibration $[i_1, i_2)$ and deployment $[i_2, i_3)$; benign subsampling is confined to $[0, i_1)$. A separate red-team task record (288 narratives, 83 machine-readable compromise reports, 295 timestamped step submissions) is used only as external ground truth in Appendices D and F, never as a feature.

AIT-LDSv2.0. The transfer tests use AIT-LDSv2.0 [20] (Zenodo 5789064, CC-BY-4.0), eight organisation testbeds whose released packet captures we convert to per-organisation tstat tcp_complete netflows following the dataset’s own recipe (45,001 attack flows). Each flow carries the dataset’s own attack-label string; we treat as malicious exactly the attack-step labels it defines (service_scan, host_discover_dmz/_local, on-line_cracking, wpscan, dirb_scan, upload_rce_shell...) by exact string match, and as benign exactly the labels the dataset marks benign; flows whose label is unknown or empty are dropped rather than assumed benign. Numeric

columns are kept per fold only if they are non-null on $> 90\%$ of rows and non-constant; all identity fields and all absolute-time fields are excluded so no fold can key on the capture epoch, and the six host-context features are the same strictly causal ones as on LSPR23. Splits are chronological within each organisation: calibration uses benign flows strictly preceding that organisation’s first attack flow, deployment is everything from that instant on.

Detectors and grouping. The detector is HistGradient-Boosting; an unsupervised Isolation Forest runs alongside as a feasibility-versus-sufficiency probe, with the same margin yet tail reach 0.000 at all ten configurations. A group’s evidence uses every flow in its time bucket, so its hypothesis is emitted only when the bucket closes, using only observations available then. Groups closing in the same bucket are sequenced by the canonical splitmix64 hash of the group’s own (SrcIP, DstIP, bucket) key, collisions broken on the key; the first-flow arrival order is retained as a labelled upper bound. The headline grouping is not tuned on its reported window: under previous-window selection and five leave-one-out folds it is near-worst of the 30 groupings feasible at every window, the cap being the selection-stable choice while the grouping does not transfer (Table XX). Every closed form is verified numerically against an independent implementation, and each measurement script carries unit tests.

Appendix C Full Feasibility Evidence

Table X gives the ten-configuration e-LOND detection matrix, Table XI the power of every procedure over the same grid, Table XII the joint (q, γ) sweep, Table XIII the smoothing arm, Table XIV the restart arm, Table XV the closure, donation and decision-deadline procedures against the horizon, and Table XVI the low-prevalence sensitivity.

Question answered. Which procedures escape the finite horizon, at what price, and does a better detector change the boundary? Takeaway. Escaping takes a different spending mechanism, not a better one, and ranking quality moves the detection counts rather than the margin behind them.

A. Feasibility versus power, in full

Under the canonical order the primary window yields three e-LOND detections out of 275 malicious episodes and the secondary window eleven; at positions 0.70 and 0.77 no episode clears its own step anywhere in the stream, so the canonical count there is zero (Table XXI carries the full five-window order matrix). The separation appears directly in paired contrasts. At position 0.70 both detector seeds share $|\mathcal{C}|$, T and the margin +0.536, yet under first-flow order one seed yields 30 detections and the other none. A host-conditioned detector raises AUROC from 0.916 to 0.954 at the primary window but reduces e-LOND detections from 18 to two (first-flow, both

legs). An Isolation Forest has the same margin as the gradient-boosted detector and not one attack flow above the calibration maximum at any of the ten window-seed cells. Ranking quality can move power; it cannot move the margin in (9).

Two configurations outside the headline pair corroborate that the grouped counts are a property of the spending sequence rather than of the ceiling: e-LORD under the horizon-aware e-GAI weighting reaches 101 detections at the same window and seed with no false discoveries, and periodic restart at the same grouping and the same deployment-wide budget reaches 55.

Low prevalence compounds the cold-start problem, and live-fire prevalence is the optimistic case. Thinning malicious episodes to a target rate while replacing each with benign evidence from the same window holds T , the margin and every survivor’s index exactly fixed: at $\pi = 10^{-4}$ the probability that e-LOND ever rejects falls to 0–40% across the evaluated cells, mostly below 10%, while the margin does not move (Table XVI). This is a sensitivity analysis, not a counterfactual deployment, but it shows why a short discovery prefix is most damaging in exactly the regime that motivates FDR control.

Prevalence as a swept parameter. LSPR23 is a live-fire exercise: 10.06% of flows and 0.48–0.81% of episodes are malicious, the optimistic case, so Table XVI treats prevalence as a swept parameter rather than asserting a figure for a typical SOC: the targets run from 0.7 to 2.7 orders of magnitude below the exercise’s episode rate, a sensitivity range, not a claim about any deployment. We thin malicious episodes — not flows, which would change group arity and evidence sums and so confound the feasibility question with the padding cost — to each target π , keeping a nested random subset so a lower target keeps a strict subset of a higher one. The primary mechanism replaces each thinned episode with benign evidence drawn from the same window, so the hypothesis stays at its own index and T , the margin and every survivor’s level are held exactly fixed; the deletion mechanism removes it instead, which shrinks T by up to 0.80% and raises the margin by up to +0.0129. The two agree to within 0.02 in bootstrap probability, so it is prevalence and not re-indexing that drives the collapse. Feasibility is untouched while detection collapses: the feasibility-is-not-detection separation, driven by base rate alone. The $\pi = 0$ arm produces no rejection on any stream, but that is 5 effective (overlapping) windows, and zero would be the likely outcome even at the 4.5% early-false rate quoted from simulation in Appendix C-B ($P = 0.79$): it is consistency, not a test of that number.

B. What escapes the horizon, and what it costs

Among the procedures we study the boundary is sharp: the horizon binds a procedure whose spending index advances on every hypothesis. One that advances it only on tested hypotheses can escape that bound; one that

defers the decision falls outside the predicate altogether (Table IX). SAFFRON sits outside the letter of the theorems, since its index counts non-candidates, but does not escape in practice (silent at 99.3% of steps at the primary window, first-flow procedure matrix). ADDIS advances its index only on hypotheses whose p-values are selected ($p \leq \tau$) but not candidates ($p \leq \lambda$); with two-point evidence at $k = 1$ no observed episode lands in $(\lambda, \tau]$, so the index stops advancing and the level stays high. That is a genuine escape from absorption, but on the real stream it coincides with a violation of ADDIS’s uniform-conservativeness premise, and Appendix G shows that conditioning the index on observed evidence also makes the state attacker-controllable. Online e-BH’s step-up threshold is a fixed point over the whole observed history [5], [9], [26], so it does not decide hypothesis t against a level fixed at t : (5), and with it the absorption predicate, is simply not defined for it. What it escapes is the permanence of an early infeasible prefix — a later observation raises the step-up count and can make a hypothesis already passed rejectable. It does not escape bounded evidence: index j is rejectable only if some attainable count R gives $R\alpha\gamma_j \geq 1/M$, so a horizon- T run still leaves an index no step-up count can reach. Decision deadlines interpolate between immediate e-LOND-style decisions and indefinite reconsideration [41], [30], and a bucket-close deadline is exactly the batched architecture a SOC receives, valid under arbitrary dependence where a batched BH is not. What deferral buys is latency, not feasibility: the fraction of hypotheses that cannot be rejected at their own arrival is 96.3% at the primary window whether decisions are immediate, held to bucket close, or never forced, because a step-up count of R needs R hypotheses to clear the bar at once and at most 0.48% of episodes carry any evidence. Under the canonical order it adds one detection in the stress window and none elsewhere (Table XV).

Four constructions alter the boundary, but none removes the tradeoff for free, and Table IX prices each; the smoothing and restart contrasts are measured under first-flow order on both legs, so the order shifts the level and not the sign. Two remove a premise. Randomised smoothing dissolves the absorbing state by replacing the discrete p-value with a $\text{Unif}(0, 1)$ draw [6], [15], but its surplus is a lottery: the episodes detected with probability ≥ 0.9 are exactly the discrete ones, and the surplus does not survive arbitrary-dependence-valid merging (Table XIII). Periodic restart renews the cold-start budget and recovers real power. Held at the headline grouping, the primary window, seed 0 and the same deployment-wide budget $\sum_i q_i \leq q = 0.05$, restarting every two hours takes e-LOND from 18 rejections to 55, recall 0.065 to 0.200, at a measured FDP of zero. Spending a fresh q per epoch instead buys more — recall 0.360 at a total budget of 0.25 — but that is a per-epoch FDR, a weaker promise than the deployment-wide one (Table XIV, proof in Appendix A).

Restart forces that choice rather than removing it, and the comparison has to hold grouping, seed and total budget fixed or it measures the wrong thing. Two work within the premises: the recent closure and compound-e donation constructions [30], [31], strict improvements over e-LOND under arbitrary dependence, are bounded on a rejection-free prefix by a constant cold-start factor ($1/(1 - \delta)$, a 5% change against a $20\times$ shortfall) and by an additive horizon offset respectively (Propositions 1 and 2 and Corollary 3), and buy +0 true detections at every window and seed we run (Table XV, seed 0 shown). Among the elapsed-time procedures bound by Theorems 1 and 2, the cold-start barrier is therefore a property of the evidence rather than of the controller’s sophistication.

The level- w_0 margin. The appendix tables report feasibility on one cross-controller scale rather than per procedure, so that a single column serves every row:

$$\text{margin} = M \frac{w_0}{T} - 1 = \frac{(|C| + 1) w_0}{kT} - 1. \quad (9)$$

It is the body’s ρ of (6) rescaled, $\text{margin} = \rho_p c_{0,p}^{-1} w_0 - 1$, and for e-LOND, whose $c_0 = 2w_0$, $\rho_{\text{e-LOND}} = 2(\text{margin} + 1)$. A negative margin therefore does not imply e-LOND infeasibility: e-LOND can still fire down to margin $-\frac{1}{2}$, the rejection rule (3) being inclusive. This is exactly why the body plots ρ instead.

Feasibility inputs, window by window. Across the five positions $|C|$ is 2,448,993, 2,449,031, 2,287,988, 2,116,418 and 1,813,113 benign calibration flows, against $T = 57,368, 49,267, 37,231, 31,672$ and $31,568$ two-hour host-pair episodes, matching Table X. Within-bucket ordering changes the sequence, not T . The two headline windows are the two largest calibration sets, and 0.62 exceeds 0.55 rather than repeating it.

Cold-start arithmetic. The count ratio of Corollary 1 is dataset-independent, but its calendar form is not: for LORD++ the horizon-uniform requirement is 1.44×10^9 calibration flows for one hour of a 10k-flows/s link and 3.46×10^{10} for a day. Under $\gamma \propto j^{-1.6}$ the feasible prefix before any rejection is 902/902/865/824/748 steps at the five windows for e-LOND. At the $\pi = 10^{-4}$ sensitivity point that window almost always closes empty: the first true attack is not expected until index $\approx 10,000$, so a rejection arrives in time to bootstrap the controller with probability only $\approx 0.3\%$, and the alternative, an early false rejection, has probability at most a few percent (4.5% for e-LOND, 2.2% for LORD++, in simulation at a stated $|C|$). The benign-only arm of Table XVI produces no rejection on any stream, which with five effective windows is consistent with that rate rather than a test of it ($\text{Pr}(\text{zero} \mid 4.5\%) = 0.79$). Deleting thinned hypotheses instead of replacing them shortens T and raises the margin by up to +0.013, yet gives the same collapse to within 0.02 in bootstrap probability, so it is prevalence and not re-indexing that does the work.

TABLE IX

Procedures by what advances their spending sequence, and how each relates to the finite horizon. “Covered by” is coverage of the procedure together with the spending sequences we run, which is what Theorems 1 and 2 quantify over; a procedure name alone does not carry the decay condition (Appendix A-B). “Absorbing” is whether a rejection-free prefix can enter the absorbing infeasible state on bounded evidence. Prices are measured on LSPR23 two-hour host pairs: detection deltas under the canonical order (Table XV), SAFFRON’s silence fraction from the first-flow procedure matrix (Table VIII). [†]Donation’s bound needs $\delta < 1$ and holds on the rejection-free prefix only, so it is not a global member of Theorem 1’s family; the closure’s additive offset needs sparse evidence ($H + P_T \leq T$). Both are stated exactly in Appendix A. [‡]Theorem 1 bounds mem-e-LORD’s feasibility, not its error rate: it controls mem-FDR rather than FDR. [§]The predicate is defined only for procedures that decide at arrival, so it does not apply to online e-BH, nor to the $d_t = \infty$ end of the deadline family, which is online e-BH (Appendix C-B).

Procedure	Evidence	Spending index	Covered by	Absorbing?	Escape mechanism / measured price
LOND, e-LOND [3], [5]	p / e	elapsed time	Thm. 1	yes	—
e-LORD, mem-e-LORD [27]	e	elapsed time (reparam.)	Thm. 1 [‡]	yes	— (α -death fix = e-LORD before the first rejection)
LORD++ [4]	p	lag over rejections	Thm. 2	yes	—
Donation, closed e-LOND [30], [31]	e	elapsed time; zero-evidence subset	App. A	yes [†]	cold-start coefficient $\times 1/(1 - \delta)$; horizon offset $\leq P_T$; +0 detections
SAFFRON [24]	p	non-candidates	no	in practice	silent at 99.3% of steps (primary window)
ADDIS [25]	p	selected non-candidates	no	no	index stops advancing; conservatism premise violated; state attackable (Appendix G)
Decision deadline, e-TOAD [41], [30]	e	deferred to d_t	no	no	latency; +1 detection, stress window only
Online e-BH [5], [26]	e	fixed point over history	no	n/a [§]	revisiting; still needs $ C \geq T/(\alpha R) - 1$
Smoothing, restart	any	—	premise removed	no	unstable surplus; budget-split keeps deployment-wide q , fresh- q gives a per-epoch guarantee

Horizon misspecification (single cell). Online e-BH’s advantage over the covered procedures is robustness to a misspecified horizon. At the stress window and one seed, the only cell we sweep the horizon on, online e-BH keeps 144 of its 152 rejections under a $100\times$ over-estimate of T , where LOND keeps all 151 at $2\times$ but none at $100\times$, and LORD++ none already at $2\times$; read this as an illustration of the mechanism, not a measured rate.

Closure and donation, measured. Runs here do pass the $1/\delta$ rejection count at which Proposition 1 stops applying (30, 31 and 73 rejections at 0.70, 0.77 and 0.85, where the realised boost reaches 1.83 against the cold-start ceiling 1.05); what holds is the weaker fact that the denominator never vanishes on these streams. Both procedures buy +0 true detections under the canonical order across the ten cells and +1 and +0 under first-flow arrival (Table XV).

Appendix D Granularity and Ordering

Table XVII gives the grouping families at the primary window, Table XVIII the grouping-independent resolution cost against the fixed atomic reference (including the malicious-flow coverage the body omits), Table XIX the external red-team denominator, and Table XX the cross-window configuration transfer.

Restart as a granularity lever. Coarsening is not the only way to divide the hypothesis count a run must survive: hourly restart at one-hour grouping reaches recall 0.406 at ≈ 125 alerts, against 0.103 for six-hour coarsening, while

preserving episode resolution. A pre-committed budget split remains deployment-wide; assigning a fresh q to each epoch changes the guarantee to per-epoch (Section III-D).

The external denominator. LSPR23 ships the red team’s task record: 288 tasks and 295 timestamped step submissions, an attacker-defined decomposition of the same campaign whose declared network segments are the dataset’s own per-flow seg vocabulary (83 compromise reports agree, 0 disagree). Counting distinct red-team steps merged into each issued alert, the external count rises with the alerting unit as the proxy does (at 0.70, $4.3 \rightarrow 27.0$ across 5 min–24 h against the proxy’s $1.0 \rightarrow 28.3$; rank agreement +1.000 over six widths, +0.800 under the canonical order). We do not claim corroboration: a null that rotates the red-team timeline against the stream reproduces the observed counts at 18 of 19 cells; the record sees the alerts at only 3 of 10 window-order cells, because issued alerts sit in the feasible prefix, the first minutes of a window, which at 0.55 and 0.62 predates the exercise’s first submission; and every attributed alert sits on a labelled-malicious episode from the same exercise. The external unit agrees in direction and is too coarse to discriminate.

Attribution rule and nulls. The 5-minute atom is our proxy for one attack action, so Table XIX counts instead how many distinct red-team steps each issued alert merges, using LSPR23’s own task record (288 tasks, 295 timestamped submissions, 2023-03-09 07:03–15:08 UTC), an attacker-defined decomposition fixed before the dataset was published. A step is attributed to an alert when its

TABLE X

e-LOND ($\gamma \propto j^{-1.6}$) across five window positions and two detector seeds, two-hour host-pair grouping. The margin is seed-independent by construction; $|C|$ is the calibration size $|C|$. “silent” is the fraction of steps at which the controller is structurally infeasible — the offered level α_t has fallen below $1/M$, so no rejection is possible even at the maximal admissible evidence M (it is not the fraction of steps without a rejection). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

Position	Seed	AUROC	T	$ C $	margin	rejections	recall	silent
0.55	0	0.916	57,368	2,448,993	+0.067	18	0.065	90.1%
0.55	1	0.927	57,368	2,448,993	+0.067	18	0.065	90.1%
0.62	0	0.799	49,267	2,449,031	+0.243	13	0.033	90.5%
0.62	1	0.871	49,267	2,449,031	+0.243	12	0.030	90.9%
0.70	0	0.914	37,231	2,287,988	+0.536	30	0.105	80.1%
0.70	1	0.847	37,231	2,287,988	+0.536	0	0.000	97.7%
0.77	0	0.828	31,672	2,116,418	+0.671	31	0.126	77.3%
0.77	1	0.833	31,672	2,116,418	+0.671	31	0.126	77.3%
0.85	0	0.999	31,568	1,813,113	+0.436	72	0.282	65.4%
0.85	1	1.000	31,568	1,813,113	+0.436	71	0.278	65.7%

TABLE XI

Procedure power over five positions $\times$ two seeds (min / median / max), two-hour grouping. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else. The uniform rows are the horizon-uniform γ , an [horizon-aware] that must know T in advance (Table VII).

Procedure	γ	rejections			recall		
		min	med	max	min	med	max
LOND	$\propto j^{-1.6}$	0	24	72	0.000	0.085	0.282
LOND	uniform	0	100	151	0.000	0.273	0.576
e-LOND	$\propto j^{-1.6}$	0	24	72	0.000	0.085	0.282
e-LOND	uniform	0	100	151	0.000	0.273	0.576
LORD++	$\propto j^{-1.6}$	0	25	72	0.000	0.089	0.282
LORD++	uniform	0	100	151	0.000	0.268	0.576
SAFFRON	$\propto j^{-1.6}$	0	21.5	70	0.000	0.069	0.275
SAFFRON	uniform	0	0	0	0.000	0.000	0.000
ADDIS	$\propto j^{-1.6}$	0	106.5	152	0.000	0.274	0.576
ADDIS	uniform	0	0	0	0.000	0.000	0.000
online e-BH	$\propto j^{-1.6}$	0	31	72	0.000	0.094	0.282
online e-BH	uniform	0	105	152	0.000	0.273	0.576

TABLE XII

Error target q vs the spending sequence (median episode recall).

Feasibility scales linearly in q : the measured (margin + 1)/base ratio tracks the prediction $q/0.05$. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else. The uniform column are the horizon-uniform γ , an [horizon-aware] that must know T in advance (Table VII).

q	median recall		(margin + 1) ratio	
	poly	uniform	measured	predicted
0.01	0.049	0.000	0.20	0.20
0.05	0.085	0.273	1.00	1.00
0.10	0.100	0.274	2.00	2.00
0.20	0.137	0.274	4.00	4.00

TABLE XIII

Randomised smoothing at the primary window (LOND, $\gamma \propto j^{-1.6}$, position 0.55, seed 0): removing the floor adds no reliable detection that survives the merge choice. Rejections are mean $\pm$ SD over 100 randomisation draws where applicable; “reliable” is the number of malicious episodes detected with probability ≥ 0.9 , “any” with probability > 0 . All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

merge	route	rejections	reliable	any
mean-e	discrete	18.0	18	18
simes	smoothed	24.1 ± 2.7	18	100
hommel	smoothed	12.5 ± 4.7	1	67
bonf-p	smoothed	23.2 ± 2.7	17	100

submission falls in the alert’s bucket span and its task’s declared network segments meet the alert’s, or one of its compromise-report IPs is an alert endpoint. The link is the segment vocabulary — the task record’s `bt_baf_int` is the dataset’s own per-flow `baf_int` — corroborated against the

machine-readable compromise reports (36/39 IPv4 appear as a flow endpoint; 83 agree, 0 disagree). Two nulls are run. The label null permutes which task each submission belongs to, holding the times fixed: it asks whether which task acted is informative, and goes degenerate at the

TABLE XIV

Periodic restart at the primary window (e-LOND, seed pair 0/1 — decision-identical to LOND here, since the group p-value is $\min(1, 1/\text{Ev})$ so $p \leq \alpha_t \Leftrightarrow \text{Ev} \geq 1/\alpha_t$): detection restored at the cost of a per-epoch guarantee (total spend $n \cdot q$). Budget reset (E2c) spends $\alpha_{\text{tot}} = n \cdot q$; index reset (E2b) keeps $\alpha_{\text{tot}} = q$ via a pre-committed allocation. Rejections are shown per seed (0/1); recall and FDP are the two-seed mean. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

arm	epoch (h)	α_{tot}	rej. (s0/s1)	recall	FDP
uninterrupted (baseline)	—	0.050	18/18	0.065	0.000
restart 2h, budget reset (E2c)	2	0.250	99/91	0.345	0.000
restart 1h, budget reset (E2c)	1	0.450	130/121	0.406	0.000
restart 2h, index reset, uniform (E2b)	2	0.050	55/52	0.195	0.000
restart 2h, index reset, geometric (E2b)	2	0.050	28/26	0.098	0.000
[horizon-aware] uninterrupted	—	0.050	104/96	0.364	0.000

daily bucket where every submission shares one bucket. The shift null rotates the whole red-team timeline against the stream, preserving task identity, ordering, attributes and inter-arrival structure: it asks the sharper question, whether the alignment in time is informative, and does not degenerate.

Reading. The result is agreement in direction without discrimination. The external count and the proxy rise together (panel (b) of Table XIX), so the resolution cost survives substituting an attacker-defined unit for ours; but the count clears the label null at only 6 of 19 cells and the shift null at only 1, so the rise is driven by a wider bucket capturing more submissions rather than by alert-level correspondence, and we report it as agreement rather than as confirmation. The record also sees the alerts at only 3 of 10 window-order cells: at 0.55 (canonical), 0.62 (canonical), 0.62 (first-flow) the issued alerts sit in the feasible prefix — the first minutes of a deployment window — which predates the exercise’s first submission, so at most 9% of them overlap it in time at all; at 0.85 (canonical), 0.85 (first-flow) they do overlap but only 2 of 17 tasks with a timed step declare a segment or file a compromise report. Four further limits. (i) At the 86,400s bucket both nulls tie the observation exactly: every submission shares one daily bucket, so no permutation and no rotation can move a step between alerts (the spatial filter is not vacuous there — at 0.70/first-flow it admits 12 of 30 temporally overlapping alerts — it is the nulls that lose their power); only sub-daily widths inform. (ii) The whole positive result rests on six coarse segment labels: across every row the compromise-IP conjunct adds no alert the segment conjunct did not already admit. (iii) Every attributed alert sits on a labelled-malicious episode and the labels came from this same exercise, so nothing here is label-independent evidence that an alert is a true positive. (iv) A submission is when the red team reported: the lag, estimable from the compromise reports’ own timestamps, has median 170s (IQR 115–279s), small against every bucket width.

A. Within-bucket order

Ordering is robust to timestamp ties (at most 1.3% of episodes share a first timestamp; all 36 reported statistics are invariant across 50 randomised within-timestamp orders). The within-bucket order, by contrast, is load-bearing, and it is where Theorem 1 becomes visible: only a prefix of 902/902/865/824/748 steps can reject at all, so a detection needs a near-ceiling episode to land inside it, which under first-flow arrival is exactly what live-fire attack traffic does. Sequencing the same episodes by their own metadata takes the counts from 18/13/30/31/72 to 3/11/0/0/34 (Table XXI). Over 50 pre-committed orders the count has median 1 at the primary window and 23.5 at the stress window; the canonical order is not a uniformly typical draw (above the ensemble median at three windows, at its maximum at 0.62, below it at the two where nothing detects), while first-flow matches or exceeds the ensemble maximum at every window and seed.

Why that canonical order. First-flow arrival is an attacker timing lever, and the damage reaches beyond the attacker’s own hypothesis: one pad at the start of its bucket moves 373 other episodes at the primary window, shifting the γ_j and hence the $\mathcal{M}_j$ each is offered. A tie-break reading only the group’s own key makes that 0 by construction, so the Assumption 1-preservation argument of Appendix A describes the order the paper reports. Three qualifications are measured rather than assumed. It removes the timing lever, not the metadata one: appending flows moves no other group, but an adversary that creates keys moves ranks. A public hash is grindable over the attacker-selectable (SrcIP, DstIP): with the feasible prefix at 1.6–2.6%, an adversary trying 100 candidate keys lands one inside it with probability 0.80–0.93, so the canonical order is a canonicalisation, not a security mechanism. A keyed hash removes grinding at the price of a seed the adversary can neither learn nor infer: blind insertion still suppresses every one of the 84 targets in every draw at a budget of 10^6 keys, at 2.1–89 $\times$ (median 19 $\times$) the instantiated-key cost, with a per-window median N_{99} of up to 3.2×10^4 endpoint pairs (Table XXII). That cost is a capability, not a computation: on a network where only a few hundred reachable pairs are available the seed is a hard stop, a regime these figures do not cover. The measurement assumes the target sits in the first bucket of its window, which holds at every cell here; where an earlier bucket exists the adversary inserts there and precedes the target with probability one. One keyed instance detects 0 at the primary window.

How the keyed-hash budget is measured. Under a public hash the adversary ranks candidate keys offline for free and instantiates only the G^* that land ahead of the target; under a keyed hash it cannot rank at all and must instantiate blind, and the cost scales as G^*/u with the target’s own hash quantile u , which is why the spread in Table XXII is wide and heavy-tailed. Every one of the 84

TABLE XV

The UAI 2026 procedures [30] against the feasibility horizon ($\gamma \propto j^{-1.6}$, seed 0, two-hour grouping; Section III-D). “det.” is true detections, “silent” the fraction of hypotheses that cannot be rejected at their own arrival even at the evidence ceiling. For the step-up procedures that fraction is computed exactly, by re-running the step-up on the counterfactual in which the arriving hypothesis carries M ; the cheaper test — does M clear $1/(\alpha\gamma_t r)$ at the largest conceivable r — is only necessary, since reaching r needs r hypotheses to clear the bar at once, and it reports 0.0% at every window here. All counts and silence fractions are under the canonical within-bucket order, as Table III is. Deferring the decision leaves at-arrival feasibility essentially unchanged — identical to $d_t = t$ at three of the five windows — and buys power only by revisiting, which is why its price is alerting latency. The last row is online e-BH, reached as the $d_t = \infty$ limit and verified rejection-identical to it at every row. Donation e-BH is the literal reading of its source’s Eq. (102); a second reading of that equation gives one more detection at 0.85 and is reported here as a sensitivity analysis, and neither is quoted as a power comparison (Section III-D).

Procedure	0.55		0.62		0.70		0.77		0.85	
	det.	silent	det.	silent	det.	silent	det.	silent	det.	silent
e-LOND (baseline), $d_t = t$	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.1%
donation e-LOND	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.1%
closed e-LOND (e-LOND)	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.0%
donation e-BH	3	—	11	—	0	—	0	—	35	—
e-TOAD, bucket-close deadline	3	96.3%	11	91.3%	0	96.5%	0	97.4%	35	77.3%
e-TOAD, $d_t = \infty$ (= online e-BH)	3	96.3%	11	91.3%	0	96.5%	0	97.4%	35	77.3%

TABLE XVI

Detection at lower base rates (item R5): malicious episodes thinned to a target prevalence π with benign replacement; e-LOND, seed 0, both within-bucket orders (canonical key-hash and first-flow). Upper panel: mean rejections R and the bootstrap probability that the controller ever rejects; lower panel: that probability at $\pi = 10^{-4}$ for the escape procedures and for the deletion mechanism. Design and reading: Appendix C-A.

pos	order	observed		$\pi = 10^{-3}$		$\pi = 10^{-4}$		$\pi = 10^{-5}$		$\pi = 0$	
		R	boot	R	boot	R	boot	R	boot	R	boot
0.55	key-hash	3.00	100	0.54	38	0.06	4	0.02	2	0.00	0
0.55	first-flow	18.00	100	1.40	56	0.08	6	0.00	0	0.00	0
0.62	key-hash	11.00	100	0.18	14	0.02	2	0.02	2	0.00	0
0.62	first-flow	13.00	100	1.36	62	0.20	18	0.04	4	0.00	0
0.70	key-hash	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0
0.70	first-flow	30.00	100	3.52	92	0.38	28	0.10	10	0.00	0
0.77	key-hash	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0
0.77	first-flow	31.00	100	2.22	70	0.06	6	0.04	4	0.00	0
0.85	key-hash	34.00	100	0.48	26	0.02	2	0.00	0	0.00	0
0.85	first-flow	72.00	100	8.28	100	0.54	40	0.28	28	0.00	0
bootstrap probability (%) at $\pi = 10^{-4}$: the escapes collapse too											
pos	order	e-LOND	online e-BH	e-TOAD	e-LOND (deletion)						
0.55	key-hash	4		4	5						4
0.55	first-flow	6		6	5						6
0.62	key-hash	2		2	0						2
0.62	first-flow	18		18	25						18
0.70	key-hash	0		0	0						0
0.70	first-flow	28		28	40						30
0.77	key-hash	0		0	0						0
0.77	first-flow	6		6	5						8
0.85	key-hash	2		2	5						2
0.85	first-flow	40		40	40						40

Notes. “pos”: the window’s chronological position; “observed”: the unthinned stream; “ R ”: mean rejections over 50 draws; “boot”: the percentage of draws in which the controller ever rejects; the observed and $\pi = 0$ arms are single deterministic streams. Lower panel: online e-BH and e-TOAD are the escape procedures of Appendix C-B; “e-LOND (deletion)” removes thinned episodes instead of replacing them. $\pi = 10^{-2}$ is unreachable: it is above the observed episode rate, and we cannot manufacture attacks. The 0.85 rows are the stress window, whose evidence is not a valid e-value.

targets falls in every draw at a budget of 1,000,000 keys, so the seed does not stop the attack; at the nearest matched reliability, N_{99} against the certain public G^* , it multiplies the instantiated-key cost by a median $19.3\times$ (range $2.1\text{--}89.1\times$). Suppression is verified by replay at every G^* , not

TABLE XVII

Grouping families at the primary window (0.55, seed 0); flow coverage is window-specific. The margin column is the level- w_0 feasibility margin (eq. (9)); the recall and coverage columns are e-LOND under the horizon-uniform [horizon-aware] γ , whose cold-start coefficient is $\alpha = 2w_0$. $\dagger$ marks a negative level- w_0 margin; e-LOND can still fire there when that margin exceeds $-\frac{1}{2}$ (its own $\alpha = 2w_0$ cold-start bound), which is why some daggered rows carry non-zero recall/coverage — a row is 0 only when e-LOND is itself infeasible. Feasibility and detection are thus not the same condition (cf. Table III). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI).

Family	bucket	margin	episode recall	flow coverage
src-dst	300 s (5 m)	$-0.637^\dagger$	0.000	0.000
src-dst	7200 s (2 h)	$+0.067$	0.378	0.276
src-dst	none	$+0.548$	0.253	0.032
src	300 s (5 m)	$+1.546$	0.501	0.408
src	7200 s (2 h)	$+20.505$	0.382	0.353
src	none	$+54.964$	0.286	0.350
dst	300 s (5 m)	$+0.085$	0.532	0.689
dst	7200 s (2 h)	$+1.736$	0.304	0.033
dst	none	$+3.646$	0.184	0.034
subnet24	300 s (5 m)	$-0.451^\dagger$	0.448	0.697
subnet24	7200 s (2 h)	$+0.501$	0.285	0.031
subnet24	none	$+1.251$	0.193	0.031
src-dport	300 s (5 m)	$-0.327^\dagger$	0.324	0.368
src-dport	7200 s (2 h)	$+3.025$	0.066	0.351
src-dport	none	$+6.189$	0.032	0.340

asserted from a closed form, and the replayed success probability is compared with the binomial lower bound $P(\text{Bin}(N, u) \geq G_{\text{targeted}}^*)$: it falls short of that bound by at most 0.030 in probability, the Monte Carlo tolerance of 400 draws. None of this touches Surface A padding (Theorem 3), which needs no key.

Appendix E

Attack Measurements and Mitigations

Table XXIII gives the five padding pools under both orders, Table XXIV the per-window suppression cost (running-level versus static) at both seeds and both

TABLE XVIII

Grouping-independent resolution cost (src-dst family, e-LOND $\gamma \propto j^{-1.6}$, seed 0). The atomic evaluation reference is the set of 5-minute src-dst malicious episodes — a fixed fine-grained denominator we choose, not externally validated incident truth, since LSPR23 carries no campaign identifier — 1899 at the primary window, 1973 at the stress-test window — and its size is fixed across every alerting bucket. “cov (fixed)” is the fraction of those atomic units covered by ≥ 1 issued alert; “blur” is the mean number of distinct atomic malicious units an issued alert lumps together (the resolution lost); “recall (moving)” is the ordinary episode recall, whose denominator changes with the bucket. As the unit coarsens the fixed coverage rises while the blur climbs from 1 to ~ 20 –40: that climb, not the moving recall, is the denominator-invariant resolution cost. $\dagger$ marks a negative level- w_0 feasibility margin; e-LOND can still fire there when that margin exceeds $-\frac{1}{2}$ (its own $\alpha = 2w_0$ cold-start coefficient), which is why some daggered rows carry non-zero coverage/recall — a daggered row reads 0 only when e-LOND is itself infeasible (cf. Tables III and XVII). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI).

bucket	cov (fixed)	blur	recall (moving)	flow cov.
primary analysis window (0.55)				
5 m $\dagger$	0.020	1.0	0.020	0.001
30 m $\dagger$	0.065	3.6	0.077	0.004
1 h $\dagger$	0.068	7.2	0.058	0.004
2 h	0.152	16.1	0.065	0.007
6 h	0.240	25.3	0.103	0.011
1 d	0.385	38.5	0.123	0.021
stress-test window (0.85)				
5 m $\dagger$	0.308	1.0	0.308	0.301
30 m $\dagger$	0.317	4.5	0.277	0.418
1 h	0.322	8.8	0.238	0.418
2 h	0.322	8.8	0.282	0.418
6 h	0.743	20.4	0.439	0.995
1 d	0.743	20.4	0.439	0.995

orders, Table XXV the real-flow padding-dilution replay on the shipped detector, Table XXVII the position-weighted schemes, Table XXVIII the aggregation-cap sweep, Table XXIX the insertion attack against padding, Table XXX the host-conditioned detector, Tables XXXI and XXXII the AIT organisation matrices, Table XXXIII the AIT ordering arms, Table XXXVII both attacks in operational units, and Table XXXVIII the joint zero-evidence controller-state rerun. Question answered. How much of the padding result survives changes of window, seed, order, pad pool and attacker knowledge? Takeaway. The costs move; the suppressibility does not, with one labelled exception.

A. Windows, seeds and orders

Priced against the running level at each detected episode’s step, the canonical primary window’s three detections cost 23, 24 and 33 added flows and the eleven canonical secondary-window detections have a median of six. Over fifty pre-committed orders at the primary window the median is one detection, the range is 0–10, and 19 of the fifty produce none at all; the canonical order’s three alerts lie within this distribution, whereas first-flow arrival produces eighteen and exceeds all fifty audited

TABLE XIX

An external denominator for the resolution cost: distinct red-team steps of LSPR23’s task record merged per issued alert, under both within-bucket orders (canonical key-hash and first-flow). (a) steps per alert against the label and shift nulls; (b) within-window rank agreement across bucket widths. Design and reading: Appendix D.

pos	order	bucket (s)	attrib.	steps/alert		label null		shift null		5-min atoms/alert
				bucket	flow	mean	>	mean	>	
0.55	first-flow	86400	12	18.33	18.33	18.33	–	18.33	–	38.47
0.62	key-hash	86400	1	35.00	35.00	35.00	–	35.00	–	35.00
0.62	first-flow	86400	6	41.00	41.00	41.00	–	41.00	–	32.00
0.70	key-hash	300	35	4.29	3.13	2.59	✓	2.02	–	1.00
0.70	key-hash	1800	7	12.57	12.57	11.06	–	5.83	–	3.00
0.70	key-hash	3600	2	18.50	12.00	14.81	✓	11.75	–	2.43
0.70	key-hash	86400	1	20.00	20.00	20.00	–	20.00	–	29.00
0.70	first-flow	300	32	4.31	3.18	2.63	✓	1.99	–	1.00
0.70	first-flow	1800	13	13.23	12.62	11.65	–	6.09	–	3.51
0.70	first-flow	3600	13	18.77	17.38	15.06	✓	11.95	✓	6.51
0.70	first-flow	7200	12	19.08	18.67	15.47	✓	11.45	–	13.00
0.70	first-flow	21600	12	19.08	18.67	15.28	✓	11.55	–	13.00
0.70	first-flow	86400	12	27.00	27.00	27.00	–	27.00	–	28.33
0.77	key-hash	3600	4	4.00	4.00	3.19	–	1.68	–	10.00
0.77	key-hash	86400	2	3.00	1.00	3.00	–	3.00	–	18.00
0.77	first-flow	3600	12	3.00	3.00	2.74	–	1.32	–	9.93
0.77	first-flow	7200	13	3.15	3.15	3.17	–	2.94	–	19.61
0.77	first-flow	21600	13	3.15	3.15	3.15	–	3.15	–	21.61
0.77	first-flow	86400	13	3.15	3.15	3.15	–	3.15	–	21.61

(b) within-window rank agreement across bucket widths						
pos	order	widths	steps vs. atoms	steps vs. bucket	atoms vs. bucket	
0.7	key-hash	4	+0.800	+1.000	+0.800	
0.7	first-flow	6	+1.000	+0.986	+0.986	
0.77	first-flow	4	+0.816	+0.775	+0.949	

Notes. “pos”: the window’s chronological position; “attrib.”: alerts with at least one attributed step. “bucket”/“flow”: the same attribution rule priced against the alert’s bucket span or its actual first-to-last flow span. “mean”: the null’s mean steps per alert; ✓ marks an observed count exceeding that null’s 97.5th percentile. At the 86,400s bucket both nulls tie the observation exactly, so only sub-daily widths inform. Panel (b): Spearman rank agreement across bucket widths within one window.

TABLE XX

Cross-window configuration transfer: the cap transfers, the grouping does not. Each cell is the previous-window normalised regret $\rho \in [0, 1]$ (0 = oracle, 1 = worst feasible config) on the evaluation window, seed 0; “–” is a flat selection window where transfer is undefined. The grouping axis is not flat — its oracle flow coverage varies by 0.676 across windows (cap: 0.253) — yet the frozen grouping still fails to transfer. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI).

Transfer fold (select $\rightarrow$ eval)	normalised regret ρ	
	cap axis	grouping axis
0.55 $\rightarrow$ 0.62	0.000	0.945
0.62 $\rightarrow$ 0.70	0.048	0.991
0.70 $\rightarrow$ 0.77	0.070	0.000
0.77 $\rightarrow$ 0.85	0.009	0.921
worst (previous-window sel.)	0.070	0.991
worst (leave-one-out)	0.048	0.945

metadata orders. At the secondary window the canonical order is the ensemble maximum, 11 against a median of 6. We use it because it reads only a group’s own metadata, so appending flows moves no other hypothesis: a validity-preserving canonicalisation, not a security mechanism, and not because it detects more. Conditional on an alert being issued, every evaluated alert is suppressible; the order changes which episodes are there to attack rather

TABLE XXI

e-LOND true detections under pre-committed, evidence-independent within-bucket orders, all five windows, detector seed 0. All orders emit hypotheses at bucket close and differ only in how groups closing in the same bucket are sequenced. The count is materially order-sensitive, and first-flow arrival is the largest over every order and seed we audit — which is why the paper reports the key-hash row, a deterministic hash of the group’s own (SrcIP, DstIP, bucket) key (collisions broken on the key), and carries first-flow only as a labelled upper bound. The mechanism is Theorem 1: before the first rejection only a prefix of 902 / 902 / 865 / 824 / 748 steps out of 57,368 / 49,267 / 37,231 / 31,672 / 31,568 can reject at all, so a detection needs a near-ceiling episode to land inside it. Under the canonical order at 0.70 and 0.77 no episode clears its own step anywhere in the stream, which is why those entries are 0. The key-hash is a timing-independent canonicalization, not a security mechanism: a public hash of attacker-selectable fields can be ground, and the keyed row is the grinding-resistant variant, itself an unpredictable draw from the random ensemble below. Conditioned on there being detected episodes, the per-order median padding cost is 109.5 / 16 / 93 / 175 / 98 flows — a survivorship-conditioned estimand. Because the key-hash reads no arrival time, it is also what makes the padding attack leave Assumption 1 intact for every other hypothesis: injecting one pad at the earliest instant of its own bucket moves 373 / 348 / 4 / 7 / 3 other hypotheses under first-flow arrival, against 0 under the key-hash at every window and seed. The canonical order is also the more seed-stable of the two: across detector seeds its count moves by at most 3, against 30 for first-flow arrival. Feasibility (C1) is order-invariant. See Appendix D-A.

within-bucket order	0.55	0.62	0.70	0.77	0.85
first-flow arrival (upper bound)	18	13	30	31	72
group-id (packed key)	0	3	0	0	24
last-flow arrival	0	0	0	0	0
fixed random permutation (one instance)	2	1	9	1	38
key-hash (canonical, reported)	3	11	0	0	34
keyed key-hash (secret seed)	0	6	3	3	24
median over 50 orders	1	6	2.5	1	23.5
range over 50 orders	0–10	0–11	0–12	0–6	0–44

than whether they can be. At the stress window, whose evidence is not a valid e-value and which is therefore a cost measurement only, the canonical median is 116 over 34 detections. The first-flow order detects more episodes and detects them deeper in the level sequence, so its costs are larger (118, 72 and 5,202 at the same three windows), but every target remains suppressible; that order-specific matrix is Table XXIV. Across 50 pre-committed orders and both detector seeds the per-order median stays at 16–178 flows, conditioned on the order having produced a detection at all — some orders produce none, so this is a survivorship-conditioned estimand.

B. Sizing without the oracle

Multiplying the attacker’s own episode by c defeats an alert whenever $c \geq 1 + r_t^*/m_t$. That overshoot factor has median 1.57 at the canonical primary window and maximum 6.17 at the stress window, and it is what the attacker must cover — never the threshold itself. Tuning c per window is cheaper than the fixed $c = 10$ the body reports ($c = 3$ suppresses all three primary alerts at a median of 80 added flows, 3.5 $\times$), but that choice reads the realised costs and is reported as an oracle-informed floor, not an attacker rule; so is the flat pad, where 33

TABLE XXII

What a secret seed for the canonical key-hash order costs the insertion attacker (e-LOND, seed 0): the public-hash cost G^* against the keyed-hash budgets $N_{50/90/99}$ over 400 draws, and their ratio (N_{99} against the certain public suppression). Design and reading: Appendix D-A.

pos	order	targets	u	public	keyed: instantiated keys			blind
				G^*	N_{50}	N_{90}	N_{99}	N_{99}/G^*
0.55	public	3	0.037–0.072	217–217	5,663	6,639	8,962	41.3 $\times$
0.55	keyed	0	—	—	—	—	—	—
0.62	public	11	0.016–0.652	70–813	825	962	996	4.6 $\times$
0.62	keyed	6	0.060–0.413	288–506	4,994	5,843	6,053	12.3 $\times$
0.70	public	0	—	—	—	—	—	—
0.70	keyed	3	0.032–0.059	257–257	8,238	9,620	9,961	38.8 $\times$
0.77	public	0	—	—	—	—	—	—
0.77	keyed	3	0.016–0.037	416–416	26,101	30,432	31,502	75.7 $\times$
0.85	public	34	0.019–0.489	30–509	8,533	11,118	12,625	24.8 $\times$
0.85	keyed	24	0.016–0.385	78–518	8,262	9,640	9,981	19.3 $\times$

Notes. “pos”: the window’s chronological position; “targets”: detected episodes under that order; “ u ”: the target’s own hash quantile, unknown to a blind adversary; “ G^* ”: hypotheses an adversary who knows the ranking must instantiate ahead of the target on that same stream; “ N_p ”: distinct (SrcIP, DstIP) pairs a blind adversary must send traffic between to suppress the target with probability p , located on a six-points-per-decade grid and so exact to within 1.48 $\times$ and no further; the N_p cells are medians over the row’s targets; last column: N_{99} against the certain public G^* , the nearest matched-reliability comparison the grid offers (99% against 100%), a reliability budget rather than a lower bound. Rows with no detection carry no cost claim.

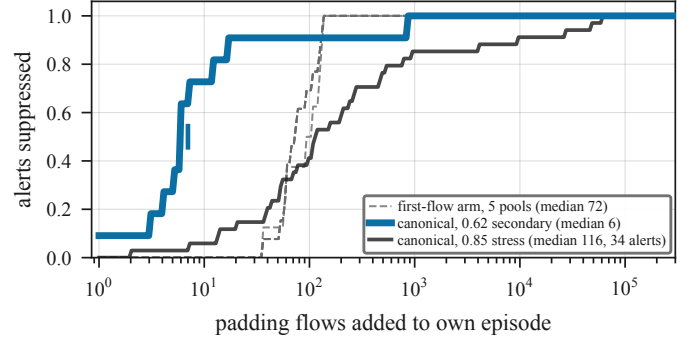


Fig. 5. The sensitivity arms the body’s Fig. 4 omits: the secondary window under the canonical order against the known-invalid stress window (canonical) and the five pad pools under first-flow arrival. Ticks are the five pools’ canonical medians at the secondary window, which span 6–7 flows — the pad-realism claim of this subsection. Costs move with window, order and pool; suppressibility does not.

flows suffice at the primary window and 17 for 90% of the secondary window’s eleven. The one exception is the first-flow stress arm, where overshoot reaches 4,302 and no multiplier we test suffices (Table XXVI). The fixed $c = 10$ the body reports costs a median 15.7 $\times$ the oracle at the primary window and 5.1 $\times$ at the stress window, so dropping to state-free sizing costs a factor of about ten, not the 10^4 – 10^6 span of $1/\alpha_t$ within a window. The resulting pad is large but not anomalous: the padded primary-window episode, arity 120, is exceeded by 0.9–9.9% of deployment episodes.

C. Pad realism and the pool construction

The realism of the pad barely matters. Generic benign, attacker-origin, protocol-matched and black-box pools

have nearly identical median costs, because the running threshold dwarfs the largest pool mean e-value (216.8); service-matched padding is dearer — $1.2\text{--}1.5\times$ under the canonical order, $1.4\text{--}2.0\times$ under first-flow, and on a smaller episode subset — and does not change the conclusion (Table XXIII). Under the headline key the adversary sends ordinary traffic between the source and destination already involved in its activity before the two-hour bucket closes. The flow-only detector reads no endpoint identity, but 17 of 33 features are backward or bidirectional, so whether a flow the victim answers transfers is empirical; the real-flow replay measures it rather than assuming it. Beyond the 3/3 and 11/11 canonical suppressions at the primary and secondary windows, the replay gives 34/34 canonical at the stress window, and 18/18, 13/13 and 72/72 under first-flow order; the measured cost equals the closed form on every episode of every arm (Table XXV).

The black-box pool and the dilution replay. The pad flows of Table XXV are real flows of the black-box pool, selected on network-observable frequency alone: the most common (proto, port) over the training prefix, which precedes both calibration and deployment — no labels, no detector output, no knowledge of the attacker’s own traffic. It picks proto 17/port 53 at both windows, the same service the label-using rule would, and the resulting pool is measured to be 0.0000% attack-labelled (an audit, not a selection criterion). The selection rule matters: taking the mode over all window traffic instead picks the attacker’s own flood at the stress window, a pool 90% attack-labelled that does not dilute. Zero pad fires out of the sampled flows at both windows is evidence for the structural premise (ordinary victim-service traffic scores like ordinary traffic), not proof that the firing probability is zero; the Clopper–Pearson bound in the table says how much that observation leaves open. Both within-bucket orders are attacked. The key-hash rows are the canonical order the paper reports, under which a pad provably moves no other hypothesis’s spending weight, so the Assumption 1 argument is clean; the first-flow rows re-run the identical chain — same detector, same pool, same pricing — on the detected set the initial first-flow order produces, which Table XXI shows is the largest over the orders we audit. Every detection is suppressible under both, and the canonical medians reproduce Table III. The canonical primary-window set is small, so read it as three individually-priced episodes (costs 23, 24, 33 flows) rather than as a rate; the canonical costs are lower than the first-flow ones because the two orders detect different episodes at different steps of the level sequence, not because the canonical order is weaker.

Residual risk at the horizon-aware arm. The pool’s zero firing count over 845,691 flows is a fact about the pool, and it leaves open a risk that replay cannot bound — so we simulate pads whose flows fire independently at the one-sided 95% Clopper–Pearson upper limit the observation admits, $p \leq 3.5 \times 10^{-6}$. Even there the

worst alert suppresses on an estimated 90.5% of draws at the primary window and 82.5% at the secondary (Section V-E). Both figures are Monte Carlo estimates over 200 draws and both treat pool flows as independent draws from a common firing rate, which the pool’s composition does not establish.

D. The AIT transfer arms

Host-conditioned detector on LSPR23. The detector of Table XXX reads the 33 flow features plus 6 strictly causal ($ts < ts(\text{flow})$) host-context features — per-host prior counts, distinct-peer counts and failed-connection fractions — and no endpoint ground truth (label_src and label_dst both have attack rate 1.0 and are excluded). It has higher AUROC than the flow-only detector at both windows, yet lower tail reach and fewer e-LOND detections (2 against 18 at the primary window): higher ranking quality buying fewer, not more, operational alerts, the detector-quality-versus-feasibility gap of Section III-C. To test the padding boundary we graft each attacked pair’s real causal context (strongest feature src-distinct-dst, $\rho = +0.34$; padding one victim cannot raise it) onto ordinary black-box pads — the most common (proto, port) on the training prefix, which precedes both calibration and deployment, so the pool is chosen on network-observable frequency alone with no labels and no detector output — and replay the append causally. The pads fire zero across the scored accumulation range at both windows — up to 100,000 appended flows per episode, 4,000,000 pad scorings at the primary window — and the suppression cost is the identical flow-level closed form ($0.55 r_t^* = 108$ host vs 118 flow; $0.85 r_t^* = 11,484$ host vs 5,202 flow), so within that range the transfer holds at both; costs above the cap are closed-form values, not replayed. A non-black-box pool rule does fire: selecting the modal service among detector-non-firing flows picks proto 6/port 80 at 0.85, a pool that is only 12% benign and fires 0.42. Grafting the attacker’s own flood onto attack context and observing the detector fire is not evidence of a defence, so that rule is reported only as a sensitivity. What LSPR23 cannot settle is the benign-inclusive question: its attacked pairs are 100% malicious, so no real ordinary-to-victim traffic exists there, which is what the AIT testbed supplies.

Resequencing the flow-only folds by first-flow arrival — the detector fit, scores, calibration and e-values are computed before ordering and are identical across the two arms — gives 84 of 85, and the canonical alerts are the cheaper to suppress: per-organisation median $r_{t,\text{emp}}^*$ falls from 212–13,142 flows under first-flow to 12–4,650 under the canonical order (Table XXXIII). On shaw, ordinary victim traffic fires often in its own context (7%), which under a static-context replay made no episode suppressible; scored in the attacked pair’s evolving context the same pool fires at most 0.5%, and all three targets suppress (Tables XXXI and XXXII). That contrast is

why the host-conditioned arm recomputes context causally rather than scoring the pool in isolation.

Design of the AIT replay. Per organisation, Table XXXII trains a leave-one-organisation-out detector whose flow features are chosen per fold from the training organisations only, calibrates on in-organisation benign flows strictly preceding the first attack, forms (SrcIP, DstIP, 2h) episodes, runs e-LOND, and suppresses each detected malicious episode by replaying real benign flows to an attacked victim of the organisation (200 draws). A firing pad contributes the maximum e-value and raises the cost, which the zero-pad closed form misses. The two arms replay differently because the two detectors read different things. Flow-only: flow features carry no context, so pads are drawn with replacement from the pool’s observed 0-or- M evidence, an empirical i.i.d. generator, so a modelled cost may exceed the finite observed pool. Host-conditioned: the six host features are causal, so appending pads changes the context every later pad is scored in, and holding them fixed would be a static-context diagnostic rather than a replay; we therefore recompute the victim’s context as pads accumulate (counts raised, distinct-peer counts pinned, failure fractions moved to the pool’s rate), score the whole pool under it, and report a success only over the accumulation range actually scored. The chain runs end-to-end at all eight organisations, in both arms: under first-flow order 84 of 85 flow-only detections are suppressible, per-draw success 0.79–1.00, and 57 of 63 host-conditioned detections are, per-draw success 0.61–1.00. Host conditioning therefore blunts the attack without stopping it. The causal replay is what shows both halves. It is why most episodes still suppress: shaw’s ordinary-to-victim flows fire 7% in their own contexts, which under a static-context replay made no episode suppressible, but a pad’s context is the attacked pair’s, and under that context the same pool fires at most 0.5% across accumulation, so all three episodes suppress. It is also why the remaining 6 do not (harrison, wardbeck, wheeler): under the accumulated context a pad can itself fire, and a firing pad adds the ceiling to the group’s evidence instead of diluting it, so on those episodes no draw brings the group mean below threshold in any of the 200 draws, within twice the zero-pad cost (at least 1,000 pads), which is the range the context curve is scored over. What decides the outcome is the pad count an episode needs against the rate at which pads fire along that accumulation path. These are censored simulations rather than proof that the episodes cannot be padded: a budget or a draw count large enough might still find a route. The pinning models a single already-active attacker–victim pair; padding from new hosts would move the distinct-peer counts and is not covered.

The joint attack on AIT. Table XXXIV re-runs e-LOND over the padded stream for each organisation, so the controller’s rejection count responds to the attack and, on the host-conditioned arm, every later flow on the

attacked pair is re-scored as the pads arrive. Each figure is the median over 100 pad-sampling trajectories; the totals below sum those medians over the eight organisations and are not counts from a single run. Two template pools are separated throughout: the deployment pool is the benign-to-victim traffic the per-alert arms of Table XXXII use, and the prior pool is restricted to flows the victim had received before the attacker’s own last flow in the episode. An episode whose prior pool holds fewer than 20 flows is never padded.

With the deployment pool the campaign costs 244,250 and 91,713 flows against per-alert sums of 607,802 and 599,967, leaving 1 and 0 of the 79 and 80 own alerts. With the prior pool 24 and 27 remain and no organisation is cleared, at 430,626 and 242,206 flows: 30 of the 95 attacker-owned episodes have too little prior traffic to imitate and 18 have none at all, and those episodes carry most of what survives. On the host-conditioned arm, shaw issues 3 and 3 own alerts and 1 and 6 non-attacker alerts; the deployment pool clears all of them for 28 and 7 flows, the victim’s prior traffic leaves 2 and 2; wilson issues 6 and 6 own alerts and 0 and 0 non-attacker alerts; the deployment pool clears all of them for 73,337 and 16,975 flows, the victim’s prior traffic leaves 1 and 1.

The state-free multiplier $c = \lfloor \rho \rfloor + 1$ of Corollary 2 leaves 65 and 60 alerts standing with the prior pool. Sizing it on the episode’s full arity rather than the attacker’s own flow count changes nothing there (65 and 60), and changes the outcome only with the deployment pool (57 and 1 against 59 and 6): what the attacker does not know about co-resident benign flows costs it nothing while cover traffic is the binding constraint.

Reliability over trajectories. A summed median can hide a bimodal outcome (a median of zero survivors is compatible with many trajectories having some), so Table XXXV reports, per organisation, spending rule and attacker arm, the probability that every attacker-owned alert is silenced and the 95th percentile of the residual count and of the cost alongside the median. The residual count is essentially not a sampling quantity here: in 58 of the 60 cells its 5th and 95th percentiles coincide, and in the remaining 2 they differ by one alert. With the deployment pool the sequential oracle silences every own alert in every trajectory in 15 of the 16 flow-only cells; the exception is russellmitchell under horizon-free spending, where exactly 1 alert survives in every trajectory ($P = 0.00$). What pad sampling does move is the cost: the 95th percentile exceeds the median by up to a factor of 1.90 (median ratio 1.00 across cells). The variation that matters is therefore across organisations, not across pad draws, and the sums of medians in Table V should be read as eight nearly deterministic per-organisation outcomes added together, not as one typical campaign.

Testing the premise of Corollary 2 rather than a proxy. The corollary needs one thing of the rest of the stream: no non-attacker hypothesis carries evidence at or above the

cold-start threshold T/α . A baseline non-attacker alert does not by itself show that this fails, because under either spending rule an alert can fire only after earlier rejections have raised the level. Table XXXVI therefore measures the premise directly on the 10 streams the joint attack ran on, rebuilt with the same chain and asserted to reproduce every stored baseline. The cold-start premise fails on 3 of them (russellmitchell at 1.50 with 1 episode at or above it; shaw at 2.29 with 2 episodes at or above it; shaw (host) at 2.29 with 4 episodes at or above it); on the others the largest non-attacker evidence is below T/α , and in 5 streams no non-attacker episode carries any evidence at all. Of the 16 baseline non-attacker alerts over both spending rules, 9 clear the $R = 0$ level at their own position and 7 fire only at a raised level. An attacker episode attains the ceiling in every cell but shaw, shaw (host), where the largest attacker evidence gives $c_{\text{crit}} = 1.53 < \rho = 2.29$ in each, so the stream-specific integer $\lfloor c_{\text{crit}} \rfloor + 1$ can be smaller than c_{int} there. The other three conditions the corollary rests on are read from the joint run itself: own episodes without a usable prior pool are never padded and account for most of what survives; admitted pads from the prior pool are not zero-evidence, and where they fire the multiplier’s dilution is partly undone; and the residual counts are the observed outcome. The corollary is stated for horizon-uniform spending, so its multiplier under horizon-free spending is a heuristic transfer, reported as such.

E. Aggregation alternatives and the capacity boundary

Caps large enough to preserve the intended validity domain yield zero detections at the median window–seed cell (first-flow, the detection-favourable order; Table XXVIII). In the primary-window weight sweep every scheme that retains the mean’s recall is defeated by one to eight leading flows, against 111 appended flows for the mean on the same episodes, and buying a larger L^* costs about 58% of the recall (Table XXVII). For the group maximum, direct group calibration replaces 1.8–2.4M calibration flows by 37–57k groups while leaving T essentially unchanged; by Corollary 1 that is 20 groups per hypothesis the construction cannot supply, so the margin collapses to about -0.97 and the primary and secondary windows issue no alerts (Appendix F-B). Appending is not the only lever against it either: creating hypotheses ahead of a target pushes it past the cold-start window, and in that construction’s short prefix every group-maximum detection falls to 62–295 inserted hypotheses (Table XXIX).

Static versus running pricing. Pricing against the static first-step threshold T/w_0 is the cheap extreme (median 34 flows at 0.85 under the canonical order); the body reports the larger running-level cost. Whole-detected-set totals are upper bounds, since suppressing an early rejection lowers R and can make the rest cheaper or moot, and are heavy-tail driven at their extreme.

Insertion against padding. Insertion beats padding for one alert at none of the twelve window–order cells, but silences the entire controller for 217–813 inserted hypotheses against $80\text{--}7.9 \times 10^8$ flows to pad every detection separately; per episode the targeted insertion cost is 434–5,674 hypotheses under the canonical and first-flow orders (down to 407 under a keyed hash). Under a public hash the ranking is an offline search we price at $1.0 \times 10^4\text{--}1.8 \times 10^5$ candidate keys, on three stated conditions (a public seed, enough endpoint pairs the adversary can send between, and a stable key encoding).

F. The joint controller-state rerun

The per-alert costs of Section V-D price each alert at the level it received on the unperturbed trajectory. Table XXXVIII instead propagates the zero-evidence pads that the real-flow replay of Section V-E validates through the controller, which is re-run over the padded stream, so that its rejection count — and with it every later level $\alpha_t = \alpha\gamma_t(R_{t-1}+1)$ — responds to the attack. The black-box pool contains no flow that reaches the conformal tail at either window, so this is the zero-evidence limit of the real-flow replay, not a second replay. Three attackers are run. Static: the per-alert oracle pads of Table II, applied to every true detection at once. Sequential: a greedy oracle that pads each of its own episodes that would fire at the live level by the minimum; its total J_{seq} is one strategy’s cost on this stream, an upper bound on the minimum joint cost over all attacker-controlled episodes in the window. State-free: every own episode is multiplied by the integer $c_{\text{int}} = \lfloor \rho \rfloor + 1$ of Corollary 2, which is 3 at the primary and 3 at the secondary window, and silences the arm with no controller state. With $R = 0$ throughout, the stream-specific boundary is $c_{\text{crit}} = \max_j \text{Ev}_j \alpha\gamma_{t_j} \leq M\alpha/T = \rho$, with equality when a malicious episode attains the ceiling, as one does at each window ($c_{\text{crit}} = \rho = 2.134$ and 2.485); rejection is inclusive, so the boundary must be strictly exceeded, the real-valued checks just above and below c_{crit} are a continuous relaxation, and only the integer c_{int} is a realisable prescription. The lower panel re-runs the sequential attacker under a per-host-pair volume cap n : it pads only if $m_t + r \leq n$ and otherwise lets the alert fire and raise R , and each surviving alert is classified as structural or as a cascade victim by whether its pad would have fitted at $R = 0$. Every padded stream was regrouped from flows and asserted against the unperturbed stream on what an append must leave unchanged — the episode count, the (source, destination, bucket) key of every controller position and hence the permutation, each group’s first timestamp and its label — and asserted to carry arities $m_t + r_t$ and exactly the padded group means $S_t/(m_t + r_t)$ that the attacker was priced on; the padded means differ from the unperturbed ones by construction.

Capacity interpretation. The lower panel constrains the attacker’s additions; it does not apply an admission policy

TABLE XXIII

Five padding pools priced against the running e-LOND level $1/\alpha_t$, detector seed 0, two-hour grouping, at the secondary window (0.62) and the stress-test window (0.85), under both within-bucket orders — the canonical metadata hash Table III reports and the first-flow arrival order kept as the optimistic upper bound. Entries are the median suppression cost r per detected episode, over the detected set that order produces (counts in the block headings); the order changes which episodes must be paid for and how deep into the level sequence they sit, which is why the two blocks differ by more than a constant. Four pools coincide within each block; service-matched is dearer, both from a larger mean e-value and from being defined on a costlier episode subset (Section V-C). μ is the pool-level mean e-value at 0.85 where defined, a property of the pool and not of the order. Protocol- and service-matched use the episode’s modal protocol/port. Costs are lower bounds.

Padding pool	μ (0.85)	median r (0.62)	median r (0.85)
canonical metadata-hash order (headline)	—	11 detected at 0.62, 34 at 0.85	
generic benign	50.9	6	116
attacker-origin	216.8	6	116
protocol-matched	—	6	116
service-matched	—	7	175
black-box	0.0	6	116
first-flow arrival (optimistic upper bound)	—	13 detected at 0.62, 72 at 0.85	
generic benign	50.9	72	5,212
attacker-origin	216.8	72	5,247
protocol-matched	—	72	5,232
service-matched	—	99	10,299
black-box	0.0	72	5,202

to the complete stream, and baseline groups are not truncated or rescored.

Appendix F Validity and Data Quality

Table XXXIX records what each result depends on; Appendix F-A audits the calibration tail against the labels, Appendix F-B calibrates the grouped unit directly, and Appendix F-C prices calibration-set contamination.

A. Calibration validity and the labels

The measured-to-nominal benign firing ratio stays near one across the ten position $\times$ seed configurations (median $1.94\times$; independently $0.86\text{--}2.03\times$ on the AIT wilson testbed), with a single exception, the stress window 0.85, where the benign firing rate runs $50.9\times$ nominal (exact interval $[37.3, 67.9]$), so $\mathbb{E}[e] \approx 51$ and the evidence is not a valid e-value there (Table XL). The excess is sharply localised: three host pairs carry 44 of the 46 firing benign flows, and removing them restores the ratio to $2.22\times$. It is best explained by label error, which distribution shift, a new benign mode, duplicated records and Mondrian conformal each fail to account for, though we do not prove it. The other four windows are not refuted ($1.07\text{--}3.79\times$, every interval containing 1, and at 0.55 the interval $[0.03, 5.96]$ excludes essentially nothing); that is a failure to reject and not support.

The marginal rate is the weakest possible evidence for Assumption 1, so we interrogate it conditionally, stratifying the firing benign flows on the observable components of $\mathcal{M}_j$ (arity, service, bucket, destination host) at rank depths $k \in \{1, \dots, 1000\}$, resampling both

TABLE XXIV

Padding suppression cost priced against the running controller level $1/\alpha_t$ (real) vs the static T/w_0 threshold. Part (a) gives both within-bucket orders at both detector seeds: the canonical metadata-hash order the paper reports, and the first-flow arrival order retained as an optimistic upper bound over the orders we audit (Table XXI). Detections and the level an alert fires at both move with the order; T , $|C|$, AUROC and the margin do not. Part (b)’s pool comparison is on the first-flow detected set, the larger of the two; Table XXIII gives the same comparison under both orders. Four pools coincide; service-matched is dearer. Medians are rounded to the nearest integer; “—” marks a window/seed/order with no detections.

(a) per window $\times$ seed $\times$ order (e-LOND poly, two-hour grouping)								
pos	seed	canonical key-hash			first-flow (upper bd.)			
		det.	pad (real)	pad (static)	det.	pad (real)	pad (static)	
0.55	0	3	24	3	18	118	4	
0.55	1	2	27	2	18	101	3	
0.62	0	11	6	4	13	72	4	
0.62	1	8	5	2	12	69	0	
0.70	0	0	—	—	30	1,851	62	
0.70	1	0	—	—	0	—	—	
0.77	0	0	—	—	31	1,165	85	
0.77	1	0	—	—	31	1,289	90	
0.85	0	34	116	34	72	5,202	33	
0.85	1	32	109	34	71	6,802	33	

(b) five padding pools, seed 0 (median suppression cost r)			
padding pool	real (0.62)	real (0.85)	static (0.85)
generic benign	72	5,212	33
attacker-origin	72	5,247	33
protocol-matched	72	5,232	33
service-matched	99	10,299	35
black-box	72	5,202	33

TABLE XXV

Controlled padding-dilution replay on the shipped detector: real flows of the black-box pool appended to every detected episode under both within-bucket orders (canonical key-hash and first-flow); src-dst two-hour unit, e-LOND $\gamma \propto j^{-1.6}$, seed 0. Design and reading: Appendix E-C.

window	within-bucket order	pool flows	pad fires	95% upper	detected	measured = c.f.	median r_t^*
primary (0.55)	first-flow (upper bd.)	845,691	0/20,000	1.5×10^{-4}	18	18/18	118
	key-hash (canonical)	845,691	0/20,000	1.5×10^{-4}	3	3/3	24
stress-test (0.85)	first-flow (upper bd.)	584,850	0/20,000	1.5×10^{-4}	72	72/72	5,202
	key-hash (canonical)	584,850	0/20,000	1.5×10^{-4}	34	34/34	116

Notes. “pool flows”: the black-box pool’s size in the window; “pad fires”: how many of n sampled pad flows fire when run through the detector, with the exact one-sided 95% Clopper–Pearson upper bound on the firing probability in the next column; “measured = c.f.”: detected episodes whose padded group e-value $S_t/(m_t + r)$ crosses the firing threshold at exactly the predicted $r_t^* = \lfloor S_t \alpha_t \rfloor - m_t + 1$; “median r_t^* ”: over the detected episodes of the row, priced at the running level.

the shared calibration tail and the test side (Table XLI). At the depth the pipeline uses the data mostly cannot speak: 129 of 134 strata carry no test at $k = 1$ (428 of 536 cells across the four depths), and none that are estimable reject at the primary or secondary window. Deeper in the tail, 32 contrasts survive correction at all five windows (25 after removing the one confounder we can identify), the strongest an arity bin at 0.77. A departure at $k = 1000$ is compatible with a valid rank-1 rate, so the guarantee-bearing claims stay stated under Assumption 1, not supported by this diagnostic. The Bates calibration-conditional adjustment (Table XLIV) is stronger than e-

TABLE XXVI

Surface A priced without oracle state: a re-analysis of the same per-episode (m, S, α_t) the oracle costs come from, with no re-simulation — the closed form (8) is re-derived from the stored fields on all 127 episodes as a read-back check. Multiplier: append $(c-1)m$ flows, that is, multiply your own episode by c . It defeats the alert iff $c \geq 1 + r_t^*/m_t$ and needs no estimate of S_t , α_t or $|C|$ — only m , which on LSPR23 is the attacker’s own flow count, since every host pair carrying attack traffic is 100% malicious. “padded arity” is cm , and the column beside it answers the deployment-side question — would a volume monitor see the pad — as the fraction of deployment episodes larger than the padded one, bracketed by the two stored group-size quantiles that straddle it. Those quantiles are not shown and are not benign deployment values: they are quantiles of source-destination group size over the calibration slice, unfiltered by label, which is why the deployment exceedance and not the quantile is what the claim rests on. Rows are given under both within-bucket orders — the canonical metadata hash the paper reports and first-flow arrival, retained as the optimistic upper bound. Absolute: append a pre-committed flat N flows, which needs nothing at all; N is reported at the fraction of that cell’s alerts it suppresses. The multiplier arm covers the two windows for which per-episode arities are stored; the secondary window appears in the absolute arm only. Both strategies are upper bounds on the oracle cost, and both are optimistic on a benign-inclusive stream, where m also contains benign flows the attacker cannot see.

multiplier — needs the attacker's own arity m and nothing else										
order	pos	alerts	med. c	max c	c used	med. added	vs oracle	padded arity	% deployment larger	
canonical	0.55	3	1.57	2.74	3	80	3.5×	120	0.9–9.9%	
canonical	0.85	34	2.76	6.17	10	792	5.1×	880	0.1–1.1%	
first-flow	0.55	18	3.54	7.44	10	414	3.6×	460	0.1–0.9%	
first-flow	0.85	72	73.19	4302.15	none ≤ 100					

absolute — a flat pad: executing it needs no state, but the N below is read off the realised r_t^* distribution										
order	pos	seed	alerts	N for 50%	N for 90%	N for all				
canonical	0.55	0	3	24	33	33				
canonical	0.55	1	2	30	30	30				
canonical	0.62	0	11	6	17	859				
canonical	0.62	1	8	6	659	659				
canonical	0.85	0	34	118	9,395	58,616				
canonical	0.85	1	32	111	9,426	58,620				
first-flow	0.55	0	18	125	645	837				
first-flow	0.55	1	18	108	645	829				
first-flow	0.62	0	13	72	127	134				
first-flow	0.62	1	12	70	93	98				
first-flow	0.77	0	31	1,165	3,736	7,886				
first-flow	0.77	1	31	1,289	4,258	7,886				
first-flow	0.70	0	30	1,877	4,145	5,445				
first-flow	0.85	0	72	6,714	69,195,773	145,525,527				
first-flow	0.85	1	71	6,802	69,086,348	145,096,310				

TABLE XXVII

Pre-committed weight schemes at the primary window, LOND, $\gamma \propto j^{-1.6}$, mean over two detector seeds. $T = 57,368$ episodes, 275 malicious, $|C| = 2,448,993$. L^* is quoted at the unperturbed level and is therefore an upper bound. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); both legs of every comparison use it, so the order moves the level and not the ranking.

Rule	pad-inv.	recall	L^*	appended	ratio
arithmetic mean (symmetric)	—	0.065	—	—	—
first-event-only	yes	0.062	1	111	112×
exp-decay $\rho = 0.5$	yes	0.062	2	111	56×
uniform over 10 slots	yes	0.062	8	111	14×
uniform over 100 slots	yes	0.027	81	492	6×
exp-decay $\rho = 0.99$	yes	0.027	164	492	3×
last-event-only	no	0.045	—	336	—
security-prior (out of class)	yes	0.000	—	—	—

LOND needs and is treated as optional.

Stratified design. Assumption 1 constrains benign flows of true-null groups conditionally on that group’s metadata, whereas the marginal diagnostic (Table XL) reads the marginal rate at the shipped depth $k=1$, where the four non-stress windows fire 0–3 benign flows. Table XLI

TABLE XXVIII

Aggregation-cap sweep (median over five positions $\times$ two seeds): the frozen $n_0 = p_{99}$ is chosen for validity and attack cost, not power. “% groups $m > n_0$ ” is the share of groups whose size exceeds the cap; “front-load cost” is the number of appended flows needed to defeat the front-loading defence. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

cap	n_0	% groups $m > n_0$	det. raw	det. trunc.	front-load cost
mean	47	5.72	39.5	35	16
p50	2	37.58	99	71	1
p90	21	10.24	68.5	60	8
p99	408	1.08	0	0	116
p999	7,258	0.12	0	0	3,560
max	97,911	0.01	0	0	—

stratifies those flows on the observable components of $\mathcal{M}_j$ — episode arity (the quantity padding manipulates), service (proto,port), the two-hour bucket that fixes a group’s position, and high-support destination host, all fixed by an outcome-blind support rule — and reads them at $k \in \{1, 10, 100, 1000\}$. Both sources of uncertainty are resampled jointly, and both are clustered: a window’s whole diagnostic depends on the calibration set only through its k -th largest score, which every stratum shares, so the calibration tail is resampled over calibration host pairs; the test side is a Poisson-multiplier bootstrap over deployment host pairs. Multiplicity is corrected with Benjamini–Yekutieli, valid under the arbitrary dependence this design has, over every cell: a cell firing fewer than ten times carries no test, but still enters the family at $p = 1$, since dropping it would shrink the denominator using the same tail count that drives significance. A contrast is a stratum departing from its own window’s marginal on the same replicates — the quantity that speaks to the conditional statement, since the shared calibration draw largely cancels in a ratio of two rates read at one threshold.

Reading, in both directions. 428 of the 536 cells carry no test at all (419 of them fire fewer than ten times), which is itself the point: at the depth the pipeline uses, the data mostly cannot speak. Of the 40 cells that do reject, none at $k=1$ rejects at the primary or secondary window — all 5 $k=1$ rejections sit at 0.85, already an unambiguous violation, which is an inconclusive non-rejection and not evidence for Assumption 1: at $k=1$ most cells are simply uninformative. Deeper in the tail the conditional statement is another matter: 32 stratum contrasts survive correction, at windows 0.55, 0.62, 0.70, 0.77, 0.85. The largest lower bound is $m=21-100$ at 0.77, $k=1000$ ($9.3 \times [6.5, 13.7]$); the largest point estimate is proto 6/port 6444 at 0.62, $k=10$ ($14.4 \times$). The arity family — the component of $\mathcal{M}_j$ the padding attack moves — carries 3 of them, the strongest being $m=21-100$ at window 0.77, $k=1000$, firing $9.3 \times [6.5, 13.7]$ its own window’s rate. One dependence remains unresampled: host-pair clustering does not absorb a shock shared across many

TABLE XXIX

Suppression by insertion, not by padding. Theorem 3 and every cost in Section V-C concern appending flows to a fixed hypothesis. This attack appends nothing: it creates hypotheses ahead of the target, pushing it out of the cold-start window past which no rejection is possible. Two models are kept apart because they are different attacks. Targeted inserts only immediately ahead of one episode, which leaves every earlier hypothesis and hence R untouched, so its cost is closed-form

$G^* = \lfloor (\alpha(R+1)E/\zeta(1.6))^{1/1.6} \rfloor - p$ (replay-verified). Window inserts ahead of everything and silences the controller outright — that is not a per-episode cost, since it also deletes the earlier detections that raised R . The split verdict is the result: insertion does not beat padding for one alert (0/12 cells), but silences the whole window for a few hundred hypotheses (10/12 cells at one flow per inserted group, 7/12 at two). Padding totals are upper bounds — the joint cost is lower, because suppressing an early rejection lowers R — and the largest ratio is heavy-tail driven. Panel (b) is the construction of Appendix F-B, which padding provably cannot suppress: calibrating on groups shrinks the ceiling, so its cold-start window collapses to 65–86 and every detection falls to a few dozen insertions. Group validity by design plus padding robustness does not buy attack resistance. Getting ahead of a target is free under first-flow arrival (send earlier in the bucket), costs an offline key-space search under a public hash, and cannot be ranked offline at all under a keyed one — which removes the search, not the attack. Both within-bucket orders are reported: the canonical metadata key-hash Table III uses, and first-flow arrival, the optimistic upper bound over the orders we audit (Table XXI).

(a) the shipped flow-level pipeline							
pos	order	det.	per episode		silence the window		cheaper
			insert	pad (flows)	insert	pad (upper)	
0.55	key-hash	3	434	24.0	217	80	–
0.55	first-flow	18	1,696	118.0	502	4,432	✓
0.62	key-hash	11	605	6.0	813	926	✓
0.62	keyed	6	452	14.5	506	126	–
0.62	first-flow	13	2,284	72.0	554	1,060	✓
0.70	keyed	3	407	97.0	257	310	✓
0.70	first-flow	30	4,088	1,850.5	706	62,250	✓
0.77	keyed	3	556	323.0	416	1,205	✓
0.77	first-flow	31	3,161	1,165.0	789	49,957	✓
0.85	key-hash	34	1,414	115.5	509	151,390	✓
0.85	keyed	24	998	160.5	518	162,496	✓
0.85	first-flow	72	5,674	5,201.5	743	786,257,264	✓

(b) the group-calibrated max pipeline — padding cannot suppress it at all						
pos	groups	firing	detections	cold start	targeted G^*	window G
0.55	57,368	110	0	81	–	–
0.62	49,267	109	0	83	–	–
0.70	37,231	69	1	86	81.0	81
0.77	31,672	60	2	78	76.5	68
0.85	31,568	152	42	65	207.0	62

pairs at once, and a window’s calibration tail concentrates in its two to four buckets, so these counts may be optimistic — dropping the one stratum family aligned with that confounder leaves 25 contrasts at windows 0.55, 0.62, 0.70, 0.77, 0.85. A departure at $k=1000$ is nonetheless compatible with a valid rank-1 rate — the excess may sit entirely in ranks 2..1000 — so this tests a necessary implication where it is estimable, not the assumption itself. Every guarantee-bearing conclusion in this paper is therefore stated under Assumption 1, and this diagnostic is not offered as support for it.

At 0.85 every FDP is measured against labels. Auditing the ADDIS ($\gamma \propto j^{-1.6}$) set’s 152 alerts against the red-team task record, 146 agree (96.1%) and six disagree, so an under-estimate is not excluded; e-LOND’s realised FDP is

TABLE XXX

Host-conditioned detector and the padding-transfer boundary on LSPR23 (src-dst two-hour unit, e-LOND $\gamma \propto j^{-1.6}$, seed 0; first-flow within-bucket order on both legs, the optimistic upper bound over the orders we audit, not the canonical order Table III reports). “h/f”: host-conditioned / flow-only detector. Design and reading: Appendix E-D.

window	AUROC h/f	tail reach h/f	det. h/f	real benign fires	graft fire	median cost h/f	transfer
primary (0.55)	0.954/0.916	0.009/0.033	2/18	0/2,287,988	0.000	108/118	holds
stress-test (0.85)	1.000/0.999	0.295/0.610	37/72	3/1,638,722	0.000	11,484/5,202	holds

Notes. “AUROC”: area under the ROC curve on the window’s flows; “tail reach”: the fraction of attack flows scoring above every benign calibration score; “det.”: e-LOND detections; “real benign fires”: host-detector firings among the window’s benign flows; “graft fire”: the grafted-pad firing rate on detected episodes across accumulation; “median cost h/f”: the median suppression cost over detected episodes under the causally recomputed host context / the flow-level closed form r_t^* ; “transfer”: whether the flow-level closed form still prices the suppression. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

0.0066 (frozen) or $[0, 0.0265]$ (strict post-hoc). The audit is partly circular (99 of 152 verdicts rest on the label-generating identities), so it indicates the error’s direction, not a value (Table XLIII).

B. Calibrating the group directly

The constructive alternative to Assumption 1 is to run split conformal on the grouped unit, same key and protocol, so that only calibration/test group exchangeability is required. Built and measured over two group statistics, five windows and two seeds (Table XLII), it discharges the premise but fails as a repair. Group exchangeability is still a deployment assumption, and our threat model is one under which test groups are partly adversary-constituted: insertion instantiates endpoint pairs that enter the stream as true nulls, so the group-formation process is something the adversary moves. The max statistic makes that fragility acute: its rank correlation with arity is $+0.23$ – $+0.41$ across the ten cells, and while marginal coverage is roughly right (0.55 – $2.32\times$ nominal), arity-conditional coverage runs from $0.13\times$ to $24.2\times$ nominal under max against 0.00 – $5.6\times$ under the mean. Stratifying calibration on arity restores a per-stratum guarantee and leaves each stratum a cold-start window of 2–52 hypotheses. The robustness and the fragility are one fact: appending cannot lower a maximum, which is why Theorem 3 does not run against it and why the arity dependence is there at all.

Grant the repair its premise and it still cannot alert. Group calibration replaces 1.8–2.4M flows by 37–57k groups, so M falls 40 – $49\times$ at unchanged T : the margin moves from $+0.067$ to -0.977 at the primary window, no window is feasible, and detection at the primary and secondary windows falls from 18 and 13 (first-flow) to zero. At 0.55 the cold-start window is 81 hypotheses while the first of 110 firing groups arrives at position 401, and feasible-step counts match $\lfloor (\alpha M(R+1)/\zeta(1.6))^{1/1.6} \rfloor$

TABLE XXXI

The padding-transfer boundary re-measured on a benign-inclusive testbed (AIT-LDSv2.0, 8 organisations, 45,001 attack flows), where victims serve heavy real benign traffic while attacked, so the padding question is answered on real ordinary-to-victim flows — no graft. Leave-one-organisation-out (train 7, test 1; a “deploy on a new network” test), in-organisation benign conformal calibration, flow features selected per fold from the training orgs with all identity and absolute-time fields excluded. “benign→victim” is the fire rate of real benign flows to an attacked host (the pad analogue), host/flow-only. Host conditioning adds little and inconsistently (median host→flow delta -0.0000 ; host higher on 1/8 orgs, max 8.9%), with shaw the one organisation where it is materially higher. That per-organisation exception is real but does not survive as a defence. A raised ordinary-to-victim fire rate is measured here in each flow’s own context; the pads an attacker actually sends carry the attacked pair’s context, and replaying the whole Surface A chain under that context suppresses every detected Shaw episode (Table XXXII). So the honest reading is that host conditioning is usually negligible here and can look decisive on an individual organisation, but replaying the chain end-to-end at all eight organisations it does not stop the attack on most episodes once the pad’s own context is modelled, and where it does the binding constraint is the number of pads required (Table XXXII). This also settles the LSPR23 side: the 0.43 grafted firing at the stress window comes from a non-black-box pad pool that is mostly the attacker’s own flood (Table XXX). AIT has no high-volume flood, which remains open. The flow-only folds are audited under both within-bucket orders in Table XXXIII: the canonical order detects fewer episodes and its alerts are cheaper to suppress, as on LSPR23. Rows sequenced by first-flow arrival are conditional on the detected set that order produces on this testbed.

organisation	AUROC h/f	atk. recall	benign→victim % (h/f)
fox	0.995/0.996	0.93	0.00/0.01
harrison	0.975/0.973	0.86	0.01/0.01
russellmit	0.988/1.000	0.47	0.00/0.02
santos	0.948/0.980	0.36	0.00/0.01
shaw	0.955/0.932	0.47	8.92/0.05
wardbeck	0.969/0.907	0.39	0.14/0.01
wheeler	0.999/0.998	0.97	0.01/0.01
wilson	0.974/0.984	0.85	0.07/0.00

everywhere. Calibration groups containing any malicious flow are excluded, dropping up to 26.1% of calibration flows in groups whose mean arity runs to $45.4\times$ the benign mean, a bias that favours firing, so the negative result survives its own bias.

C. Calibration contamination

At $k = 1$ the conformal rule is one order statistic, the calibration maximum, so contamination acts only through it. Injecting j attack flows above that maximum hands the threshold to the attacker once $j \geq k$, so tolerable adversarial mislabels are exactly $k-1 = 0$. One mislabelled flow above the calibration maximum takes episode recall to 0.000, from 0.011 at 0.55 and 0.133 at 0.85 under the canonical order (0.065 and 0.282 under first-flow arrival). A random mislabel is milder but not much: median recall 0.004 at 0.55 and 0.035 at 0.85, with zero recall in 22.5% and 32.25% of draws, at a scale $\varepsilon^* = 1/(Nq_0)$ of 1.11–1.31 flows. A contaminating flow below the maximum only inflates the p-value denominator, degrading the guarantee to at most $q(1 + \varepsilon)$, so power dies at one flow while validity barely moves. The feasibility margin is not a safety

TABLE XXXII

Full suppression pipeline on the benign-inclusive AIT testbed: per organisation, a leave-one-organisation-out detector, in-organisation benign calibration preceding the first attack, two-hour host-pair episodes, e-LOND, and each detected malicious episode suppressed by replaying real ordinary-to-victim flows against the running controller level (200 draws). Rows are sequenced by first-flow arrival and are conditional on the detected set that order produces; the flow-only folds under both orders are Table XXXIII. Design and reading: Appendix E-D.

org	det.	pad fire	$r_{t,emp}^*$	r_t^* (closed)	suppr.	succ.
flow-only detector						
fox	13/14	1.3e-04	277	277	13/13	1.00
harrison	10/14	1.8e-04	1,412.5	1,412.5	10/10	1.00
russellmitchell	14/14	5.2e-04	3,551.5	3,365.5	14/14	0.97
santos	9/9	4.7e-04	212	212	8/9	0.79
shaw	3/6	0.0e+00	404	404	3/3	1.00
wardbeck	9/9	3.2e-04	13,142	8,294	9/9	0.90
wheeler	15/16	2.1e-04	1,609	1,609	15/15	1.00
wilson	12/13	4.3e-05	793	793	12/12	1.00
host-conditioned detector, end-to-end						
fox	9/14	0.0e+00	115	55	9/9	1.00
harrison	7/14	2.9e-04	792.5	692	6/7	0.86
russellmitchell	9/14	0.0e+00	1,140	1,140	9/9	1.00
santos	5/9	0.0e+00	43	43	5/5	1.00
shaw	3/6	7.0e-02	269	269	3/3	1.00
wardbeck	9/9	1.3e-03	8,471.5	8,263.5	8/9	0.74
wheeler	15/16	3.6e-04	1,609	1,486	11/15	0.61
wilson	6/13	0.0e+00	355	355	6/6	1.00

Notes. “det.”: e-LOND true detections over malicious episodes; “pad fire”: the real ordinary-to-victim firing rate as observed; “ $r_{t,emp}^*$ ”: the empirical median replay cost over suppressible episodes; “ r_t^* (closed)”: the zero-pad closed-form lower bound over the same episodes ($\leq$ empirical by construction); “suppr.”: detected episodes for which at least one of the 200 draws brought the group below threshold inside the scored range — a possibility count, not a reliability one; “succ.”: the mean per-draw success, which is the reliability the count does not carry. An episode outside “suppr.” had no successful draw within that budget, which does not establish that it cannot be padded.

indicator: it stays at $+0.436$ throughout the sweep that destroys detection.

Appendix G Adaptive Controller-State Extension

Question answered. A controller can escape the finite horizon by conditioning its spending index on observed evidence (Appendix C-B). Does that escape create a new attack surface? Takeaway. It does: the attacker supplies the evidence that drives the state. The result is mechanism-specific and conspicuous in volume, which is why it is an extension rather than one of the paper’s main empirical claims.

The padding attack of Section V manipulates the composition of one hypothesis. A distinct surface appears when a controller escapes Theorem 1 by conditioning its spending index on observed evidence. ADDIS advances its index on selected non-candidates, whose p-values lie in $(\lambda, \tau]$, and never rejects them. Concretely it offers $\alpha_t = \min\{\lambda, (\tau - \lambda)[W_{\gamma S^t - C_{0+}} + \dots]\}$ at $\lambda = 0.25$, $\tau = 0.5$, where $S^t - C_{0+}$ counts exactly those selected non-candidates and γ is the 0-indexed $(j+1)^{-1.6}/\zeta(1.6)$; the attack moves that index and nothing else (Appendix B). An attacker who prepends groups in the interval therefore

TABLE XXXIII

The AIT transfer under both within-bucket orders, flow-only folds.

This is a resequencing; the leave-one-organisation-out feature selection, the per-fold detector fit, the scores, the chronological in-organisation calibration and the e-values are all computed before episodes are ordered, so the two arms differ by the permutation alone; the canonical arm uses the same key-hash rule as the LSPR23 results. The stored first-flow arm is re-derived here as a read-back check that this is the same chain and not merely a similar one (it reproduces 84/85). The canonical order detects fewer episodes, as on LSPR23, and the alerts it does issue are cheaper to suppress — per-organisation median $r_{t,\text{emp}}^*$ falls from 212–13142 to 12–4650 flows. The host-conditioned arm runs at all eight organisations and is left on first-flow (Table XXXI). “det./mal.” counts true detections against malicious episodes; the eight per-organisation runs issue 83 rejections under the canonical order and 89 under first-flow, carrying 4 and 4 false discoveries respectively (pooled realised FDP 0.048 and 0.045); costs are empirical replay medians $r_{t,\text{emp}}^*$, at or above the zero-pad lower bound r_t^* (Table XXXII lists both for the first-flow arm).

organisation	det./mal.	median $r_{t,\text{emp}}^*$	suppressible	per-draw succ.
canonical order — 78/79 suppressible				
fox	11/14	12	11/11	1.00
harrison	9/14	50	9/9	1.00
russellmitchell	14/14	4,650	13/14	0.93
santos	8/9	35.5	8/8	1.00
shaw	3/6	26	3/3	1.00
wardbeck	8/9	65	8/8	1.00
wheeler	15/16	192	15/15	1.00
wilson	11/13	274	11/11	1.00
first-flow order — 84/85 suppressible				
fox	13/14	277	13/13	1.00
harrison	10/14	1412.5	10/10	1.00
russellmitchell	14/14	3551.5	14/14	0.97
santos	9/9	212	8/9	0.79
shaw	3/6	404	3/3	1.00
wardbeck	9/9	13,142	9/9	0.90
wheeler	15/16	1,609	15/15	1.00
wilson	12/13	793	12/12	1.00

advances the index without spending any discovery of its own, and can force the future level below the conformal floor.

Theorem 6 (ADDIS state budget). Under ADDIS’s 0-indexed $\gamma_j = (j+1)^{-1.6}/\zeta(1.6)$ with $0 < \lambda < \tau < 1$, rank- k conformal evidence, precursors injected at index 0, and a cap at or above the conformal floor, $\lambda \geq k/(|\mathcal{C}|+1)$, the number of precursor hypotheses that silences every later target is B^* , the smallest D with

$$D+1 > \left\lceil \frac{(\tau-\lambda)W(|\mathcal{C}|+1)}{k\zeta(1.6)} \right\rceil^{1/1.6},$$

where $W = w_0$ before the first rejection. After $R \geq 1$ rejections ADDIS’s reward terms sit at their own selected-non-candidate lags instead of collapsing to one index. Their coefficients still sum to exactly αR , but each lagged term carries $\gamma_{D-\text{lag}} \geq \gamma_D$ because γ decreases, so the driving level is at least $(\tau-\lambda)\alpha R\gamma_D$ and feasibility survives longer. The displayed expression evaluated at $W = \alpha R$ is therefore a lower bound on B^* , not an upper one — with a lag of 100 at $R = 1$ it returns 313 where $B^* = 373$. The exact case is $R = 0$, which is the state the attack is mounted in, since the precursors precede every target. The cap hypothesis is what makes the driving term rather than

the cap binding: if $\lambda < k/(|\mathcal{C}|+1)$ then $\alpha_t \leq \lambda$ is already below the floor and no target can be rejected at any D , so the budget is trivially zero and the displayed expression overstates it. Our configuration satisfies it by six orders of magnitude ($\lambda M = 4.5 \times 10^5$). Each precursor can be a group with one firing flow and enough zero-evidence flows to place its p-value in $(\lambda, \tau]$; the smallest such group has $m^* = \lfloor \lambda(|\mathcal{C}|+1)/k \rfloor + 1$ flows, which is a valid precursor whenever $m^* \leq \tau M$ — guaranteed by $(\tau-\lambda)M \geq 1$, and satisfied by six orders of magnitude here. The total injected volume is B^*m^* .

At the stress window (position 0.85, first-flow order) the closed form is verified against the implementation on the real stream: 202 precursors leave all 152 ADDIS rejections intact and 203 eliminate all of them, permanently (Fig. 6B, Table XLV). Because that window’s evidence is not a valid e-value, and the two-point structure that lets ADDIS escape is the same property that violates its conservativeness condition, this demonstrates the mechanism and prices it; it is not an FDR-valid real-data result. The attack survives a grey-box setting: the admissible group size is a τ/λ -wide interval, so a $\pm 33\%$ estimate of $|\mathcal{C}|$ suffices and unknown controller state costs about $36\times$ over-provisioning.

To isolate the mechanism from that validity failure we generate streams with genuinely conservative independent $\text{Unif}(0,1)$ null p-values and 120 strong targets. ADDIS’s empirical all-null FDR is 0.010 and its mixed-stream mean FDP is 0.047 at recall 1.00, both below $q = 0.05$. We then prepend alternative, non-null precursors in $(\lambda, \tau]$, so the assumptions on true nulls are untouched and the guarantee holds on the attacked stream. The attack silences all 120 targets at a median $B^* = 140$ over 300 streams (range 140–255; Fig. 6C, Table XLVI). The guarantee therefore remains valid while the security objective is defeated: controlling the fraction of false discoveries does not require making any discovery.

The real-stream cost is conspicuous, not cheap. At $|\mathcal{C}| = 1,813,113$ each precursor needs 453,279 flows and 203 precursors need 92,015,637, about 36.1 GB or 33.4 Mbit/s over the deployment span and $37.5\times$ the window’s own flow count; the total scales as $\lambda|\mathcal{C}|[(\tau-\lambda)w_0|\mathcal{C}|]^{0.625}/k^{1.625}$, so the calibration budget that buys feasibility also buys resistance. Suppressing one canonical e-LOND alert at the same window costs a median of 116 flows. The result is also mechanism-specific: online e-BH decides by a fixed point over the whole history rather than by a spending index, so this attack has no state variable to move. We do not claim that online e-BH is immune to every manipulation, only that Theorem 6 does not cover it. Where both surfaces apply to ADDIS they coincide: when its level cap λ binds, the minimal suppressing pad also lands in $(\lambda, \tau]$ and advances the index (101 of ADDIS’s 147 true detections; Appendix A). Against ADDIS the state attack is also the cheaper one: its level is five orders of magnitude above e-LOND’s, so padding one episode costs a median 3.4×10^7

TABLE XXXIV

Joint attack on the benign-inclusive AIT stream, flow-only detector, canonical order, seed 0, medians over 100 pad-sampling trajectories. “own” and “other” are the baseline alerts on attacker-owned and other episodes, horizon-free/horizon-aware. “no pool” counts own episodes whose prior traffic to the victim is under 20 flows. For each spending rule the attacker is run twice: from the deployment pool the per-alert arms use, and from the victim’s prior traffic only; each pair of columns gives the alerts still standing and the flows spent. Reading and the host-conditioned arm: Appendix E-D.

org	own	other	no pool	horizon-free				horizon-aware			
				deployment		prior		deployment		prior	
				left	flows	left	flows	left	flows	left	flows
fox	11/11	0/0	4/14	0	972	2	601	0	5,610	3	61,704
harrison	9/9	0/0	5/14	0	4,144	3	11,624	0	21,741	3	69,940
russellmitchell	14/14	1/1	4/14	1	68,638	4	157,922	0	5,433	4	8,677
santos	8/9	0/1	4/9	0	397	4	2,718	0	2,003	6	4,267
shaw	3/3	3/3	4/6	0	536	2	906	0	353	2	659
wardbeck	8/8	0/0	3/9	0	225	3	550	0	4,728	4	15,198
wheeler	15/15	0/0	2/16	0	93,221	1	93,830	0	32,676	1	33,121
wilson	11/11	0/0	4/13	0	76,116	5	162,476	0	19,169	4	48,641

TABLE XXXV

Reliability of the joint AIT attack over the 100 pad-sampling trajectories, canonical order, seed 0. For each organisation and spending rule, and for each attacker arm of Table V: $P(\text{all})$, the fraction of trajectories in which every attacker-owned alert is silenced; “left”, the median and 95th percentile of the residual own-alert count; “flows”, the median and 95th percentile of the flows spent. The 95th percentile is the upper quantile the stage stores (linearly interpolated, so a count can be fractional); the trajectories themselves were not retained, so no 90th percentile is available. Flow-only detector on all eight organisations; the two host-conditioned rows are the joint host arm.

org	spending	sequential oracle, deployment pool			sequential oracle, prior pool				state-free $c = \lfloor \rho \rfloor + 1$, prior pool			
		$P(\text{all})$	left med/p95	flows med/p95	$P(\text{all})$	left med/p95	flows med/p95		$P(\text{all})$	left med/p95	flows med/p95	
fox	horizon-free	1.00	0/0	972/1,366	0.00	2/2	601/601	0.00	5/5	8,896/8,896		
fox	horizon-aware	1.00	0/0	5,610/5,617	0.00	3/3	61,704/62,444	0.00	11/11	8,896/8,896		
harrison	horizon-free	1.00	0/0	4,144/4,538	0.00	3/3	11,624/11,671	0.00	8/8	29,217/29,217		
harrison	horizon-aware	1.00	0/0	21,741/21,755	0.00	3/3	69,940/70,865	0.00	8/8	29,217/29,217		
russellmitchell	horizon-free	0.00	1/1	68,638/130,582	0.00	4/5	157,922/168,760	0.00	14/14	1,242/1,242		
russellmitchell	horizon-aware	1.00	0/0	5,433/5,496	0.00	4/4	8,677/8,725	0.00	10/10	1,242/1,242		
santos	horizon-free	1.00	0/0	397/401	0.00	4/4	2,718/3,064	0.00	6/6	2,916/2,916		
santos	horizon-aware	1.00	0/0	2,003/2,039	0.00	6/6	4,267/6,307	0.00	6/6	2,916/2,916		
shaw	horizon-free	1.00	0/0	536/536	0.00	2/2	906/906	0.00	3/3	382/382		
shaw	horizon-aware	1.00	0/0	353/353	0.00	2/2	659/659	0.00	3/3	382/382		
wardbeck	horizon-free	1.00	0/0	225/225	0.00	3/3	550/594	0.00	4/4	5,400/5,400		
wardbeck	horizon-aware	1.00	0/0	4,728/4,789	0.00	4/4	15,198/16,887	0.00	8/8	5,400/5,400		
wheeler	horizon-free	1.00	0/0	93,221/93,382	0.00	1/1	93,830/94,031	0.00	14/14	40,524/40,524		
wheeler	horizon-aware	1.00	0/0	32,676/32,712	0.00	1/1	33,121/33,149	0.00	3/3	40,524/40,524		
wilson	horizon-free	1.00	0/0	76,116/76,144	0.00	5/5	162,476/162,476	0.00	11/11	20,334/20,334		
wilson	horizon-aware	1.00	0/0	19,169/19,175	0.00	4/4.05	48,641/48,641	0.00	11/11	20,334/20,334		
shaw (host)	horizon-free	1.00	0/0	28/28	0.00	2/2	258/258	0.00	2/2	382/382		
shaw (host)	horizon-aware	1.00	0/0	7/7	0.00	2/2	144/144	0.00	2/2	382/382		
wilson (host)	horizon-free	1.00	0/0	73,337/73,397	0.00	1/1	73,454/73,505	0.00	6/6	20,334/20,334		
wilson (host)	horizon-aware	1.00	0/0	16,975/17,010	0.00	1/1	17,187/17,261	0.00	3/3	20,334/20,334		

flows and state manipulation wins from the third episode on.

Appendix H

Secondary Operational Baselines and Negative Results

A. Matched operating points

Comparing a $q = 0.05$ procedure against an arbitrary classifier threshold measures nothing, so Tables XLVII and XLVIII place every method on the achievable frontier of the episode-max-score threshold family at its own budget, an ex-post oracle whose thresholds use the test labels. Every method sits within 0.047 recall of the best achievable point at its budget, so the comparison that matters is choice of point: at the primary window e-LOND

operates at 3 alerts, recall 0.011, FDP 0.000 under the canonical order (18 and 0.065 under first-flow arrival), while an ex-post oracle at the same zero error reaches recall 0.378 on 104 alerts. Sweeping q twenty-fold moves median recall by 0.088, while switching the spending sequence at fixed q moves it by 0.188 (first-flow paired sweep, Table XII): formal error control can reduce the operator’s freedom over the workload–recall tradeoff.

B. Analyst-feedback controllers

The three feedback baselines of Table XLIX are threshold controllers, not FDR procedures. Each carries a single state $u_t \in [0, 1]$, a position in the calibration score distribution: it alerts on episode t when that episode’s maximum flow score exceeds the u_t -quantile of the calibration scores,

TABLE XXXVI

Corollary 2’s premise tested on the joint-attack AIT cells, canonical order, seed 0. “max other $\div T/\alpha$ ”: the largest non-attacker group evidence divided by the horizon-uniform cold-start threshold (≥ 1 means the premise fails on that stream); “ $n \geq T/\alpha$ ”: non-attacker episodes at or above it. “other alerts (at $R=0$)”: baseline non-attacker alerts under horizon-free / horizon-aware spending, with the number that clear the $R = 0$ level at their own position in parentheses; the rest fire only at a level raised by earlier rejections. “no pool”: own episodes with fewer than 20 prior templates, never padded. “pads fired”: median share of admitted pads that carry the ceiling rather than zero, for the state-free multiplier (regime-independent stream) and the sequential oracle (horizon-free / horizon-aware), both from the prior pool. “left”: median residual own alerts, horizon-free / horizon-aware. Under horizon-free spending the multiplier is a heuristic transfer of a result proved for horizon-uniform spending.

org	ρ	premise: other hypotheses			own no pool	pads fired (prior pool)		left (prior pool)	
		max other $\div T/\alpha$	$n \geq T/\alpha$	other alerts (at $R=0$)		c_{int}	sequential	c_{int}	sequential
fox	1.61	0	0	0 (0) / 0 (0)	4/14	3.76%	0.00% / 3.91%	5/11	2/3
harrison	3.48	0	0	0 (0) / 0 (0)	5/14	3.50%	2.51% / 3.87%	8/8	3/3
russellmitchell	2.99	1.497	1	1 (1) / 1 (1)	4/14	0.00%	0.01% / 0.05%	14/10	4/4
santos	4.03	0.336	0	0 (0) / 1 (0)	4/9	5.97%	11.35% / 8.57%	6/6	4/6
shaw	2.29	2.289	2	3 (1) / 3 (2)	4/6	0.00%	0.00% / 0.00%	3/3	2/2
wardbeck	5.48	0	0	0 (0) / 0 (0)	3/9	2.12%	7.73% / 4.57%	4/8	3/4
wheeler	3.31	0.019	0	0 (0) / 0 (0)	2/16	0.05%	0.05% / 0.05%	14/3	1/1
wilson	2.95	0	0	0 (0) / 0 (0)	4/13	0.13%	0.31% / 1.05%	11/11	5/4
shaw (host)	2.29	2.289	4	1 (0) / 6 (4)	4/6	0.00%	0.00% / 0.00%	2/2	2/2
wilson (host)	2.95	0	0	0 (0) / 0 (0)	4/13	0.00%	0.00% / 0.01%	6/3	1/1

TABLE XXXVII

Both attacks in operational units (position 0.85 — the known-invalid stress window; these are attack costs, which are measured against the controller’s level and do not rest on the evidence being a valid e-value — black-box padding pool). Wire bytes are the per-flow median; rate is priced over the relevant span (7200s window for padding, the position’s own deployment span for the state attack). “ $\times$ window” and “ $\times$ dataset” are flow-count ratios. Each padding row names the procedure whose level it is priced against, whether that level is the running $1/\alpha_t$ at the episode’s own step or the static cold-start τ , and the detected set it is summed over, because these differ by two orders of magnitude and are easily confused, and the within-bucket order the detected set was produced under. Row 1 is the headline cost: the median over e-LOND’s own 34 detections at 0.85 under the canonical order, priced at the running level — the value Table III quotes. Row 2 is the same quantity over the 72 detections of the first-flow arrival order, the optimistic upper bound over the orders we audit; it is $45\times$ dearer because that order detects more episodes and detects them deeper into the level sequence, where $1/\alpha_t$ is larger. Both appear in Table XXIV. Rows 3–4 are the cheap extreme: r_{90} over the 257 episodes of positions 0.62 and 0.85 (two seeds) at which all five pools are defined, priced against the static level- w_0 threshold T/w_0 , first-flow. Row 5 multiplies that static median by ADDIS’s 147-episode detected set (Table XLV), so it mixes an e-LOND-scale per-episode cost with an ADDIS-scale count; it is the cheapest reading of “suppress everything” and is reported only for contrast with the last two rows. Every row below the first is under first-flow arrival.

attack budget	flows	wire bytes	rate	$\times$ window	$\times$ dataset
e-LOND, running $1/\alpha_t$: one alert (median), canonical	116	45.5 KB	50.5 bit/s	4.7×10^{-5}	7.1×10^{-6}
same, first-flow (optimistic upper bd.)	5,302	2.0 MB	2265.8 bit/s	2.1×10^{-3}	3.2×10^{-4}
level-up static $\tau=T/w_0$: one alert (median)	34	13.3 KB	14.8 bit/s	1.4×10^{-5}	2.1×10^{-6}
level-up static $\tau=T/w_0$: one alert (p90)	227	89.0 KB	98.9 bit/s	9.3×10^{-5}	1.4×10^{-5}
static median $\times$ ADDIS’s 147 detections (mixed)	4,998	2.0 MB	2176.9 bit/s	2.0×10^{-3}	3.1×10^{-4}
ADDIS, running $1/\alpha_t$: one episode (median)	34,465,311	13.5 GB	15.0 Mbit/s	$14.05 \times$	$2.11 \times$
ADDIS spending-state attack (B^* precursors)	92,015,637	36.1 GB	33.4 Mbit/s	$37.51 \times$	$5.63 \times$

and moves u_t only when an analyst’s disposition of an earlier alert arrives, after a wall-clock delay L . Writing $\widehat{\text{FDP}}_t$ for the false fraction over the last 200 dispositions received so far and $\varepsilon_t = \widehat{\text{FDP}}_t - q$,

$$\text{P: } u_{t+1} = u_t + \eta_P \varepsilon_t,$$

$$\text{PI: } u_{t+1} = u_t + \eta_P \varepsilon_t + \eta_I \sum_{r \leq t} \varepsilon_r,$$

$$\text{AQT: } u_{t+1} = u_t + \eta [(1-q)\mathbf{1}\{\text{false}\} - q\mathbf{1}\{\text{true}\}],$$

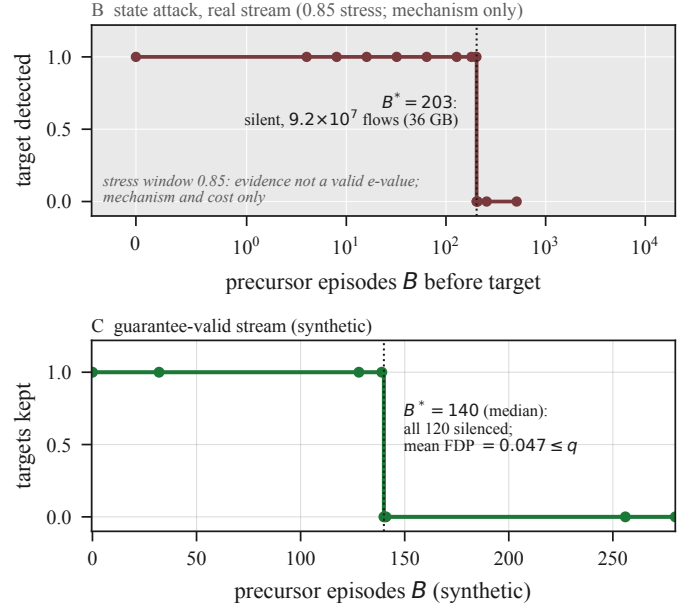


Fig. 6. Manipulating ADDIS’s spending state. B: on the real stress window, precursors silence every rejection permanently at exactly the Theorem 6 threshold — on evidence that is not a valid e-value, so this measures the mechanism and its cost, not an FDR-valid outcome. C: the same poisoning on synthetic streams where ADDIS’s guarantee genuinely holds still silences every target.

with the PI integral clamped against windup and u clipped to $[0, 1]$. Gains are chosen per window, delay and controller by grid search ($\eta_P \in \{0.005, 0.02, 0.05, 0.2\}$, window $\in \{50, 200\}$, $u_0 \in \{0.99, 0.999\}$, AQT $\eta \in \{10^{-4}, 10^{-3}, 5 \times 10^{-3}, 0.02\}$, $\eta_I = \eta_P/10$) minimising ($|\text{FDP} - q|$, –recall) on the realised test labels, an ex-post oracle, so the reported performance is an upper bound, the conservative direction for the negative conclusion. Under first-flow order they reach FDP 0.402 (P, PI) and 0.523

TABLE XXXVIII

Joint zero-evidence controller-state rerun: LSPR23, detector seed 0, canonical order, two-hour host-pair episodes, e-LOND, both windows and both spending regimes. Upper panel: the static, sequential and state-free attackers of Section V-D; lower panel: the per-host-pair volume cap against the sequential attacker. Design and reading: Appendix E-F.

window	regime	true det. (FD)	median $r_t^* / \sum_{t \in \mathcal{D}} r_t^*$	static: remain	$J_{\text{seq}} (\div \sum_{t \in \mathcal{D}} r_t^*)$	med. pad (padded)	spared	padded arity (per-alert)	% episodes $\geq$	c_{int}
0.55	horizon-free	3 (0)	24 / 80	0	30 (0.375)	15	1	44.5 (63)	6.1%	2
0.55	horizon-aware	105 (0)	2,946 / 627,495	0	5,650 (0.009)	48.5	13	91 (2,989)	3.8%	3
0.62	horizon-free	11 (0)	6 / 926	0	859 (0.928)	859	10	881 (20)	0.5%	41
0.62	horizon-aware	107 (1)	3,908 / 977,567	0	9,992 (0.010)	33	4	55 (3,928)	4.3%	3

0.55 horizon-aware (105 true det.)				0.62 horizon-aware (107 true det.)			
cap n	fire (struct.+casc.)	per-alert fire	benign $> n$	fire (struct.+casc.)	per-alert fire	benign $> n$	
30	105 (87+18)	105	7.36%	107 (85+22)	107	7.30%	
100	96 (44+52)	100	3.40%	79 (33+46)	103	2.86%	
300	19 (1+18)	91	1.04%	60 (22+38)	95	1.59%	
1,000	0 (0+0)	80	0.50%	0 (0+0)	82	0.41%	
3,000	0 (0+0)	52	0.21%	0 (0+0)	59	0.19%	

Notes. “true det. (FD)”: true detections and false discoveries on the unperturbed run; “median $r_t^* / \sum_{t \in \mathcal{D}} r_t^*$ ”: the per-alert oracle cost and the per-alert sum of Table II. “remain”: true detections still rejected after the rerun (the one 0.62 horizon-aware false discovery also disappears). “padded” statistics are over the alerts the sequential attacker actually padded; “spared” alerts fire only at a level raised by earlier rejections and need no pad once those are gone. “padded arity (per-alert)”: the sequential attacker’s median padded arity, with the per-alert reading of Section V-D in parentheses; “% episodes $\geq$ ”: the share of deployment episodes at least that large. $c_{\text{int}} = \lfloor \rho \rfloor + 1$ (Corollary 2). Lower panel: “fire (struct.+casc.)” splits the surviving alerts into structural (the pad would not fit even at $R = 0$) and cascade victims (it would have fitted at $R = 0$); “per-alert fire” is $m_t + r_t^* > n$ on the unperturbed pads; “benign $> n$ ”: the share of benign deployment episodes whose arity exceeds the cap, i.e. the episodes an implemented admission policy would alter; no baseline episode is truncated or rescored in this experiment. Caps are the sampled grid; nothing is claimed between grid points.

TABLE XXXIX

What each result depends on. “Valid e ” is whether the claim needs the evidence to be a valid e-value; “A1” is Assumption 1; “labels / ownership” is whether dataset labels enter, including as the oracle for which episodes the attacker owns. Windows are named by position: 0.55 primary, 0.62 secondary, 0.85 stress (evidence not a valid e-value; mechanism and cost only); AIT is the second dataset.

Result	valid e	A1	labels / ownership	windows
Finite horizon, exchange rate (§III)	no	no	no	all (algebraic)
Granularity buys feasibility (§IV)	no	no	no	all
Alert blur (§IV)	no	no	yes	0.55, 0.85
Padding impossibility (Thm. 3)	no	no	no	—
Suppression and r_t^* of a specified alert (§V)	no	no	no	0.55, 0.62, 0.85, AIT
Aggregates over $\mathcal{D}$, incl. $\sum_{t \in \mathcal{D}} r_t^*$ (§V)	no	no	yes	0.55, 0.62
Joint rerun J_{seq} , whole-window multiplier (§V)	no	no	yes	0.55, 0.62
Replay suppression counts, recall, FDP (§V)	no	no	yes	0.55, 0.62, 0.85, AIT
Nominal e-LOND FDR guarantee	yes	yes	no	0.55, 0.62
ADDIS state attack, real stream (Appendix G)	no	no	yes	0.85 (mechanism)
ADDIS silenced, guarantee valid (Appendix G)	yes	—	no	synthetic

(AQT) at fifteen-minute disposition delay, agree with one another by four hours, and reach the no-feedback threshold (0.841) only by eight: a statement about delay on these controllers, not about feedback-aware testing [42], whose input our measurement prices.

C. Negative results

Inflating a benign group cannot create a discovery at a fixed level. The argument is pathwise, not distributional: for the arithmetic mean with evidence sum S over m members, appending r zero-evidence flows gives $S/(m+r) \leq S/m$, so a group that is not rejected at a fixed level with its existing evidence is not rejected after padding either. The bound $\mathbb{E}[\text{Ev}(G) \mid \mathcal{M}_j] \leq 1$ of Assumption 1 does not by itself give this: an expectation bound permits large individual realisations (an e-value equal to 100 with probability 0.01 and zero otherwise

has mean one and crosses a threshold of 20), which is exactly why a benign group can be rejected and why the FDR guarantee is a rate, not an exclusion. Fragmentation leaves expected group evidence unchanged, and no more. For a pure-attack group whose m flows fire independently with probability ϕ , $\mathbb{E}[\text{Ev}(G)] = \phi M$ does not depend on m ; the rejection probability does, $\Pr(\text{Ev}(G) \geq \theta) = \Pr(\text{Binomial}(m, \phi) \geq \lceil m\theta/M \rceil)$, and splitting a campaign also changes the number of hypotheses and the controller state, so we make no analytic claim about its effect on detection. Boosting recovers nothing for two-point evidence: the optimal e-to-e boosting factor for a $\{0, M\}$ -valued e-value is $b^* = 1$ by construction. Raising k removes the operating point rather than trading power for validity; randomised smoothing removes the floor but not the reliably-detected set (Appendix C-B); asymmetric

TABLE XL

Benign firing rate vs nominal by window (seed 0). “ratio” is the measured/nominal benign firing rate at $k = 1$, and “95% CI” is its exact Clopper–Pearson binomial interval given the observed firing count out of the benign flows. The interval separates the theoretical guarantee from the empirical diagnostic: at the four non-stress windows only 0–3 benign flows fire (expectation ≈ 1), so the interval is wide and includes 1 — the diagnostic cannot prove validity, only fail to reject it; at position 0.85 the interval excludes 1 by orders of magnitude, a clear violation. The last four columns are the ratio at rank depth $k \in \{1, 10, 100, 1000\}$. This diagnostic is order-invariant: it scores benign flows against the calibration threshold, with no controller and so no stream sequence to permute.

position	AUROC	fired benign	ratio	95% CI	ratio at rank depth k			
					$k=1$	$k=10$	$k=100$	$k=1000$
0.55	0.916	1	1.07	[0.03, 5.96]	1.07	0.64	0.51	3.85
0.62	0.799	1	1.16	[0.03, 6.45]	1.16	7.17	0.91	5.09
0.70	0.914	3	3.79	[0.78, 11.06]	3.79	0.50	0.09	0.11
0.77	0.828	2	2.72	[0.33, 9.82]	2.72	0.95	0.98	2.41
0.85	0.999	46	50.90	[37.26, 67.89]	50.90	9.18	4.49	6.80

TABLE XLI

Metadata-stratified interrogation of Assumption 1 (seed 0; order-invariant: per-flow firings against a fixed threshold, not controller decisions). Rows are windows; cells are (stratum, depth) pairs read at $k \in \{1, 10, 100, 1000\}$; “surviving BY” counts Benjamini–Yekutieli rejections over all cells, at $k=1$, and of stratum contrasts. Design and reading: Appendix F-A.

position	strata	cells		surviving BY			marginal ratio [95% CI]	
		total	untested	ratio	at $k=1$	contrast	$k=1$	$k=1000$
0.55	28	112	90	6	0	5	1.07 [0.00, 4.56]	3.85 [1.69, 8.70]
0.62	28	112	82	13	0	9	1.16 [0.00, 94.53]	5.09 [1.74, 11.26]
0.70	26	104	98	0	0	3	3.79 [0.00, 9.02]	0.11 [0.04, 0.28]
0.77	26	104	86	5	0	9	2.72 [0.00, 12.90]	2.41 [1.45, 3.80]
0.85	26	104	72	16	5	6	50.90 [3.70, 204.68]	6.80 [3.38, 13.04]

Notes. “untested”: cells firing fewer than ten times, which carry no test but enter the Benjamini–Yekutieli family at $p = 1$. Ranks break ties conservatively, so each marginal ratio is a lower bound on the anti-conservatism; the interval is a 95% percentile interval from the joint resampling. “contrast”: strata departing from their own window’s marginal on the same replicates.

weighting is not a free mitigation (Section VI); stratified (Mondrian) conformal does not repair the position-0.85 anomaly; and the published α -death fix (mem-e-LORD) is identical to e-LORD before the first rejection.

TABLE XLII

What calibrating the grouped unit costs (seed 0; Appendix F-B). Left: the shipped construction, calibrated on benign flows. Right: split conformal on the group, same key and protocol, which would need only calibration/test group exchangeability. The evidence ceiling falls by the flows-per-group ratio ($40\text{--}49\times$) with T unchanged, so the margin collapses and detection at the primary and secondary windows goes to zero. That is the feasibility boundary, not a weak statistic: the cold-start window is tens of hypotheses and the first firing group arrives hundreds of positions later (at 0.55, 110 groups fire and none lands inside the window); feasible-step counts match $\lfloor (\alpha M(R+1)/\zeta(1.6))^{1/1.6} \rfloor$ at every configuration. The last two columns qualify the repair without refuting it: the max statistic — the padding-robust one — is strongly arity-dependent, and arity is what the adversary controls, so group exchangeability may be fragile under group-formation shift, though the correlation alone neither refutes it nor shows that it fails; the mean is arity-invariant to within noise and is exactly the rule Theorem 3 defeats. Position 0.85 is the stress window and its detections are not guarantee-bearing (Appendix F-A). The flow-calibration columns use the first-flow within-bucket order (the optimistic upper bound), not the canonical order of Table III; the group arm matches its tie-break to the same order, so the two remain comparable (Table XXI).

Position	flow calibration			group calibration						arity corr.		
	C	margin	det.	C	margin	det.	max	det. mean	cold start	1st fire	max	mean
0.55	2,448,993	+0.067	18	52,279	−0.977	0		0	81	401	+0.35	+0.04
0.62	2,449,031	+0.243	13	53,897	−0.973	0		0	83	349	+0.36	+0.01
0.70	2,287,988	+0.536	30	57,093	−0.962	1		0	86	6	+0.23	-0.12
0.77	2,116,418	+0.671	31	48,869	−0.961	2		1	78	11	+0.34	-0.01
0.85	1,813,113	+0.436	72	36,944	−0.971	42		42	65	4	+0.30	-0.07

TABLE XLIII

Adjudicated alert audit against the red-team task record (position 0.85, the known-invalid stress window). Table XL is what establishes that invalidity, from the benign firing rate; this is a follow-up label-quality audit at the same window, not a guarantee-bearing result and not the diagnostic that made the call. Panel (a): adjudicated verdict counts with their label split. Panel (b): false-discovery proportion by method under the label, the pre-registered audit lower bound, and the strict lower/upper bounds. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

(a) adjudicated verdicts			
verdict	n	labelled mal.	labelled ben.
clearly malicious	49	46	3
probably malicious	102	100	2
ambiguous	0	0	0
probably benign	1	1	0
clearly benign	0	0	0

(b) false-discovery proportion					
method	R	$\text{FDP}_{\text{label}}$	$\text{FDP}_{\text{audit}}^{\text{lo}}$	$\text{FDP}_{\text{strict}}^{\text{lo}}$	$\text{FDP}_{\text{strict}}^{\text{hi}}$
e-LOND/uniform	151	0.026	0.007	0.000	0.026
ADDIS/poly	152	0.033	0.007	0.000	0.026
union	152	0.033	0.007	0.000	0.026

TABLE XLIV

Bates calibration-conditional adjustment ($n_{\text{cal}} = 1000$): it fixes conditional validity at a power cost set by the same quantity. The first three columns are the conditional exceedance probability — analytic prediction, measured nominal, and β -adjusted; the last two are the e-LOND detection cost (median rejections, nominal vs β -adjusted). All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else. The β -bound requires $0.22\text{--}0.62$ of $|C|$ across error targets $\delta \in [0.01, 0.2]$.

k	conditional exceedance			e-LOND rej. (med)	
	predicted	nominal	β -adj.	nominal	β -adj.
1	0.368	0.370	0.092	100	62.5
10	0.459	0.439	0.084	0	0
100	0.489	0.483	0.092	0	0

TABLE XLV

ADDIS controller-state attack: $B^* = 203$ precursor episodes permanently silence the controller at position 0.85 — the known-invalid stress window, which is where the real-stream demonstration must sit, since ADDIS's horizon escape and its guarantee-invalidity are the same property on two-point conformal evidence (Table XLVI reproduces the mechanism where the guarantee does hold). The break-even is the number of detected episodes whose per-episode padding cost equals the whole one-off state attack. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

quantity	value
B^* (precursor episodes, $R = 0$)	203
per-precursor cost $\lfloor \lambda(C + 1)/k \rfloor + 1$	453,279 flows
total attack flows	92,015,637 ($\approx 9.2 \times 10^7$)
level-saturated coincidence	101 of 147
minimal pad lands in window	101 of 147
median pad per episode (alternative)	34,465,311 flows
break-even vs padding	2.67 episodes
grey-box landing band ($ \hat{C} / C $)	[0.67, 1.33]

TABLE XLVI

The ADDIS state attack on a synthetic stream where ADDIS’s guarantee genuinely holds, isolating the mechanism from the real-window validity failure. The two-point conformal evidence used elsewhere cannot supply this: its ADDIS horizon-escape (the spending index never advances) is the same property that makes its null p-values non-conservative, so no two-point stream is both escape-feasible and guarantee-valid. We therefore use genuinely uniformly-conservative null p-values ($\text{Unif}(0, 1)$, independent, the assumption ADDIS needs). The stream is synthetic, so LSPR23’s grouping plays no part and the result is order-invariant by construction. Top: ADDIS controls FDR — an expectation, not a per-realisation bound (the mixed arm’s worst single realisation is $\text{FDP} = 0.104$, above q , which the guarantee permits) — on this stream: the honest all-null test (every rejection false) gives mean $\text{FDP} = 0.010 \leq q=0.05$, and with 120 strong targets present mean $\text{FDP} = 0.047$ at recall 1.00 (200 seeds each). Bottom: the attack is adversarial state poisoning — prepending B attacker-generated alternative (non-null) precursor episodes with $p \in (\lambda, \tau]$. ADDIS constrains only true-null p-values, so alternatives may take any value: the legitimate nulls stay $\text{Unif}(0, 1)$ and ADDIS’s guarantee holds on the attacked stream, not merely the clean one. These selected non-candidates advance the spending index and silence every rejection at a median $B^* = 140$ over 300 guarantee-valid streams (range 140–255; the tail from a rare null false positive that transiently restores wealth). So even where ADDIS’s guarantee holds after the attack, an adversary who only adds traffic drives the controller to silence — a property of the mechanism, not an artefact of the dataset already violating the assumption.

	mean FDP	$\leq q=0.05?$
all-null (every rejection false)	0.010	yes
mixed (120 targets, recall 1.00)	0.047	yes
state poisoning: median $B^*=140$ (range 140–255) silences all		

TABLE XLVII

Matched operating points at the primary analysis window (position 0.55, two-hour grouping, $T = 57,368$ episodes of which 275 are malicious; every FDR statement conditional on Assumption 1 and Lemma 1). Rows are under the canonical within-bucket order, as Table III is, with the first-flow arm — the optimistic upper bound — beneath for the four rows the order moves. The stress-test window 0.85 is Table XLVIII; the negative result, every method within 0.047 recall of the frontier, holds at both and under every order audited. The frontier is an ex-post oracle (thresholds use the test labels) and is measured order-invariant here, so it is quoted once. The slot rule is an expectation over slots and the no-feedback threshold never updates, so both are order-free by construction and verified unchanged.

Method	alerts	FDP	recall	gap
canonical metadata-hash order				
online FDR (mean rule)	3	0.000	0.011	+0.000
online FDR (slot rule)	91.0	0.000	0.331	+0.000
feedback, $L=0$	117	0.068	0.396	+0.000
feedback, $L=20$	137	0.204	0.396	+0.015
feedback, $L=50$	163	0.331	0.396	+0.044
fixed threshold	1,047	0.836	0.625	+0.000
first-flow arrival (optimistic upper bound; unmoved rows omitted)				
online FDR (mean rule)	18	0.000	0.065	+0.000
feedback, $L=0$	119	0.076	0.400	−0.004
feedback, $L=20$	135	0.185	0.400	+0.004
feedback, $L=50$	150	0.267	0.400	+0.040
frontier, $\text{FDP}=0.000$	104	0.000	0.378	—
frontier, $\text{FDP}=0.050$	110	0.009	0.396	—

TABLE XLVIII

Matched operating points at the stress-test window (position 0.85, two-hour grouping, $T = 31,568$ episodes of which 255 are malicious). FDP is measured against labels at a window where the evidence is not a valid e-value (Appendix F-A). Rows are under the canonical within-bucket order, with the first-flow arm beneath for the 4 rows the order moves; the frontier is measured order-invariant and quoted once. The companion primary-window table is Table XLVII.

Method	alerts	FDP	recall	gap
canonical metadata-hash order				
online FDR (mean rule)	34	0.000	0.133	+0.000
online FDR (slot rule)	126.6	0.001	0.496	−0.003
feedback, $L=0$	160	0.069	0.584	+0.004
feedback, $L=20$	171	0.117	0.592	+0.024
feedback, $L=50$	190	0.184	0.608	+0.047
fixed threshold	875	0.763	0.812	+0.000
first-flow arrival (optimistic upper bound; unmoved rows omitted)				
online FDR (mean rule)	72	0.000	0.282	+0.000
feedback, $L=0$	161	0.068	0.588	+0.000
feedback, $L=20$	160	0.081	0.576	+0.012
feedback, $L=50$	171	0.140	0.576	+0.039
frontier, $\text{FDP}=0.000$	117	0.000	0.459	—
frontier, $\text{FDP}=0.050$	156	0.045	0.584	—

TABLE XLIX

Analyst-feedback controllers under wall-clock disposition delay ($q = 0.05$) at the primary analysis window (0.55); FDP is the two-seed mean. “open-loop” and “at limit” are for the P controller (open-loop is identical for PI). The final row is the long-span window at a 24 h delay, where feedback collapses to the no-feedback open-loop baseline. All counts here use the first-flow within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table III reports (Table XXI); the order shifts detections and the level an alert fires at, and shifts nothing else.

delay (h)	FDP (P)	FDP (PI)	FDP (AQT)	% open-loop	% at limit
0	0.053	0.049	0.172	0.9%	95.0%
0.25	0.402	0.402	0.523	51.4%	87.3%
1	0.514	0.514	0.592	60.5%	73.4%
4	0.671	0.671	0.671	77.8%	44.7%
8	0.834	0.834	0.834	99.8%	12.0%
long-span, 24 h	0.877	0.877	0.876	no-feedback FDP = 0.887	